

and could not spread. Difficulties of this kind arise if genes affecting behaviour have pleiotropic effects, or if they are in linkage disequilibrium with other genes.

6. Genetic relatedness is the only cause of genetic similarity between interacting individuals. Thus an allele A for altruistic behaviour could spread if individuals carrying A tend to congregate, regardless of relatedness. Note that this, too, requires pleiotropy, or linkage disequilibrium.

Box 9.2—

Relatedness in Haplo-diploids

How should Hamilton's rule be applied to haplo-diploids? There are two preliminary questions that must be dealt with:

1. How shall we value genes in males and females? In a diploid species, a gene transmitted to a son, and to a daughter, contribute equally to the fitness of a parent. Is the same true for haplo-diploids? The answer is yes. I will show this for the special case of a population with a 1:1 sex ratio. In such a population, females have, on average, one son and one daughter, and males have one daughter: females have twice as many offspring as males. However, a gene in a female is transmitted to only half her offspring, whereas a gene in a male is transmitted to all his daughters. These two effects exactly cancel out, and the expected number of copies, in any future generation, of a gene in a male equals the expected number of copies of a gene in a female.
2. How shall we estimate b and c ? It turns out that the most convenient measure is in terms of the expected number of offspring gained and lost. This has the rather odd consequence that a male wasp that sacrificed itself to ensure the survival of a female would suffer a cost of -1 , and confer a benefit of $+2$. (It would be possible to measure b and c in lifetimes, so that the values of a male and a female life were equal, but this would force us to redefine the coefficient of relatedness in an unintuitive way.)

Thus, in applying Hamilton's rule, we count costs and benefits in terms of offspring gained and lost, and we value sons and daughters equally. As for diploids, r_{IS} is the proportion of genes in the recipient, S , that are IBD to genes present in the actor, I . Figure 9.6 gives some values of r_{IS} .

Notice that $r_{\text{sister, brother}} = 1/2$, and $r_{\text{brother, sister}} = 1/4$. Let us check that Hamilton's rule works for these two cases. First, consider a gene A causing a female to lost c offspring, and causing her brother to have an additional b offspring. The loss of A genes is $c/2$ (because a female transmits the A gene to

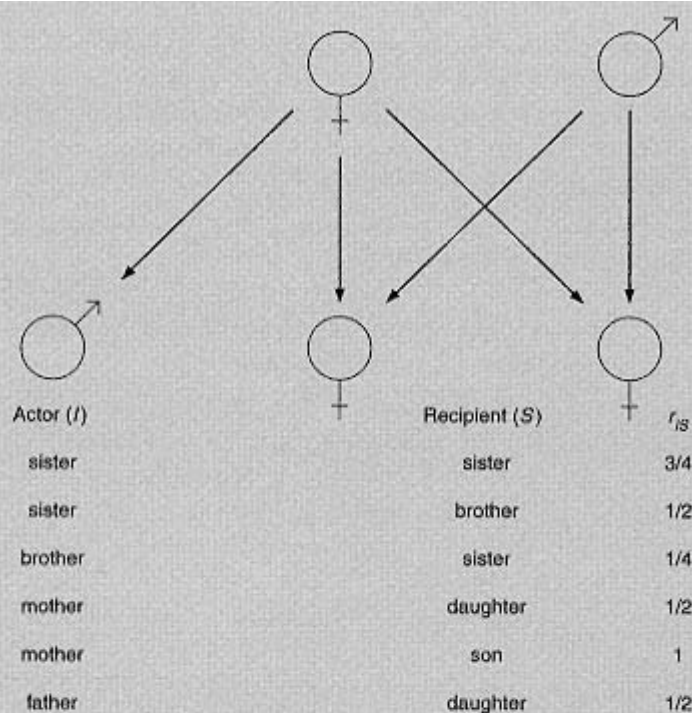


Figure 9.6 Coefficients of relatedness for haplo-diploids. As an example, consider the calculation of $r_{\text{brother,sister}}$. A gene in a female came from her mother with probability 1/2, and from her father with probability 1/2. In the former case, there is a probability 1/2 that an identical gene was transmitted to her brother; in the latter case, it certainly was not. Hence $r = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times 0 = \frac{1}{4}$.

Now consider a gene A causing a male to lose c offspring, and causing his sister to have b additional offspring. The loss of A genes is c (because all the male's gametes carry A). The female has a probability of 1/2 of carrying a gene IBD to A (thus A was certainly present in her mother, and had an even chance of being transmitted): if she does have it, she transmits it to half her gametes. Hence the gain of A genes is $b/4$, and A increases in frequency if $b/4 > c$. Again, this agrees with $rb > c$.

Thus we have a method of applying Hamilton's rule to haplo-diploids. A feature of Fig. 9.6 that has attracted attention is that a female is more closely related to her sisters than to her daughters. Hence a gene causing a female to raise sisters rather than daughters would be favoured by selection. How important has this been for the evolution of eusociality in the hymenoptera? Some people have argued that it has little relevance, because, at least in the early stages of the evolution of eusociality, a female would not distinguish the sexes of her siblings, and would raise both brothers and sisters. Her *average* relatedness to her siblings is the same as to her own daughters, and so the

genetic system does not give any special bias towards sociality. According to this argument (Andersson 1984), the reason why sociality has evolved repeatedly among the hymenoptera, and not, for example, among the diptera, is that prolonged maternal care is common even in solitary hymenoptera. The alternative view is that, when allowance is made for a bivoltine life history and a fluctuating bias in the sex ratio, haplo-diploidy can still provide a bias towards the evolution of sociality. The argument (Seger 1983; Grafen 1986) is difficult, but interesting.

The rule, $rb > c$, was given by Hamilton in a paper in which he introduced the concept of **inclusive fitness**, as an alternative to the classical fitness of population genetics. This concept is briefly explained in Box 9.3. In thinking about particular cases, however, it is usually easier to apply the rule $rb > c$, or to use a more directly gene-centred approach. Very often, a helpful way of thinking about the evolution of some behavioural trait is to imagine a gene of very low penetrance, and to ask, on those occasions when the gene is expressed (that is, when the trait appears) is the result an increase or a decrease in the number of copies of the gene?

In Box 9.4, Hamilton's rule is applied to the evolution of social behaviour in the wasp *Polistes*.

The Group as the Unit of Evolution

According to the theory of evolution by natural selection, any population of entities with the properties of multiplication, variation, and heredity will evolve in such a way that the component entities will acquire characteristics ensuring their own survival and reproduction. The relevant entities may be RNA molecules in a test tube (p. 5), or individual genes (as in the 'gene-centred' model of p. 168), or individual organisms (as in the phenotypic models of Chapter 7). If a species is divided into demes, can we regard the demes as entities that will evolve by natural selection? Thus a successful deme may split into two, and, if so, the 'daughter' demes will have some of the properties of the parent. If demes are units of evolution in a Darwinian sense, then we can expect them to evolve characteristics ensuring their own survival and reproduction. For example, demes might evolve mechanisms preventing them from outgrowing the available food, as proposed by Wynne-Edwards (1962).

The reason why we cannot safely regard demes as units of evolution is that they are themselves composed of entities that evolve: they are composed of organisms. The only reason why a deme has heredity (if we ignore cultural inheritance) is that its component organisms have heredity. All genetic differences, therefore, are subject to selection at two levels: selection between organisms within a deme, and

Box 9.3— Inclusive fitness

In population genetics, the fitness of a genotype, say AA , is the expected number of offspring produced by individuals of that genotype. Provided that we correctly ascribe fitness to genotypes, we can calculate how the population will change. Why, then, do we have to take into account the offspring produced by relatives?

One answer is that we should not, but if so, we must be very careful to calculate fitnesses correctly. Thus suppose that A is an allele for sib altruism. Then AA individuals are likely to have sibs carrying the A allele, and hence are likely to receive help: in calculating the fitness of AA , we must allow for this help received from relatives. Hamilton called the fitness calculated in this way the **neighbour-modulated** fitness. It often turns out to be difficult to calculate: the calculation in Table 9.2 was of neighbour-modulated fitness. He therefore introduced an alternative, the **inclusive fitness**. For a given genotype, say AA , the inclusive fitness is the expected value of:

(number of offspring produced by AA) - (offspring produced by AA only because of help received from relatives) + (additional offspring produced by relatives of AA , because of assistance provided by AA , weighted by the coefficient of relatedness).

It is often easier to calculate the inclusive fitness, because we only have to take into account the direct effects of individuals. Hamilton showed that, if fitnesses are defined in this way, the direction and, approximately, the rate of gene frequency change predicted is the same as that predicted by the neighbour-modulated fitness.

selection between demes. If the same traits are favoured at both levels, no difficulty arises. But if a trait is disadvantageous to the individual, but favourable to a deme whose members have the trait, the relative effectiveness of selection at the two levels can be hard to estimate.

One way of thinking about the problem is illustrated in Fig. 9.7. The environment is supposed to be divided into 'patches': these could be food plants for a herbivorous insect, or hosts for a parasite, or cowpats for a dung fly. (In fact, resources need not be patchily distributed, provided that organisms remain in coherent groups for some other reason.) Individual organisms are of two kinds A , or altruistic, and S , or selfish. A group composed of A s is more successful than one composed of S s, but, within a group, S replaces A . A patch can be in one of three states: E , or empty; S , containing a population of S s; and A , containing a population of A s. For simplicity, I omit patches containing a mixture of A and S individuals.

Box 9.4—

An Example of Kin Selection

In the social wasp, *Polistes metricus*, nests are sometimes founded by a single female, and sometimes by a pair of females. In the latter case, the co-foundresses come from a single nest in the previous summer. One, the α female, is dominant, and lays most of the eggs; the β female does most of the foraging. The following account is based on the work of Metcalf and Whitt (1977a,b), but I have omitted some complications.

First, what question are we asking? In general, we want to understand the selective forces responsible for the evolution of the observed behaviour. This requires that we know what alternative behaviours are possible, and what would be the fitness of individuals adopting them. In this particular case, we want to understand the behaviour of the β females: why do they not abandon the joint nest, and establish a nest on their own? We know that this behaviour is possible, and can estimate the fitness of individuals adopting it, because some nests are founded by single females.

Metcalf and Whitt estimated that the number of reproductives produced by a pair of foundresses is 3.1 times that produced by a single foundress: this allows for nest survival, and for the success of nests that do survive. Next, we need to know the coefficient of relatedness, r , between co-foundresses. Consider first the simple case in which only the α female lays eggs, and in which females mate only once. Then co-foundresses are full sisters, and $r = 0.75$. We can then calculate the relative numbers of genes IBD to those of an α or β female that are transmitted to the next generation, as follows:

	Genes IBD to	
	α female	β female
α and β female together	3.1	$3.1 \times 0.75 = 2.325$
α and β female act alone	$1 + 0.75 = 1.75$	1.75

Hence, if the β female abandons the joint nest and acts on her own, the number of genes IBD to hers that are transmitted falls from 2.325 to 1.75.

Things are in fact more complicated. From electrophoretic studies, r between co-foundresses was only 0.63. This was partly because approximately 20 per cent of the eggs in a joint nest were laid by the β female (I omit the complication that the fraction is different for male and female eggs), and partly because foundresses usually mate at least twice. The latter fact can be deduced by examining the offspring of a solitary foundress homozygous for a polymorphic enzyme: if she produces both homozygous and heterozygous daughters, she has mated twice. The same data showed that, even though a female had mated twice, 90 per cent of daughters had the same father.

Accepting that β females produce 20 per cent of the offspring, and that $r = 0.63$, we can recalculate as follows:

	Genes IBD to	
	α female	β female
α and β female together	$3.1 (0.8 + 0.2 \times 0.63) = 2.87$	$3.1 (0.8 \times 0.63 + 0.2) = 2.18$
α and β female act alone	1.63	1.63

Again, if a β female abandons the joint nest, this reduces the number of genes IBD to her own that are transmitted.

In Box 9.2, I discussed whether haplo-diploidy has predisposed the hymenoptera to evolve sociality. I there considered the origin of eusociality to lie in a female helping her mother to raise her sibs. In *Polistes*, a female helps her sister to raise nephews and nieces. In this case, the haplo-diploid system does have a predisposing effect. In a diploid, the productivity of a joint nest, relative to a nest with a single foundress, would have to be higher before it would pay a β female to remain and raise her sister's children rather than her own, because r between sisters is lower.

The figure shows the kind of state transition that can occur:

1. An E patch can be converted to an A or an S patch, according to whether it is colonized by an A or an S individual.
2. A and S patches may become E , because of population extinction. I have shown the arrow $S \rightarrow E$ as stronger than $A \rightarrow E$, to indicate that A populations survive better than S populations.
3. A patches are converted into S patches, if there is an S immigrant into an A patch, or (less frequently) if an A individual mutates to S .

We want to know whether A can survive in competition with S . For this to be so, the number of S patches must not increase. This in turn requires that $M < 1$, where M is the number of migrants from an S patch that successfully invade (and transform) an A patch, or which colonize an E patch, during the lifetime of an S patch, from the time it arises (by colonization of an E patch, or conversion of an A patch), to the time it goes extinct. In effect, M is the 'reproductive rate' of S patches.

For this condition to hold, the rate of migration must be low, and the survival time of patches must be short. It must also be true that new patches are colonized by one or a few individuals: if new patches were multiply colonized, the colonists would usually include an S individual, and the patch would become S .

These conditions are not impossible. They may hold, for example, for some parasites. If so, we can expect parasites to evolve traits that maximize the number of new parasites produced by a host, and not traits that make a parasite successful.

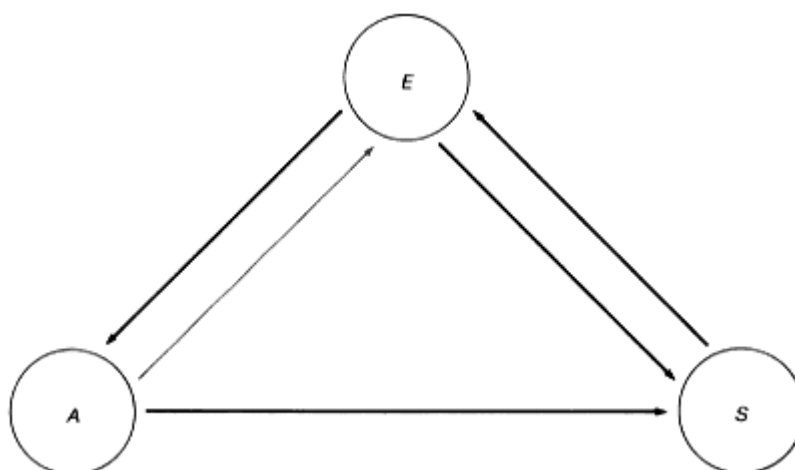


Figure 9.7

A model of group selection. Resource patches can be in one of three states: *A*, containing a population of altruists; *S*, containing a population of selfish individuals; or *E*, empty. The arrows indicate state transitions. Note that (1) an *A* patch can be converted into an *S* patch by a selfish immigrant, and (2) *S* patches are more likely to go extinct than *A* patches.

in competition with others in the same host. This may require that the host be kept alive, or at least not killed too quickly. Box 9.5 gives an example of parasite evolution.

There may be other cases in which demes are sufficiently isolated and short-lived, and are founded by sufficiently few colonists, for properties advantageous to the deme to evolve. But in most structured populations the degree of isolation is too small (see Table 8.3). Before accepting group advantage as the selective explanation of some trait, therefore, one must be satisfied that the population structure meets the rather stringent requirements.

Box 9.5—

The Evolution of the Myxoma Virus

Figure 9.8 shows the frequencies of myxoma virus strains of different virulence in Australia in 1950–51, when the virus was introduced to control rabbits, and at intervals up to 1981. Similar changes have occurred in the virus in Britain. These changes in the virus were measured by testing in a standard strain of rabbits: there has also been a slight increase in the resistance of the wild rabbits to the virus.

Evolution has led to the establishment of virus strains of an intermediate degree of virulence. The reason is as follows. The reproductive rate, R , of a virus infecting a rabbit is defined as the number of additional rabbits infected by viruses from that rabbit. R is the product of two terms:

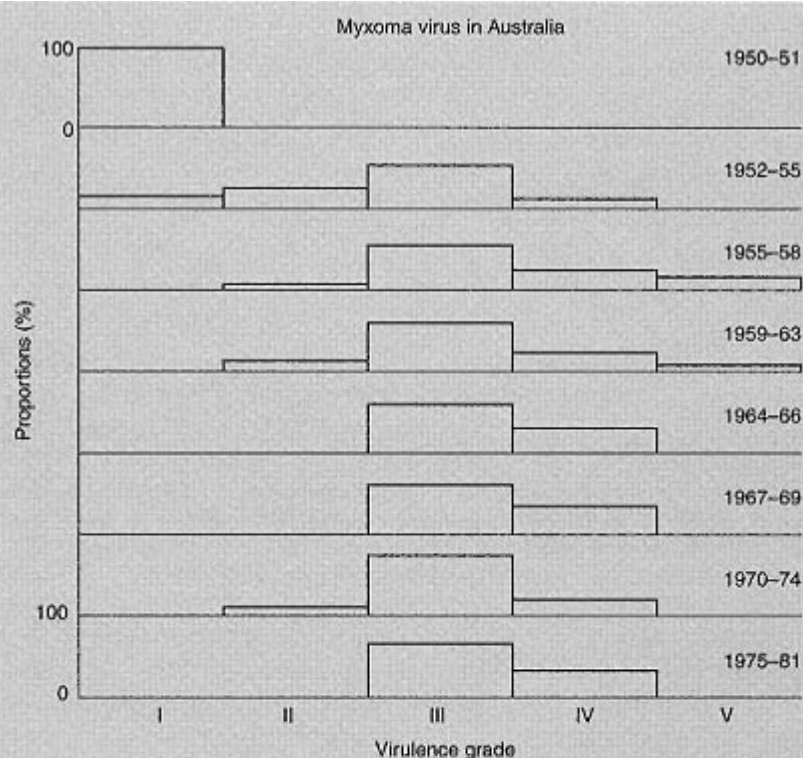


Figure 9.8 Proportions of the various grades of myxoma virus in wild populations of rabbits in Australia, from 1950–81 (data from Fenner 1983, quoted by May and Anderson 1983).

1. the rate, per day, at which new rabbits are infected: this will be an increasing function of the virulence of the virus; and
2. the time in days for which the original rabbit survives after being infected: this will be a decreasing function of the virulence.

Natural selection will favour the virus strain with the maximum value of R , which turns out to be a strain of intermediate virulence: highly virulent strains kill their hosts too quickly, and non-virulent strains have a low rate of further infection.

Notice that the argument that selection will maximize R assumes that all selection is between viral populations in different rabbits. This will only be true if, typically, a rabbit is infected only once. If a rabbit is infected by two different strains, between-strain selection within a host becomes important. This will tend to favour more virulent strains: there is no advantage to one strain that permits prolonged host survival if a second, more virulent, strain kills the rabbit quickly.

The Shifting Balance Theory

Sometimes, genotype G_1 is fitter than G_2 , yet evolution from G_1 to G_2 cannot occur in a large random-mating population. There are two simple cases ($>$ should be read as 'fitter than', and A as AA or Aa):

1. *Heterozygous disadvantage*. $AA > aa$, but $Aa < aa$. Then, starting with an aa population, allele A cannot invade, so AA cannot evolve.
2. *Epistatic fitness interactions*. For example, suppose that A is dominant to a , and B to b , and that $aabb < AB$, but $aaB < aabb$, and $Abb < aabb$. Then, starting from $aabb$, neither A nor B can invade.

Such situations can be described by saying that there are two adaptive peaks, separated by a valley (Fig. 9.9). In a large population, natural selection drives a population uphill: therefore, with the landscape of Fig. 9.9, a population can be trapped on a local peak and be unable to evolve to a higher one. If epistatic interactions are common, there can be many local peaks.

Sewall Wright's (1931) shifting balance theory of evolution was proposed to overcome this difficulty. Although a large random-mating population cannot evolve from a lower peak to a higher one, such a peak shift can occur by chance in a small population. Suppose that a population consists of a large number of small,

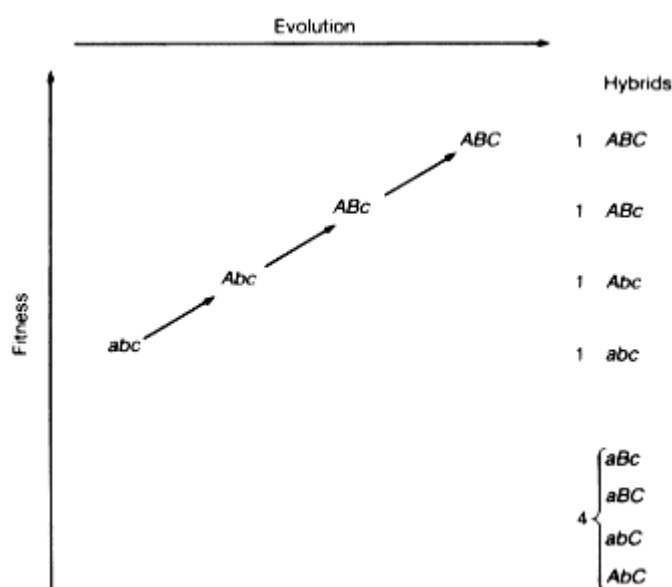


Figure 9.9

A case in which evolution could proceed by natural selection in a large population, and yet hybrids between the initial and final forms would show lower mean fitness than their parents. The organisms are haploid, with three relevant loci. It is assumed that fitness is increased if A replaces a ; if B replaces b only if A is present; and if C replaces c only if A and B are both present. The relative numbers of the different types of F_1 hybrid are shown on the right. Their fitnesses are indicated by their vertical positions; note that half the F_1 hybrids are less fit than either parent.

partially isolated demes. Occasionally, a deme will move to a higher peak. When this happens, the deme may be more successful in one of two ways: it may produce more migrants that enter other demes, or it may be more likely to split into two, or to colonize new patches. Thus, once a peak shift has occurred in one deme, the new genotype may spread through the population. Accordingly, Wright argued, the population structure most favourable for rapid adaptive evolution is that of a large population divided into many small, partially isolated demes. The small deme size makes peak shifts possible, and the large number of demes ensures that such shifts occur reasonably often.

There has been much debate about this idea. Three arguments are worth mentioning, although none is decisive:

1. The quantitative argument against group selection, based on Fig. 9.7, does not apply to Wright's model. The difficulty in explaining the evolution of altruism by inter-demic selection arises because a single 'selfish' immigrant can convert an altruistic into a selfish population. The analogous difficulty does not arise in Wright's model. Thus, for the epistatic case above, if a or b allele, or both, are introduced into an $AABB$ deme, they will usually be eliminated by selection.
2. There is one reason to think that peak shifts may have occurred in the evolution of chromosome structure (see p. 220): this is that structural heterozygotes usually have lowered fertility. If population structure has been such as to permit these peak shifts, then the same structure would permit the crossing of valleys arising from epistatic interactions.
3. An argument against the importance of peak shifts is as follows. People have sometimes been led to think that peak shifts are important by evidence from inter-population hybrids. It is certainly true that hybrids between geographically isolated populations are often of lowered viability or fertility. There can be every intermediate between fully fertile hybrids to the infertility characteristic of species hybrids. Unfit hybrids are sometimes seen as evidence that a valley must have been crossed in the evolution from one state to the other. This conclusion, however, is unjustified. It is quite possible for genotype G_1 to be connected to genotype G_2 by a series of steps, each of which would occur by natural selection in a large population, and yet for the F_1 hybrids between them to be of low fitness. A simple example involving three loci in a haploid is shown in Fig. 9.9. It is possible to construct an example involving only two loci in a diploid with dominance. Of course, this does not prove that peak shifts do not occur. But it does show that one of the arguments in favour of peak shifts is invalid: incidentally, it is not an argument that was ever used by Wright.

Further Reading

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Trivers, R.L. (1971). The evolution of reciprocal altruism *Quarterly Review of Biology* **46**, 35-57.

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Problems.

1. The lizard species, *Cnemidophorus uniparens*, consists of a clone of genetically identical females, which is thought to have originated a few thousand years ago from a single cross of a male of one sexual species and a female of another. Would you expect the females of this species to be particularly altruistic towards one another?

2. In armadillos, a female produces a litter of four genetically identical offspring, that arise by the splitting of a single zygote. Would you expect armadillos to be particularly altruistic towards their sibs?

3. In the Ani (a cuckoo-like bird), two females sometimes lay in a single nest. One of the pair is dominant, and can throw out the eggs of her partner. Pairs can successfully raise 10 offspring. A single female can raise three offspring. How many eggs laid by the subordinate (out of 10) would you expect the dominant to leave in the nest if (a) the birds are unrelated, (b) they are full sibs? (Assume that, if the subordinate bird leaves, the dominant will not find another partner.)

4. In *Tribonyx mortieri* (a Tasmanian moorhen), breeding groups consist of a male and an unrelated female, or of two males and an unrelated female. Pairs raise on average five offspring, and trios raise seven offspring. In trios, one of the two males is dominant, but the two have equal sexual access to the female. If the dominant drives out the subordinate, the latter does not find another mate. Would you expect the subordinate to be driven out if: (a) the males are unrelated; (b) the males are full sibs; (c) the males are half-sibs?

5. In red deer, males hold harems. Imagine that a female can recognize her full brother, and can join his harem. Would you expect her to do so if the viability of an offspring from a brother-sister mating is half that of an outbred offspring? Would you expect the male to accept her into the harem if she tried to enter? (Assume that a male can fertilize all the females in her harem.)

6.* In an (imaginary) species of *Polistes* (see Box 9.4), nests with two foundresses produce three times as many reproductive offspring as nests with one foundress. In joint nests, 80 per cent of the eggs are laid by the α female. Females mate only once. Joint foundresses were raised in the same nest. In a particular population, 30 per cent of all nests have two foundresses. At equilibrium, what is the coefficient of relatedness, r , between joint foundresses?

Computer Projects

A species of gull typically lays two eggs. The young have a probability P of fledging. If one egg is lost, the single surviving young is certain to fledge. A dominant gene A , if present in the first young to hatch, causes it to eject the second egg, and hence to be certain to fledge. Write an exact genetic simulation, for an infinite, random-mating population, and use it to find for what value of P the gene A will spread. Does it depend on the frequency of A ? Compare the simulation results with the predictions using inclusive fitness. (Start with frequencies of AA , Aa , and aa among adults, which need not be in Hardy-Weinberg ratio. Set up the mating table, listing first and second young to hatch separately, and follow the fate of all the offspring.)

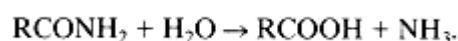
Chapter 10— The Evolution of Prokaryotes

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The prokaryotes are cellular organisms which lack a nucleus, and whose DNA exists as a circular molecule. They belong to two major kingdoms, the eubacteria, including most bacteria and blue-green algae, and the archaebacteria, including methane-producing bacteria, and bacteria living in hot springs and highly saline environments. They lack a sexual cycle involving meiosis and gametic fusion, but there are a number of other ways in which DNA can be transferred from one cell to another. Most of this chapter is concerned with these processes and their evolutionary consequences. However, in the first section I describe experiments aimed at altering the function of single genes. These experiments were performed on bacteria because of the practical advantages these organisms offer, but the processes they reveal may well go on also in eukaryotes.

The Evolution of Gene Function

The prokaryotes offer unique opportunities to study the evolution of new functions in the laboratory, because of the very large numbers of individuals that can be maintained (10^8 cells per ml), the short generation time (≈ 30 min), and the possibility of selecting the one individual out of many millions that has the required phenotype. Several studies have been made of the acquisition by bacteria of the ability to use unfamiliar substrates. Clarke (in Bendall 1983) has studied the growth of *Pseudomonas aeruginosa* on various amides as sole nitrogen source. Growth depends on the action of an amidase in hydrolysing amides to produce ammonia:



The wild-type enzyme can hydrolyse the 2-carbon amide, acetamide (CH_3CONH_2), and the 3-carbon amide, but not the 4-carbon amide, butyramide, or more complex amides. By selecting on a medium in which butyramide was the sole nitrogen source, a strain B6 was obtained that could grow on this medium. This

strain has a mutant enzyme differing by a single amino-acid substitution from the wild-type. Using B6 as a starting point, further strains were obtained that could grow on other amides, including phenylacetamide. Two points are worth noting. First, several different evolutionary pathways were found, all ending in enzymes able to hydrolyse phenylacetamide, but differing in thermal stability and in other ways: an adaptive problem need not have a unique solution. Secondly, the mutant enzymes had lost most, and in some cases all, of their activity for acetamide, the substrate of the wild-type enzyme.

Typically, bacteria do not produce an enzyme specific for a particular substrate unless that substrate is present in the medium: to do so would be wasteful. Instead, the presence of the substrate induces the synthesis of the corresponding enzymes, as first demonstrated for the *lac* operon of *E. coli* by Jacques Monod and François Jacob. Hence, to grow on a new substrate, a bacterium must have an enzyme able to catalyse that substrate, and the enzyme must be induced by the substrate. The experiments on amidases were performed on strains 'constitutive' for the enzyme: that is, mutant strains that produced the enzyme without being induced. However, in further experiments a strain was produced that could grow on butyramide, and was induced by it. The history of this strain is shown in Fig. 10.1.

After obtaining strain 3, which was constitutive and possessed amidase B, two further steps were needed. First, genetic recombination with the original wild-type strain gave strain 4, which was induced by acetamide (like the wild-type), but had amidase B (like strain 3). This new strain could no longer grow on butyramide: it had the necessary enzyme, but was not induced. Further selection for growth on butyramide produced a number of constitutive mutants (i.e. reversion to the state of strain 3), and one mutant that was induced by butyramide. Thus it was possible to produce a strain that had the new enzyme, appropriately regulated.

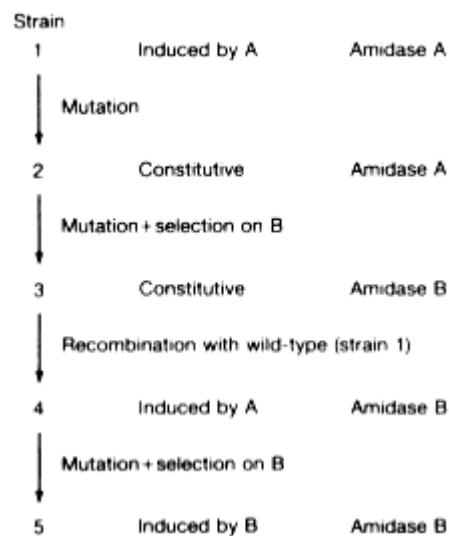


Figure 10.1

The history of a strain of *Pseudomonas* able to grow on butyramide, and induced by it (after P.H. Clarke, in Bendall 1983).

Similar experiments have been performed on other bacteria, for other enzyme functions. *Klebsiella aerogenes* selected for growth on ribitol responded, not by mutations improving the efficiency of the enzyme, but by duplicating the gene, and so increasing the amount of enzyme synthesized. The ability of *E. coli* to utilize lactose depends on the enzymes coded for by the *lac* operon. When the structural genes were deleted, and selection for growth on lactose applied, evolution took place in another operon, *EBG*, which previously had performed no essential function.

It is clear, then, that selection can alter the specificity and control of genes coding for enzymes. It may well be that, in nature, the common way in which bacteria acquire new metabolic abilities is by acquiring a plasmid carrying the relevant genes, and not by mutation in genes already present. This process is discussed later in the chapter. However, all genes, whether or not introduced by a plasmid, must at some time have acquired their present functions by mutation, recombination, and selection, and the experiments described in this section illustrate that process.

Phages, Plasmids, and Transposable Elements

This section gives a brief description of the parasites and symbionts that are relevant to an understanding of prokaryote evolution, and defines a number of useful terms.

1. **Bacteriophages**, or **phages** for short, are viruses parasitic in bacteria. When outside the cell, they consist of a DNA (or occasionally, a single-stranded RNA) core, surrounded by a protein coat that protects the DNA, and helps to introduce it into a new host cell. Once inside a cell, the phage DNA redirects the host metabolism into the synthesis of new virus particles, which are released on the death of the host cell, and which must then infect a new host. This cycle is known as the **lytic cycle**. In **virulent** phages, only the lytic cycle occurs. In contrast, the DNA of **temperate** phages can be inserted into the host chromosome as a **prophage**. It then replicates with the host: no new virus particles are produced, and the host cell is not damaged. A bacterium carrying a prophage is said to be **lysogenic**. Such a cell is immune to further infections by the same type of phage. Occasionally, a prophage is induced to enter the lytic cycle.

2. **Plasmids** are extra-chromosomal circular DNA molecules. They differ from phages in that they do not have an extra-cellular stage in their life cycle, and hence do not have genes coding for coat proteins. A major distinction is between **conjugative** plasmids, which cause effective contact to take place between their host cell and other cells, usually by the production of long thread-like pili extending from the cell wall, and non-conjugative plasmids.

Conjugation enables copies of a plasmid to pass to another cell. Some plasmids are non-conjugative but **mobilizable**, in that they can pass to another cell if conjugation is caused by another plasmid. For obvious reasons, a plasmid that is both conjugative and mobilizable is said to be **self-transmissible**.

3. **Transposable elements, transposons**, are pieces of DNA that can transpose from one site in a chromosome or plasmid to another. Two points about transposition are important. First, when a copy of a transposon is inserted at a new site, the original copy usually remains at the old site: transposition is therefore a form of replication. Secondly, although transposition involves the breaking and rejoining of DNA, it does not require sequence homology between the transposon, and the chromosome: in this respect it differs from the homologous recombination that causes crossing over between chromosomes in both prokaryotes and eukaryotes. Transposons are typically several thousand bases long, and often carry antibiotic-resistance genes: **insertion sequences** are shorter transposable elements, of approximately 1000 bases.

The widespread occurrence of phages, plasmids, and transposons means that pieces of DNA are rather readily transferred between bacteria, even between rather distantly related bacteria. This raises two questions about the reality of, say, *Escherichia coli* as a species.

First, suppose that we could, in imagination, trace the ancestry of the genes present in *E. coli* today back for a million years, or for 50 million years. Would these ancestral genes be found in a single kind of bacterium? Alternatively, would they be found in many different, unrelated bacteria? If this question was asked, not of *E. coli*, but of the house mouse, the answer is that all, or almost all, the genes would be found in a single ancestral species (although, if we went back 50 million years, that ancestor would not be a mouse). One reason for thinking this is that phylogenies based on the amino-acid sequences of different proteins give, at least approximately, the same tree. The answer is less obvious for *E. coli*. Some present-day genes have, in all probability, come from different and unrelated ancestors. But the majority have probably been together for a long time. Perhaps the strongest evidence for this is the close similarity between the linkage maps of *E. coli* and *Salmonella typhimurium*, which could hardly be so if horizontal transfer of genes was a common event in evolution.

The answer to our first question, then, is that many, and perhaps most, of the genes of *E. coli* have been together for a long time, although some have recently been acquired via phages or plasmids. A second question is as follows. Do all *E. coli* belong to a single species, separate from other species, as mistle thrushes belong to a species separate from song thrushes, fieldfare, and redwings? The answer is probably no. Thus a related 'species', *Shigella*, is distinguished from *Escherichia* by the fact that it causes dysentery, but DNA hybridization suggests

that some *E. coli* are more similar to some *Shigella* than they are to other *E. coli*. This is not surprising: in the absence of sexual reproduction, there is no reason to expect organisms to fall into well-defined species (see p. 267).

The Evolution of Phages and their Hosts

Bacteria can evolve resistance to attack by particular phages in three ways:

1. Exclusion: change in the bacterial surface prevents entry of the phage. Phage mutants that overcome such exclusion can occur.
2. Restriction: restriction endonucleases are enzymes that cleave DNA at particular sites, specified by short sequences of 4-6 nucleotides. These enzymes have been widely exploited by molecular biologists. Again, phages can evolve resistance to particular endonucleases. Just as vertebrates must not make antibodies against their own tissues, the host bacterium must not succumb to its own enzymes: one of the ways in which this is achieved is by the action of further enzymes that methylate specific cleavage sites.
3. Incompatibility: replication of the foreign DNA is slowed down or stopped.

The coevolution of virulent phages and their hosts is not in principle hard to understand. Genes in the host that confer resistance will spread if phages are present: there is, however, evidence that resistance genes lower fitness in the absence of phage, so they will not become fixed in the host population. At the same time, phage genes that overcome resistance will be favoured. The result will be an evolutionary arms race. The evolution of temperate phages is more puzzling. A bacterium may gain some benefit from the presence of prophage. Most obviously, it gains protection against further infection by the same phage. Some phages carry genes for antibiotic resistance, or other traits favourable for the host. It is common to find that some advantage is conferred on the host, even when no specific mechanism is known: for example, in a chemostat experiment phage λ was found to confer a selective advantage on *E. coli* of 3 per cent per hour. Hence, we would not in general expect to find host genes that resist the integration of the phage.

But why should phage genes evolve that mediate integration into the host chromosome? Would not the phage increase in numbers more rapidly if it remained in the lytic cycle? The answer is clearly yes, but only if the local density of bacteria is high. If host density is low, a prophage that was induced to enter the lytic cycle would kill its host, but none of the daughter particles might find a new host. There are good theoretical reasons, therefore, to expect that temperate phage should be integrated as a prophage when host density is low, and enter the lytic cycle at high densities. As yet, little evidence exists to test this prediction. We shall meet other cases in which we would expect host density to be important in the next section.

The Evolution of Plasmids

In general, it is best to think of phages as parasites, and plasmids as endosymbionts which survive by benefiting their hosts. It is in principle possible for a self-transmissible plasmid to maintain itself in a bacterial population, even though it lowers the fitness of the host cell, but in fact most plasmids carry genes that benefit their hosts, for example by conferring resistance to antibiotics, drugs, or heavy metals, by producing toxins, by coding for restriction enzymes, or by utilizing novel substrates.

We can therefore expect selection on both host and plasmid genes to favour the regular transmission of plasmids to daughter cells at cell division. Plasmid copy number is in fact regulated: in some there are 1-2 copies per cell, and in others 10-100. Even in the former kind, daughter cells usually receive a copy: the frequency with which plasmid-free cells arise after cell division is as low as 1 in 10^6 .

Three kinds of plasmid will be discussed in more detail:

Plasmids Conferring Resistance to Drugs and Antibiotics

These plasmids are usually self-transmissible. They are formed (Fig. 10.2) by the union of a 'resistance transfer factor', or RTF plasmid, carrying the genes needed for conjugation and replication, and an 'r-determinant', consisting of an insertion element and genes determining drug resistance.

The spread of these plasmids has been the most dramatic, and from a human point of view the most damaging, evolutionary event of recent years. Bacterial strains isolated before 1940 have plasmids, but none carrying antibiotic resistance. When R plasmids were first discovered by T. Watanabe in the late 1950s, most carried only one or two drug resistances. Today, it is typical to find four or five drug resistance genes carried by a single plasmid. This shows how bacteria can evolve rapidly to meet changed circumstances by acquiring plasmids. R-plasmids have evolved recently by the insertion of resistance genes into pre-existing plasmids.

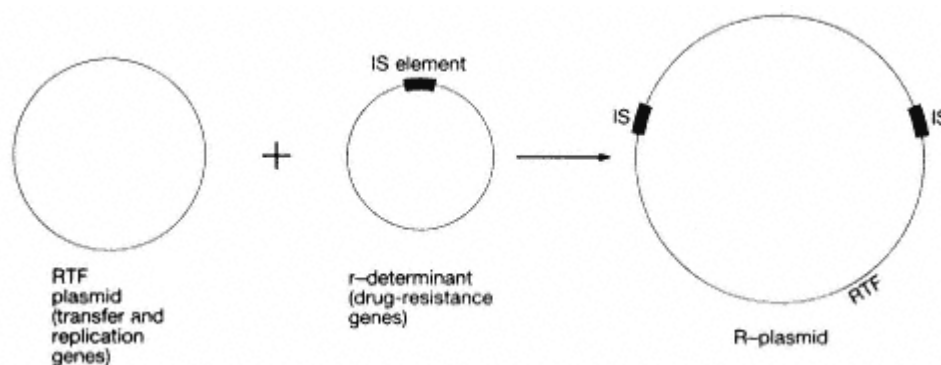


Figure 10.2
The origin of a drug-resistant plasmid.

However, there is no reason to think that the resistance genes themselves have originated since 1940. Antibiotics are produced naturally by other organisms as weapons in inter-specific competition, and genes giving resistance to them are probably ancient. The existence of plasmids means that such genes can be acquired by bacteria that need them.

It is not only antibiotic resistance that can be acquired in this way. Bacteria such as *Pseudomonas* that can utilize a wide range of substrates often acquire the necessary enzymes from plasmids.

Plasmids Producing Proteins that Kill other Bacteria.

ColE1 is a plasmid that codes for a 'colicin' that kills other bacteria. It is mobilizable, but requires the presence of a conjugative plasmid, such as the F plasmid described below, before it can transfer to a new cell. In most cells, the colicin-producing gene is repressed, and a gene that produces a protein protecting the cell against colicins is switched on. Colicin production can be induced, for example by UV light. When this happens, the cell is killed, and colicins are released. These kill other bacteria that do not carry the plasmid: bacteria with the plasmid are protected.

This is an entertaining example of kin selection. A colicin gene that is expressed kills its host and itself, but in so doing kills other bacteria that do not carry an identical gene. Clearly this is advantageous only if there is resource competition between the bacteria. At high densities, the death of competing bacteria leads to increased multiplication of the survivors. Hence a gene that produces colicins has two effects: it destroys itself, but it increases the multiplication of bacteria carrying identical copies of itself. The net effect may be to increase the number of copies of the gene. But this will be true only at high densities. As in the case of temperate phages entering the lytic cycle, we can expect the induction of colicin production to be influenced by population density, but evidence is lacking.

The F Plasmid

The F plasmid is a conjugative plasmid that has been of particular interest to geneticists because it can mediate recombination between bacterial genes. Its life cycle is shown in Fig. 10.3. F⁻ cells lack the plasmid. In F⁺ cells the plasmid is present as a separate DNA circle. If F⁻ and F⁺ cells are mixed, conjugation takes place, and copies of the plasmid enter the F⁻ cells, which become F⁺. At first sight, therefore, it seems puzzling that most *E. coli* in nature are F⁻. The explanation seems to be that expression of the genes mediating conjugation, and in particular the production of pili, is costly to the bacterium. At low densities, therefore, F⁺ bacteria are at a selective disadvantage. In fact, in bacteria carrying conjugative plasmids, the genes mediating conjugation are repressed much of the time.

Occasionally, the F plasmid is integrated into the bacterial chromosome, to give

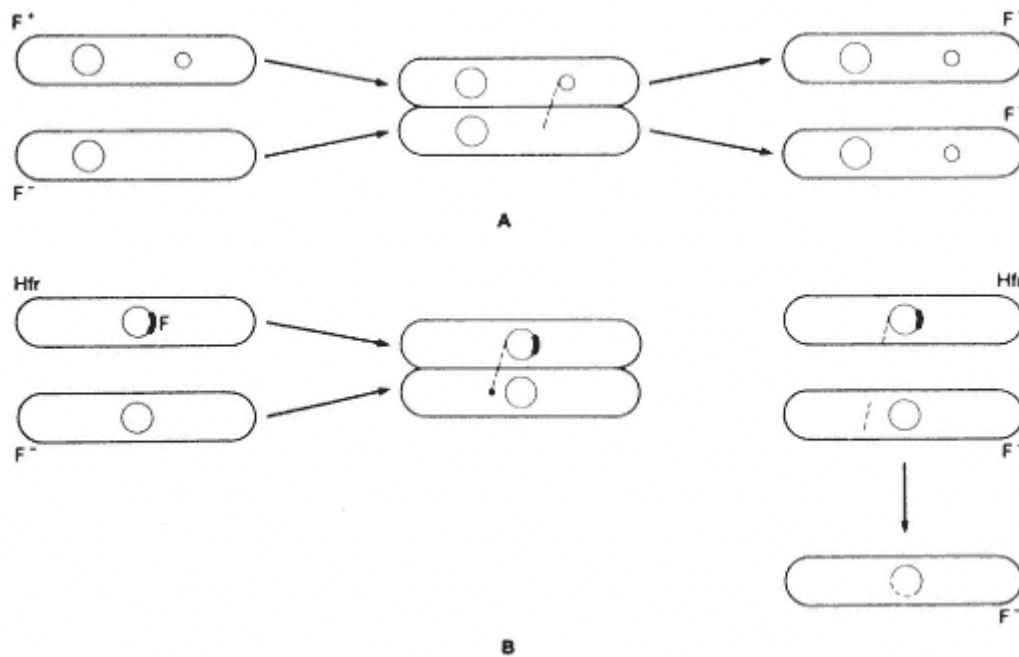


Figure 10.3

The life cycle of the F plasmid. **A** Conjugation between F^+ and F^- bacteria: both exconjugants are F^+ . **B** Conjugation between Hfr and F^- bacteria: the F^- ex-conjugant remains F^- but its chromosome contains regions from both original conjugating bacteria.

an Hfr ('high frequency of recombination') cell. If conjugation occurs between an Hfr and an F^- cell (Fig. 10.3), a small part of the F plasmid, together with part of the bacterial chromosome, enters the F^- cell. It is rare for the whole of the F plasmid to be transmitted, so the F^- cell remains F^- . After conjugation, however, the F^- cell contains its own original bacterial chromosome, and a substantial part of the donor chromosome. Homologous recombination then occurs, giving rise to a new bacterial chromosome, carrying genes from both the F^+ and the F^- cell. The residual chromosome fragments are lost.

The evolutionary significance of these events is unclear. Their relevance to bacterial evolution is considered on pp. 196-7. Here we will note only that conjugation by Hfr cells confers no advantage on the genes of the F plasmid, which are not transmitted.

The Evolution of Transposons.

The number of copies of an insertion sequence can increase in one of three ways:

1. increase, by transposition, of the number of copies in a given replicon (i.e. bacterial chromosome or plasmid);

2. transfer, by transposition, to a new replicon; and 3. increased multiplication of the replicon in which it is inserted.

A high copy number per replicon does not in itself increase the chances of long-term survival of the element, and may reduce it if the fitness of the replicon is reduced by the presence of many copies of the element. It is therefore not surprising that transposons carry genes regulating their own copy number. (The selective forces affecting copy number may be more complex, and more interesting, than this see Box 10.1.) The mechanism in IS50 is ingenious. The element has two

Box 10.1—

Natural selection on transposition rate

It seems likely that selection on a transposon can alter its overall rate of transposition, but probably not the precise sites to which it is transposed: this would be the case for the mechanism of regulation in IS50. For reasons given in the main text, we would expect selection to reduce transposition rate as copy number increases, essentially because high copy number reduces the fitness of the host organism. There will, however, be counter-balancing selection for increased transposition, because of the potential advantages of insertion into a new replicon. Actual transposition rates will reflect a balance between these two selective forces.

This balance may work out differently in prokaryotes and eukaryotes. Consider first the extreme case of a bacterium with no plasmids or phages, and lacking recombination. Selection would tend to reduce the transposition rate almost to zero, because there is never a second replicon in the cell to which the transposon might move. In the real world, there will be selection for non-zero transposition rates, because there is often more than one replicon per cell (chromosome + plasmids), and, probably less important, when conjugation leads to recombination between bacterial chromosomes, a transposon is more likely to be present in the recombinant chromosome if it is initially present in several copies. However, the balance of selective forces seems to have favoured restricted transposition rates, and transposons that enhance the fitness of their hosts.

Contrast this with the situation in eukaryotes. Consider a transposon present in a single copy in a zygote. The number of next-generation zygotes to which it is transmitted can be increased by transposition in two ways. Transposition to a different chromosome will increase the proportion of gametes carrying it. So with transposition to a different site in the same chromosome, if crossing over takes place between the new and old sites. For this reason, selection favouring high transposition rates is strong. This makes sense of the fact that transposons usually confer fitness benefits on their prokaryote hosts, but seem, on what little evidence we have, to lower the fitness of eukaryotes.

This argument is speculative, but suggests further questions. For example, how do transposons behave in asexual eukaryotes?

regulating genes. One, which codes for a protein that activates transposition, is *is*-acting: that is, it activates only the element of which it is part. The second codes for a diffusible inhibitor of transposition. Hence, as copy number increases, the local strength of activation remains constant, but of inhibition increases: this results in self-limitation of copy number.

If, as seems plausible, selection on IS elements will lead to self-regulation of copy number, and if, as is certainly the case, IS elements are often linked to genes conferring an advantage on the host cell, then there is no selection on the host to evolve resistance to transposition, and such resistance is not observed.

The Population Genetics of *E. coli*

E. coli inhabits the lower intestine of warm-blooded vertebrates. In this section, I review data on the genetic variability and population structure of natural populations.

Geographic Variation, and Variation Between Hosts

Genetic variation in surface antigens, and in the electrophoretic properties of enzymes, have been studied from different parts of the world, and from different hosts. Of 20 enzymes studied, 18 were polymorphic, with from 2 to 19 alleles per locus. This indicates that *E. coli* genes are substantially more variable than human ones. Clones isolated from humans and other animals did not differ significantly. Most of the genetic variability is contained within samples from a single locality.

An important question concerns the role of genetic recombination. If recombination is frequent, we would expect genotypes to occur approximately in linkage equilibrium (p. 82). This is not what is found. Thus, of the 109 strains in which the 20 enzymes mentioned above were recorded, three pairs of clones of electrophoretically identical type were found in geographically distant locations, and seven more pairs were found that were identical except for one locus. Given the observed degree of genetic variability per locus, the chance of finding even a single pair of clones that were identical is small, unless the two clones are derived from a common ancestor without recombination. Since the members of a pair were geographically distant, this implies that clones are long-lasting and widely distributed. However, DNA sequencing has shown that horizontal transfer of chromosomal DNA fragments between strains of *E. coli* is relatively frequent. The most likely mechanism for such transfer is **transduction**, a process in which a phage accidentally incorporates host DNA, and transfers it to a new cell, where it may be incorporated into the chromosome by recombination.

Populations from a Single Host

In a heroic experiment, 550 clones were isolated from a single human over a period of 11 months. In all, 53 distinct electrophoretic types were identified. Of

these, two were present for most of the period; the others were transient, appearing for a few weeks only. This means that the host was repeatedly infected by new clones, most of which failed to establish. As in the study of geographical variation, there was no evidence of genetic recombination.

Plasmids in Natural Populations

Most *E. coli* in nature carry plasmids. In all, over 250 different plasmids have been identified. Some carried drug resistance genes, or other genes conferring recognizable phenotypes, but most were cryptic, in the sense that they produced no detectable phenotypic effect: of course, this does not mean that they had no effect.

The picture, then, is of a highly variable population, with little geographic or host-specific differentiation, and consisting of a large number of long-lasting clones. The ubiquity of plasmids, and the wide range of functions they are known to perform, shows their importance in the adaptation of *E. coli* to particular environments.

The frequency of horizontal gene transfer varies greatly between different types of bacteria. It is particularly common in species such as *Bacillus*, *Streptococcus*, and *Neisseria*, which are competent for **transformation**, a process in which DNA is actively taken up from the surrounding medium and incorporated into the chromosome. Usually there is selective uptake of DNA with a particular species-specific recognition sequence, which is widely distributed on the chromosome of the species in question, thus ensuring preferential uptake of homologous DNA.

The Evolution of Viruses

One example of viral evolution has already been discussed. In Box 9.5, I described the evolution of the myxoma virus. In doing so, I took as the measure of fitness the reproductive rate R of a virus infecting a rabbit, defined as the number of additional rabbits infected by viruses from that rabbit. Such a definition ignores selection between viruses in the same rabbit. It is therefore only justifiable if, typically, a rabbit is infected by only one clone of viruses, and if mutation does not generate new variation too rapidly. This section discusses a quite different problem in viral evolution, in which it is essential to take into account differences between viruses in a single cell population. This is the evolution of **multi-compartment** viruses.

The genome of many plant viruses is of single-stranded RNA. Some of these are multicompartment viruses: that is, a single type of protein capsule encloses one of two different genomes *A* and *B*. These genomes have complementary functions: for example, one may code for the capsule protein, and one for an RNA replicase. If growth is to take place, therefore, a cell must be infected by at least one particle of each kind.

Why should such an odd state of affairs have evolved? One suggestion is that it makes possible recombination between different *A* and *B* genomes, and so confers a long-term advantage on the population. However, Nee (1987) has suggested a much simpler explanation, in terms of selection operating within a viral population. He imagines that the original state was a complete virus *C*, carrying both *A* and *B* genes. Mutation would give rise to incomplete viruses *I*, carrying a functional *A* gene and a non-functional *B* gene, or vice versa, and also to junk viruses *J*, carrying no functional genes. In a population of *C*, *I*, and *J* viruses, *C* viruses would have a selective advantage, because each *C* virus is certain to replicate if it enters a cell, whereas an *I* virus will replicate only if complemented by a virus carrying the other function, and a *J* virus only if complemented by viruses carrying both functions. If these were the only fitness differences, *C* would be the typical form, and *I* and *J* particles would be maintained by recurrent mutation, as deleterious mutations are always present in natural populations.

However, *I* and *J* also have a selective advantage, in being more likely to produce replicas belonging to the same class as themselves. Because of the low accuracy with which RNA is replicated, this may be an important effect. For simplicity, suppose that the *A* and *B* genomes contain the same number, n , of bases. If the error rate per base is u , the probability that an *I* virus will give rise to an *I* virus of the same kind is $(1 - u)^n$ and the probability that a *C* virus will give rise to *C* is $(1 - u)^{2n}$. We can probably ignore back mutation (that is, an *I* virus giving rise to a *C*), because, once a non-functional mutation has occurred, further mutations, including deletions, can occur without having any further deleterious effects. There may be additional fitness differences arising because *C*, *I*, and *J* viruses are replicated at different rates within a cell (the minivariant discussed in Chapter 1 can be thought of as a particle which, freed from any constraints imposed by the need to code for a protein, has evolved a sequence that is replicated particularly rapidly).

There is, therefore, a balance of selective forces. *C* viruses are always replicated, but with lower accuracy: *I* viruses need complementation, but are replicated with greater accuracy, and perhaps more rapidly. The evolutionary dynamics of a population of *C*, *I*, and *J* viruses is analysed further in Box 10.2. The crucial parameters turn out to be the accuracy of replication $Q = (1 - u)^n$ and the

Box 10.2—

The Evolution of Multicompartment Viruses

For simplicity, suppose that the two kinds of incomplete virus, *A* and *B*, are identical in accuracy and rate of replication. They will then be present in equal frequency.

Let x_C , x_I , and x_J be the frequencies, in one generation, of *C* viruses, of each type of *I* virus, and of *J* viruses, respectively. Then $x_C + 2x_I + x_J = 1$.

Let $Q = (1 - u)^2$ be the accuracy of replication of an I virus. Let R be the probability that an incomplete virus is complemented. Let r be the probability that a junk virus is complemented.

Ignoring differences in replication rate, Table 10.1 gives the kinds and numbers of replicas produced by each viral type. Hence we can write down the following equations for the frequencies after one cell generation:

$$\begin{aligned}x'_C &= Q^2 x_C / T \\x'_I &= [Q(1 - Q)x_C + QRx_I] / T \\x'_J &= [(1 - Q)^2 x_C + 2(1 - Q)Rx_I + rx_J] / T,\end{aligned}\quad (10.1)$$

where T is the total number of copies. That is

$$T = x_C + 2Rx_I + rx_J. \quad (10.2)$$

Table 10.1

Evolution in multicompartiment viruses. Number of replicas of various types from complete (C), incomplete (A and B), and junk (J) viruses

Parental type	Offspring type				Total
	C	A	B	J	
C	Q^2	$Q(1 - Q)$	$Q(1 - Q)$	$(1 - Q)^2$	1
A	—	RQ	—	$R(1 - Q)$	R
B	—	—	RQ	$R(1 - Q)$	R
J	—	—	—	r	r

Q is the probability that a single genome, A or B , will be replicated accurately. R is the probability that an incomplete virus, A or B , will be complemented; r is the probability that a junk virus will be complemented.

We now need to know R and r . Suppose that a total of N cells are infected by NM particles. M is the multiplicity. Consider an incomplete A virus. R is the probability that the cell it infects is also infected by a B or a C virus. The number of such viruses is $NM(x_C + x_I)$. The chance that any one of them does *not* infect a particular cell is $1 - 1/N$. Hence, if infection is random,

$$R = 1 - (1 - 1/N)^{NM(x_C + x_I)} = 1 - e^{-M(x_C + x_I)}. \quad (10.3)$$

If x_C is small, $r = R^2$; if x_C is not small, r is a little more difficult to calculate.

These equations are easy to simulate. Some results are shown in Fig. 10.4. They illustrate the general conclusion that complete viruses are eliminated

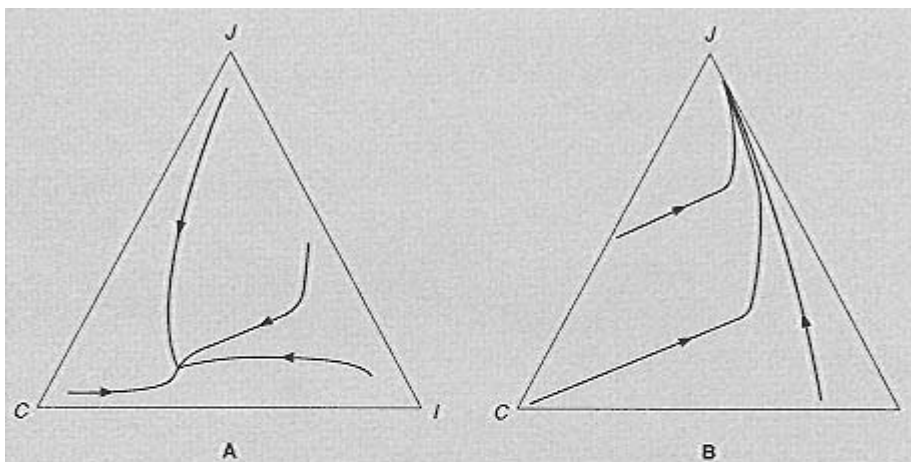


Figure 10.4 The evolution of multicompartiment viruses. **A** Multiplicity $M = 1$; accuracy of replication $Q = 0.9$. **B** $M = 10$, $Q = 0.5$. In case **A**, complete viruses, C , are present at equilibrium, but incomplete and junk viruses are also present. In case **B**, complete viruses are eliminated. The population evolves to the multicompartimental state.

if the multiplicity is not small, and if Q is not too high. This makes intuitive sense. High multiplicity increases the likelihood of complementation, and so favours incomplete viruses; a low accuracy of replication handicaps complete viruses.

number of particles infecting a single cell, called the **multiplicity of infection**, M . If a cell is usually infected by a single particle, then C is the typical form, and I and J are maintained by recurrent mutation (although they may be present in high frequency). But if M is not small, C viruses may be eliminated entirely from the population, leaving only I and J . In effect, a population of multicompartiment viruses will evolve. There is no need to invoke the advantages of recombination, or indeed any advantage to the population as a whole.

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Computer Projects

1. Write a program to simulate the evolution of multicompartiment viruses, representing the results as trajectories in an equilateral triangle (or simplex), as in Fig. 10.4. Extend the results by simulating a case in which two types of incomplete virus, *A* and *B*, have different replication accuracies: for example, let them have accuracies Q_A and Q_B , and the complete virus an accuracy of $Q_A Q_B$. (You now have four variables, so you cannot plot all the necessary information in a triangle.)

Chapter 11— The Evolution of the Eukaryotic Genome.

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The Nature of the Genome

Before the structure and role of DNA was discovered, the eukaryotic chromosome was pictured as a series of genes strung together like beads on a string, with a centromere whereby it could be moved about during mitosis and meiosis. Genetic recombination was thought to occur by breaking and rejoining the string between the beads. With the discovery of DNA, the distinction between beads and string disappeared. The genes are arranged end to end, so that the whole chromosome is one immense DNA molecule, and recombination occurs within genes. However, the precise order of genes on the chromosome is probably not needed to enable them to function properly during development: one good reason is that, in *Drosophila* and many other organisms, extensive chromosome inversions have little or no effect on the phenotype. Genes are strung together to form chromosomes because otherwise they could not be properly partitioned during cell division.

One reason for doubting the correctness of this picture of the chromosome as a string of unique genes is the so-called 'C-value paradox'. Different species have strikingly different amounts of DNA per nucleus. Typical values are given in Table 11.1. Some of the differences make sense in terms of the complexity of the adult organism. For example, it is not surprising that there is more DNA in the nucleus of a human than of a nematode, or of a nematode than a yeast. But can it really require five times as much DNA to specify a newt than a man, or 500 times more DNA to specify a fritillary than a cress plant? DNA content per cell does have some phenotypic effects. In both plants and animals, cell size increases linearly with DNA content per nucleus, and cell cycle time also increases. But the need for larger cells (even if it exists) large cells may be the unselected consequence of a high DNA content) could be brought about without a corresponding increase in the number of unique genes.

Table 11.1 also gives estimates of the proportion of the DNA that codes for proteins. These estimates are based on identifying corresponding RNA molecules

Table 11.1

DNA content per haploid genome (data from Cavalier-Smith 1985)

	Genome size (pg)	Percentage of genome coding for protein
<i>Escherichia coli</i> (a bacterium)	0.0036 .005	~ 100
<i>Saccharomyces cerevisiae</i> (a yeast)	0.009	69
<i>Caenorhabditis elegans</i> (a nematode)	0.088	25
<i>Drosophila melanogaster</i>	0.18	33
<i>Homo sapiens</i>	3.5	9 7
<i>Triturus cristatus</i> (a newt)	19	1.5 .5
<i>Protopterus aethiopicus</i> (a lungfish)	142	0.4 .2
<i>Arabidopsis thaliana</i> (an annual weed)	0.2	31
<i>Fritillaria assyriaca</i> (a monocot)	127	0.02

1 pg of DNA corresponds to approximately 10^9 base pairs.

in the cytoplasm. They are not very accurate, because not all genes are expressed in all tissues, but they are probably of the right order. It is clear that in some organisms a large part of the DNA codes for nothing.

Much of the DNA in the genome is present in many copies. There are many different kinds of repeated DNA. A rough classification is as follows:

1. **Gene clusters.** Many proteins are present in an individual in several different forms, coded for by distinct but similar genes. An example, haemoglobin, is discussed below.
2. **Tandemly repeated genes** with identical functions. An example, discussed below, is the set of several hundred genes coding for ribosomal RNA.

Although interesting, these two types of repeated DNA do not substantially alter the picture of a chromosome consisting of a string of genes, each with a unique role. Gene clusters evolve, presumably, because of the need for several similar but not identical products. Tandemly repeated genes are a response to the need for a large amount of a particular product (or, in the immune system, of a large number of similar but not identical products). The next two categories, however, are not so easily explained in terms of the needs of the organism.

3. **Middle-repetitive, dispersed DNA.** These are DNA sequences, varying in length from a few hundreds to thousands of nucleotides, usually present in only tens or hundreds of copies per genome, and dispersed throughout the genome, often with only a single copy at each site.

4. **Highly repetitive DNA.** These are short sequences, each present in very large numbers, often in tandemly arranged blocks. A particular sequence may be present on all chromosomes, but be concentrated in particular regions, for example near the centromere.

The presence of the last two kinds of repetitive DNA raises two questions:

1. Is repetitive DNA present because it increases the fitness of the individual organism, or is it in effect parasitic, as suggested by Doolittle and Sapienza (1980) and by Orgel and Crick (1980)?
2. How does the 'concerted evolution' of repetitive DNA occur, so that one species comes to have many copies of sequence A, whereas a closely related species has many copies of a slightly different sequence, B?

The Haemoglobin Gene Family

In vertebrates, oxygen is transported in the blood bound to a protein, haemoglobin. The protein is tetrameric. In adult humans, it is formed of two α and two β peptides: in fetal humans of two α and two γ peptides: and in embryos, of two ϵ and two ζ peptides. The tetrameric structure facilitates reversible combination with oxygen. The presence of three distinct proteins also makes functional sense, in terms of the different oxygen requirements of adult, fetus, and embryo, and the need to transfer oxygen across the placenta.

All the peptides have common features, in amino-acid sequence and tertiary structure, sufficient to indicate that the genes coding for them have a common origin, and have arisen by the duplication of a single original globin gene. As we shall see, study of the genes themselves confirms this view. In humans, the genes are arranged in two clusters, the α and β clusters, on different chromosomes. Figure 11.1 shows the human β cluster, and, for comparison, the cluster in mouse and rabbit. The following features are important:

1. Each structural gene consists of three **exons** and two **introns** (Fig. 11.2). An RNA transcript of the whole gene is made, but the two sections homologous to the introns are spliced out before the mRNA passes to the cytoplasm, where it is translated. Thus, only the exons code for the amino-acid sequence of the peptide. Introns occur in all the structural genes, in both α and β clusters, at exactly homologous positions. Introns are typical of structural genes in eukaryotes. There are two possible views about their origins. One is that introns are degenerate transposable elements (see p. 188), which were inserted into the gene in the very distant past, but which have now lost the capacity for trans-

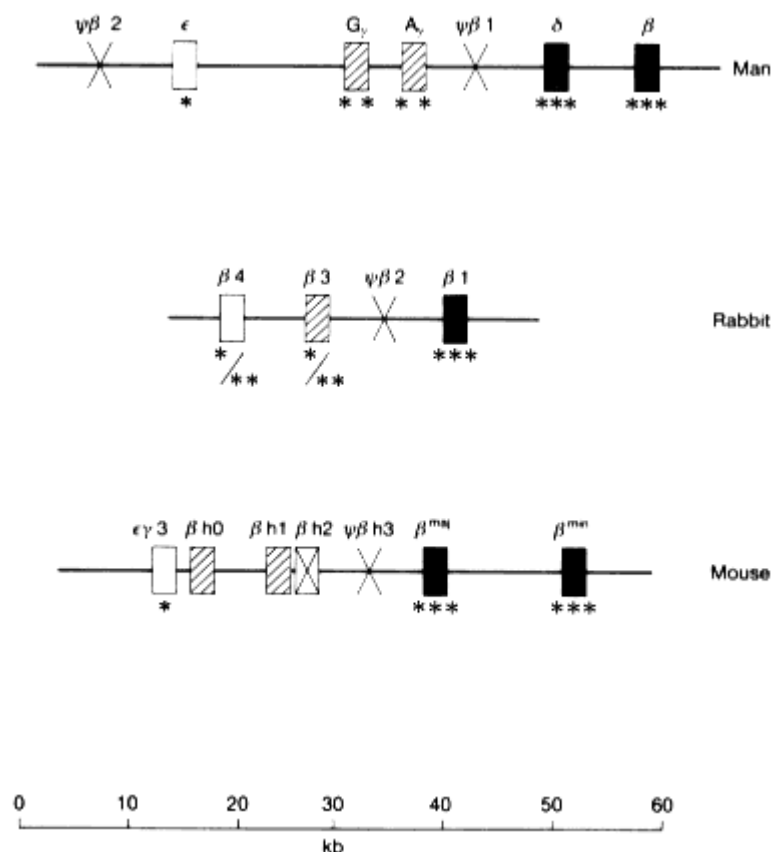


Figure 11.1

The β -haemoglobin gene cluster (after Jeffreys *et al.*, in Bendall 1983).

■ genes homologous to the human β ; □ genes homologous to the human ϵ ;
 ▨ genes homologous to the human γ ; pseudogenes. Stages of expression:
 *, embryonic; **, fetal; ***, adult. Scale bar is in kb (that is, base pairs $\times 10^{-3}$).

position. The only sequence specificities they have retained are the terminal sequences that make splicing-out possible: the loss of these would be lethal for the organism, and hence for the intron itself.

An alternative view, suggested by Gilbert (1978), is that the exons represent ancestral 'minigenes', coding for small but functional peptides, and that modern genes arose by the bringing together of several such units. The idea is attractive. If it is correct, we would expect each exon to code for a different functional region of the protein.

2. In addition to the functional genes, there are **pseudogenes**. These have sufficient sequence homology, and the same exon-intron arrangement, to show that they have arisen by the duplication of functional genes. However, they contain frame-shift mutations that make it certain that they are not translated

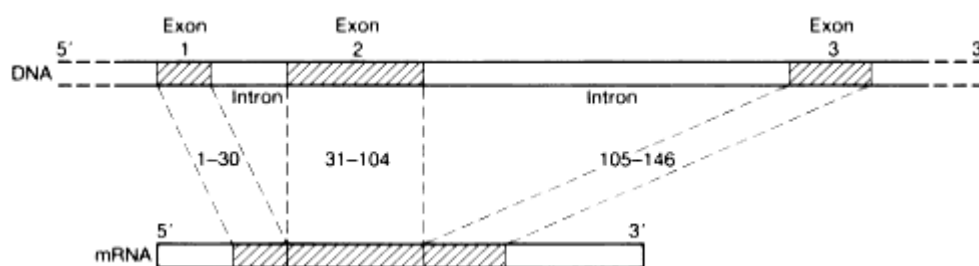


Figure 11.2

The mouse β -globin gene, and its messenger RNA (after Phillips *et al.*, in Bendall 1983).

to functional peptides. They are 'fossil' genes that have lost their function. They diverge rapidly in sequence: the genome may contain many pseudogenes that have diverged so far that they can no longer be recognized.

The genome contains a secondary category of pseudogene differing from those in the β -haemoglobin cluster in two ways: they lack introns, and are unlinked to the functional genes to which they are homologous. They are thought to have arisen by reverse transcription from mRNA, followed by insertion into the chromosome.

3. The exons and introns each make up about 8 per cent of the DNA in the cluster. The remaining 84 per cent, consisting of a mixture of unique and repeated sequences (including an Alu sequence see below), is of unknown function. Surprisingly, however, it diverges rather slowly in nucleotide sequence, at some 20 per cent of the rate of introns and silent sites in coding regions (see p. 149).

Figure 11.3 gives a phylogenetic tree of the human globin genes, since the original α - β duplication some 500 million years ago. This event can be dated from the fossil record. Thus hagfish and lampreys and therefore, presumably, the ancestral jawless vertebrates have a monomeric haemoglobin, whereas bony fish and tetrapods have α and β peptides. The duplication, therefore, must have taken place after the origin of the jawed vertebrates, but before the split between the ancestors of the tetrapods and of the bony fishes: this gives us a date of about 500 million years. The other dates in this figure are based on the assumption that the different genes arose by duplication at the nodal points, and have subsequently diverged at a constant rate: for example, a figure of 200 million years implies a number of substitutions about 2/5 of that separating the α and β chains.

In fact, the assumptions on which Fig. 11.3 is based turn out to be false in at least one respect. The difference in sequence between the two human genes is small, suggesting a very recent origin of the duplication. Yet we know that this duplication must be at least 40 million years old. It is present in the great apes and Old World monkeys, but absent in New World monkeys and lemurs, giving a date of 40 million years or more. The explanation is probably as follows. The duplica-

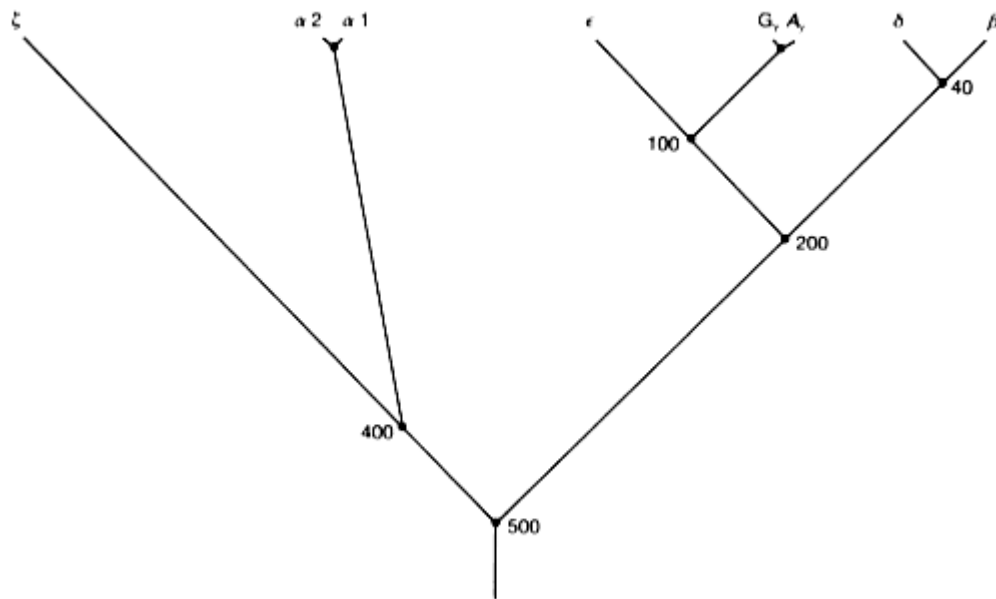


Figure 11.3

The phylogeny of the human haemoglobin genes (after Jeffreys *et al.*, in Bendall 1983), showing the estimated times of divergence in millions of years. The value of 500 MY for the duplication giving rise to the α and β genes is deduced from the fossil record. The other values are deduced from divergences in amino-acid or DNA sequences, as fractions of the divergence between the α and β chains.

tion did indeed take place some 40 million years ago, but in the human lineage a recent recombinational or gene-conversion event has homogenized the sequences. This possibility of homogenization means that we must treat phylogenies of the kind shown in Fig. 11.3 with caution.

The lower vertebrates provide more direct evidence that the α and β genes did indeed arise by duplication. In the toad *Xenopus tropicalis* the α and β genes are closely linked, as would be expected on the duplication hypothesis. How did they come to be on separate chromosomes in mammals? One possible explanation is that a chromosome doubling event (allopolyploidy) took place during the origin of the mammals, giving rise to two α - β clusters. If the α gene was silenced in one of these clusters, and the β gene in the other, this would give rise to unlinked α and β genes, each of which could then diversify by further duplications. As it happens, something of this kind has recently happened in the genus *Xenopus*: the tetraploid species, *X. laevis*, has two unlinked α - β clusters, which have already undergone some functional differentiation.

There are haemoglobins in various invertebrates (including some annelids, insects, and molluscs). More surprising, leguminous plants (and several unrelated flowering plants) have a haemoglobin which acts to reduce oxygen pressure in the nitrogen-fixing root nodules. The gene coding for this leghaemoglobin has the two

introns characteristic of vertebrate haemoglobin, and an additional third intron. Does this mean that haemoglobin has evolved independently in plants and animals (if so, why the same introns?), or that it was present in the common ancestor of plants and animals (if so, why has it been lost from most animals and plants?), or that the gene has been transferred horizontally from animals to plants? Further work on the invertebrate and plant haemoglobins should answer these questions.

Duplication and the Increase in DNA content

The DNA content, and in particular the amount of coding DNA, has increased during evolution. Where has the additional DNA come from? There are, in principle, two possible answers: by duplication of pre-existing DNA, followed by sequence divergence, or by the *de novo* synthesis of sequences not homologous to any pre-existing DNA. We have no reason to think that the latter process occurs. However, effectively random sequences that could subsequently be programmed by selection could arise by duplication followed by a frame-shift mutation.

Duplication can happen in several ways:

1. **Unequal crossing over.** Figure 11.4 shows how unequal crossing over can lead to the production of tandemly arranged copies of the same DNA sequence. We have already seen evidence for the existence of such tandem copies in the haemoglobin gene family. Unequal crossing over can also give rise to repeated segments within a gene. An extreme example is the gene for collagen. The $\alpha 2(I)$ collagen gene from the chicken is approximately 38000 nucleotides long, has more than 50 introns (the largest number so far observed), and is largely composed of a (slightly variable) repeated sequence of nine bases, coding for glycine-X-Y, where X and Y are often prolines. When thinking about the evolution of highly complex repeated sequences, whether the repeats are of whole genes or parts of genes, it is important to remember that, once crossing over has given rise to a tandem repeat, further duplication events are more likely, because chromosomes can pair 'out of register' (see Fig. 11.4).

2. **Transposition.** A DNA sequence is replicated, and the copy is inserted at a new site in the chromosome. This sometimes involves transcription to RNA, and reversed transcription to DNA, before insertion.

3. **Polyploidy.** If chromosome replication takes place without cell division, the result is a tetraploid cell. As explained in Chapter 14, this is more likely to give rise to a fertile tetraploid if it happens in a diploid species hybrid. An example of tetraploidy in the toad, *Xenopus*, was discussed in the last section. It has also happened in two groups of bony fish, the Cyprinidae and the Isospondylidi: some evidence is given in Table 11.2.

For some enzymes, the $4n$ species have more distinguishable loci than the $2n$ species. For example, *Salmo* has two distinguishable loci coding for SDH, and four

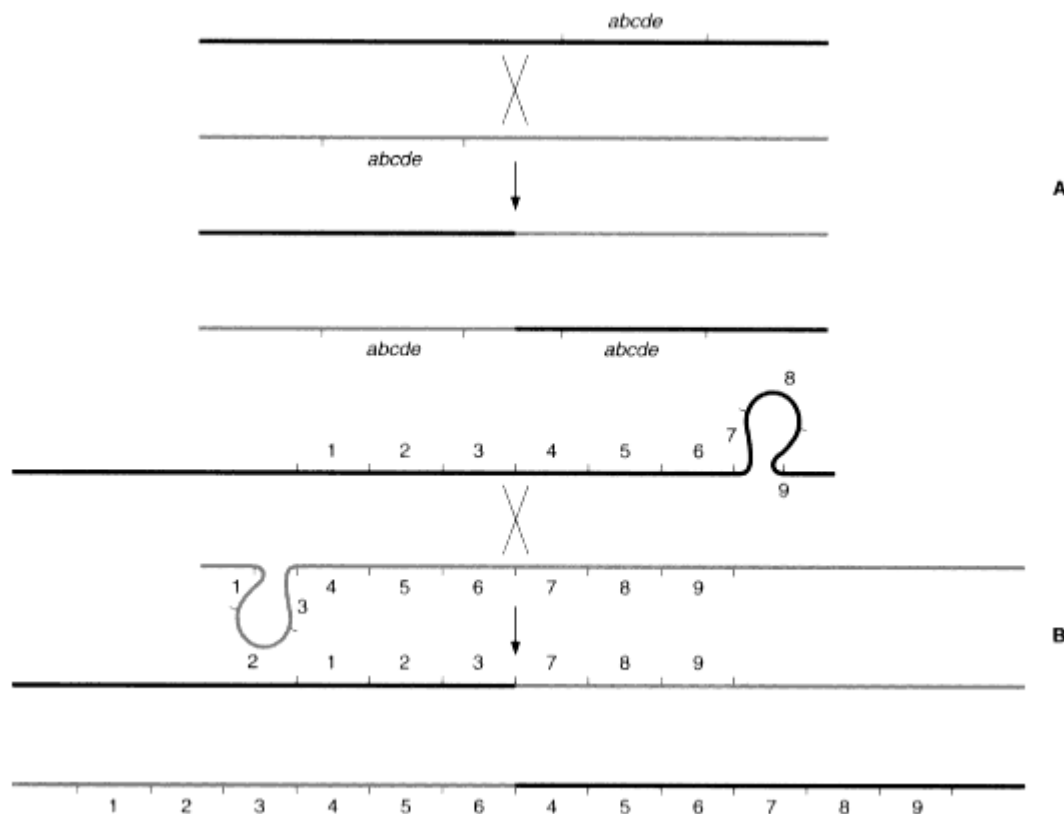


Figure 11.4

Unequal crossing over. **A** Origin of a tandemly arranged DNA sequence, *abcde*. In **B**, two chromosomes each carry nine tandemly arranged copies of the same sequence. They pair out of register, and crossing over gives rise to a chromosome with six copies, and one with 12 copies.

for LDH, compared to one and two loci, respectively, in its $2n$ relatives. However, there is one distinguishable 6-PGD enzyme, and three α -GPDH enzymes, in both diploid and tetraploid isospondylids. This may be because the duplicated loci have not diverged sufficiently for their products to be distinguished by electrophoresis, or because one of the duplicates has been inactivated by nonsense mutations.

It is worth listing the possible fates of a duplicated locus:

1. One copy may be inactivated.
2. The two loci may diverge, while maintaining the same basic functions, as has happened in the haemoglobin gene family.
3. The two loci may acquire different functions, while retaining a similar three-dimensional structure, and some degree of amino-acid homology. It is not easy to detect this when it happens. One example is afforded by the proteins lysozyme and α -lactalbumin, which have similar three-dimensional structures and about 35 per cent amino-acid homology. Lysozyme, which is present in all

Table 11.2

A comparison of diploid and tetraploid fish

		DNA content (% of human leucocytes)	Chromosome number	Number of chromosome arms
Cyprinidae				
2n	<i>Barbus tetrazona</i>	20	50	84
	<i>Tinca tinca</i>	30	48	80
4n	<i>Barbus barbus</i>	49	100	144
	<i>Cyprinus carpio</i>	50	104	150
	<i>Carassius auratus</i> (goldfish)	53	104	168
Isospondyli				
2n	<i>Clupea harengus</i> (herring)	28	52	60
	<i>Osmerus esperlanus</i>	19	54	70
4n	<i>Salmo irideus</i> (rainbow trout)	80	56 8	104
	<i>S. trutta</i> (brown trout)	80	77 2	100
	<i>S. salar</i> (Atlantic salmon)	103	54 0	72

vertebrates, is an enzyme that destroys the mucopolysaccharide component of bacterial cell walls. Lactalbumin is a major component of milk. On its own it has no known enzymic activity, but in combination with another protein it catalyses the synthesis of lactose.

4. One duplicate may acquire a new function, after undergoing frame-shift mutations that ensure that there will be no amino-acid homology, and no similarity in three-dimensional structure, between the new product and the old. No example of such a process is known. It would be even more difficult to spot than a change of function without loss of amino-acid homology. The only trace would be homology between the nucleotide sequences of the genes.

The Ribosomal Genes

Figure 11.5 shows an element coding for ribosomal RNA in *Drosophila melanogaster*. This element is tandemly repeated, with minor variations, about 200 times on both the X and Y chromosomes. This gene family illustrates the phenomenon of 'concerted evolution'. Other *Drosophila* species have a similar repeated unit. Within each unit, they have a shorter repeated unit, similar in sequence to the 250-nucleotide repeat in *D. melanogaster*, but one that lacks the *AluI*-sensitive sequence. How, then, did the *AluI* sequence spread, to the different short repeats within a unit, to the different units, and to the X and Y chromosomes? It is not

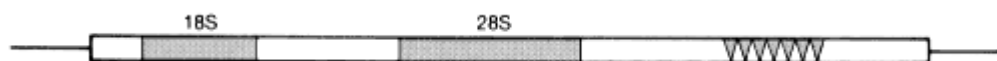


Figure 11.5

A repeating unit of DNA coding for ribosomal RNA in *Drosophila melanogaster*. The shaded regions, 18S and 28S, are the coding regions; the remainder of the unit consists of spacers. The small inverted triangles represent a region of internal repetition with a 250 bp periodicity, each containing a sequence recognized by the endonuclease.

AluI (after Dover, in Dover and Flavell 1982).

plausible that it should have arisen by mutation independently several thousand times, each mutation being established in the population. We seek mechanisms whereby a sequence that arises by mutation at one site can spread to other sites. Three such mechanisms have been proposed: unequal crossing over (the likely explanation in the case of the ribosomal genes), gene conversion, and transposition.

Unequal Crossing over and Gene Conversion

When the same DNA sequence is repeated in tandem, pairing can occur 'out of register' (either in meiosis, or between sister strands in mitosis), as illustrated in Fig. 11.4. Crossing over can then give rise to new chromosomes, one of which has two copies of elements that were present only once in the parental chromosomes. If this process continued without check, it would lead to a steady increase in the variability of copy number. If, however, individuals with too many copies, or with too few, are of lowered fitness, variation would not increase indefinitely. The result would be that, in time, all the repeated units would be copies of one original unit. This possibility is investigated further in Box 11.1. Note that unequal crossing over

Box 11.1—

Unequal Crossing Over

In principle, unequal crossing over can result in the homogenization of tandemly repeated sequences. The rate of homogenization may, however, be very slow. An analytical treatment is difficult, so I have simulated the process, with the following assumptions. In each generation, the population consists of N haploid gametes. To produce the next generation, two gametes are chosen at random, and a cross-over point is chosen randomly in each, giving two resultant gametes (see Fig. 11.6). If the number of copies of the gene in a gamete produced in this way lies within the permissible range, it becomes a member of the next generation. This process is repeated until N gametes have been produced. In the first generation, all chromosomes were supposed to have the same number of copies, the copies at a given site were identical, and the copies at different sites were different. The simulation was continued until all copies were identical, with the results shown in Table 11.3.

The Table gives estimates of the number of generations before the genes at

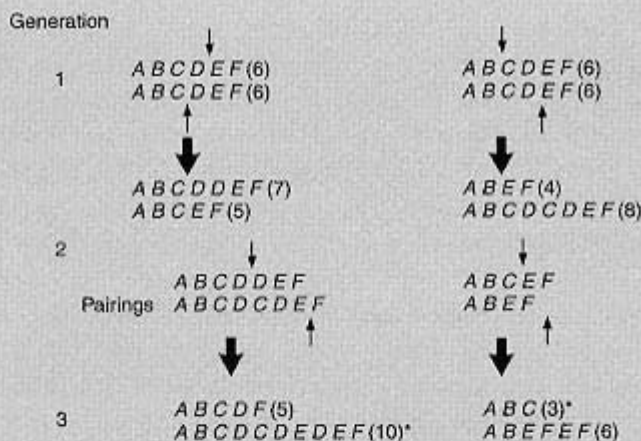


Figure 11.6 A simulation of unequal crossing over. In generation 1, all chromosomes have the same copy number (6 in this case), and all copies are different. The diagram shows two pairings, and the gametes produced. Two F_2 pairings are also shown, with the gametes passed to the F_3 ; two of these gametes exceed the permissible range of 4–8 copies, and are eliminated (*).

all sites are descended from the genes at a single site in an ancestor. When it is remembered that copy number for ribosomal RNA in *Drosophila* is about 200, and that the population size may be large (the relevant number is, of course, the number for the whole species), it will be clear that the process is very slow. To make matters worse, an appreciable genetic load is involved in maintaining the copy number within permissible limits: for these simulations, the load was 0.57, which is far higher than could possibly be the case for a single gene family. Of course, the genetic load would be lower if, as is plausible, unequal crossing over does not occur in every generation, and, when it does occur, the mispairing is less extreme. However, if this were so, the rate of homogenization would be still slower.

Table 11.3
Concerted evolution by unequal crossing over—number of generations before all of a set of tandemly repeated genes are copies of a single ancestral gene

Permissible number of gene copies	Number of haploid genomes		
	10	20	40
4–8	170±15.5	287±24.4	379±40.5
8–16	398±36.5	—	—
16–32	882±87.5	—	—

The idea that the homogenization of tandemly arranged sequences is brought about solely by unequal crossing over must therefore be treated with caution. The process would be greatly speeded up if there is some selective process acting. This could act at the level of the organism, favouring individuals with a larger proportion of a particular sequence. Alternatively, selection may act at the level of the genes, favouring the spread of some particular sequences at the expense of others.

will not result in a set of identical units, because mutation continuously generates differences between them.

The best evidence that unequal crossing over does, in fact, occur comes from *Drosophila*. The sex-linked mutation *bobbed* is known to be due to a deficiency in copy number of rRNA genes, and to arise by unequal crossing over. In a selection experiment for reduced abdominal bristle number in *Drosophila*, Frankham *et al.* (1978) found that a rapid response that occurred in females but not in males was caused by chromosomes with a reduced copy number of ribosomal genes. These chromosomes were not present in the foundation population from which the selected lines were derived, so they presumably arose by unequal crossing over during the experiment. A minimum estimate of the rate of unequal crossing over was 3×10^{-6} , per gamete, per generation.

Unequal crossing over does provide a mechanism for the concerted evolution of tandemly repeated units. However, it cannot provide an explanation for the concerted evolution of a set of similar units distributed around the genome, because, if pairing and crossing over took place between such units, the result would be a major chromosomal rearrangement (in fact, a translocation if the elements were on different chromosomes), which would usually lead to lowered fertility. An alternative process is 'gene conversion'. It is known from the study of meiosis in fungi that, when two alleles pair at meiosis, occasionally one of the alleles is converted to resemble the other: from the pairing of alleles A_1 and A_2 , there emerge two copies of A_1 , or two copies of A_2 . The process can be biased, in that allele A_1 more often converts A_2 than vice versa. Gene conversion involves similar molecular processes to recombination, but need not lead to crossing over, and hence could occur between similar genes at different chromosomal locations without leading to infertility. The role of gene conversion in evolution is hard to evaluate.

Repetitive DNA

In this section, I first describe some well-studied examples of repeated sequences in eukaryotes, and then discuss their evolutionary significance. Before reading it, it would help to remind yourself of the description of transposition in prokaryotes in Chapter 10.

The Element Copia in *Drosophila*

About 15 per cent of the genomic DNA in *D. melanogaster* consists of moderately repetitive DNA (i.e. of the order of 100 copies); of this, one-half consists of 30 or so families of 'copia-like' elements. Figure 11.7 shows copia itself: the qualitative features are common to all the families.

There are some 30-50 copies of copia in a haploid genome of *D. melanogaster*. However, the sites occupied are different in different flies. A survey of geographical races found over 200 sites of possible integration, and there may be many more. In tissue culture cells, up to 150 sites may be occupied. These facts suggest

1. that the element can move to new sites; and
2. that the total number of elements per genome is regulated.

In these respects, copia resembles the insertion sequences and transposable elements found in prokaryotes. The mechanism of regulation is not known: however, copia DNA is transcribed, so it is possible that copia regulates its own copy number.

The mechanism whereby copia moves to new sites is also unknown. However, the structure of copia is very similar to that of retroviruses: all three kinds of repeats shown in Fig. 11.7, and their arrangement, are found in retroviruses. This strongly suggests a similar process of transposition. In retroviruses, the cycle is: DNA provirus - RNA transcript - extra-cellular virion - infection of new cell - reverse transcription and insertion of provirus into chromosome. There is no evidence for an extra-cellular phase in the life cycle of copia, but otherwise the cycle may be similar, and involve reverse transcription.

When a copia-like element moves to a new site, it can cause a mutation. This was

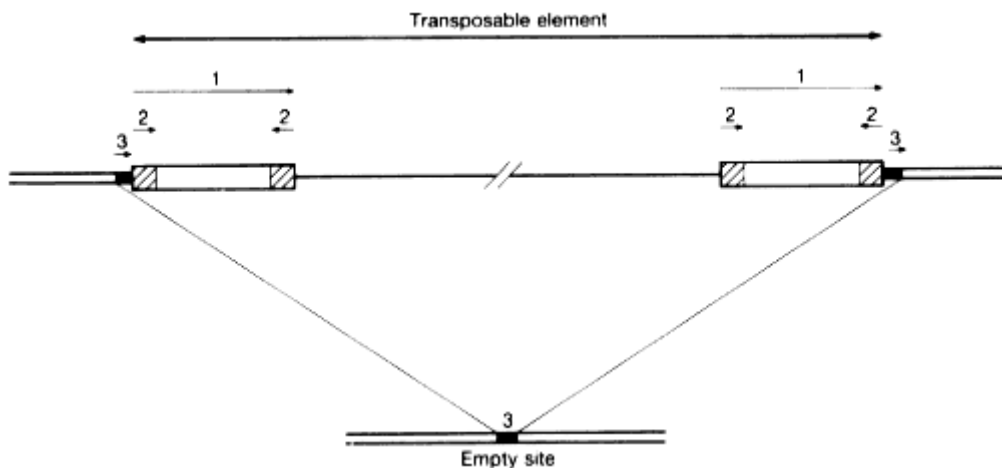


Figure 11.7

The copia element in *Drosophila* (after Finnegan 1985). The element contains the following repeats: 1, long direct repeat; 2, short inverted repeat; 3, repeat of a host sequence present only once in an empty site.

first realized when it was found that the *white-apricot* allele consists of a normal gene with a copia-like element inserted. A transposable element may also alter the expression of neighbouring genes. In vertebrates, transposable elements can cause cancers by inappropriate activation of genes. However, there is no reason to think that these elements play a necessary role in normal development. Thus copia is present at different sites in different flies, but at the same site in different tissues of the same fly: this is the opposite of what we would expect if these elements played a role in the normal control of gene activity.

***P* Factors and Hybrid Dysgenesis**

P factors in *Drosophila melanogaster* were discovered because of their effects on the fertility of certain crosses. If freshly trapped males are crossed to females from a long-established laboratory stock, the hybrids show a number of abnormalities, including lowered fertility, recombination in males, and a high rate of visible mutations and chromosomal abnormalities among their progeny. The reciprocal cross, of laboratory males to females recently derived from the wild, showed no such abnormalities. This phenomenon of **hybrid dysgenesis** is now known to be caused by transposable *P* factors.

Flies from wild populations carry 30-50 copies of a sequence of about 3000 nucleotides. This DNA codes for two proteins, a transposase that mediates their movement to new sites in the genome, and a regulatory protein. In a P^+ strain, carrying such factors, transposition is an infrequent event. Laboratory stocks carry no factors—they are P^- . When chromosomes from a P^+ male are introduced into eggs from a P^- female, the transposition rate increases dramatically, in the absence of regulatory factors in the egg cytoplasm. It is this transposition that causes the phenomena associated with hybrid dysgenesis.

In addition to *P* factors, the genomes of P^+ flies contain deficient 'P elements', which cannot cause their own transposition, but which transpose in the presence of *P* factors. They are shorter than *P* factors, and are probably derived from them by deletion. Interestingly, they are absent in P^- laboratory strains.

There are two possible evolutionary explanations of these facts. Either *P* factors have been lost from laboratory strains, or they were absent from wild populations 40-50 years ago, when these strains were established, and have spread rapidly through the species since. A strain established by crossing P^+ and a P^- fly becomes P^+ , as the *P* factors spread by transposition. Even if *P* factors were ultimately lost from such a strain, we would expect some inactive *P* elements to remain, yet they are absent from old laboratory stocks. It is therefore hard to see how *P* factors could have been lost by laboratory strains: in time we shall know whether more recently established strains lose their *P* factors.

The rapid spread of *P* elements in the wild is easier to understand. It does, however, have one puzzling feature. Is it not rather a coincidence that such an

event should have occurred in the one species geneticists have chosen for study? The puzzle is made worse by the fact that a second factor, I, also appears recently to have spread through the species. It surely cannot be the case that elements of this kind, causing serious inviability and infertility, sweep through most species every 20 or 30 years. We are therefore driven to seek some reason why *D. melanogaster* should have been subject to these dramatic events. One possible explanation is that *D. melanogaster*, like the house mouse and the house sparrow, is a human commensal that has enormously extended its geographical range in recent years. This will have brought it into contact with many new species (although none, so far as we know, with which it forms fertile hybrids). This range extension may have enabled the P and I elements to invade *D. melanogaster* by horizontal transfer from another species.

The Alu Sequences of Primates.

In humans, there are some 3-5 × 10⁶ copies of a DNA sequence, *Alu* (the name derives from the fact that the sequence includes a cleavage site for the endo-nuclease *AluI*). *Alu* is 282 nucleotides long, is distributed throughout the genome, a copy occurring every 5-10 000 bases, and constitutes in all some 5 per cent of the genomic DNA. The *Alu* sequence is a dimer, consisting of two parts, each homologous to a small functional RNA molecule, 7SL RNA, present in birds and mammals, and involved in the synthesis of secreted proteins. The sequences are not identical: individual copies differ at about 10 per cent of sites from a typical 'consensus' sequence. An essentially similar *Alu* family of sequences exists in chimpanzees and in owl monkeys; in the galago, a prosimian, there is a similar sequence, but differing by a number of substitutions. These sequences are thought to transpose by the production of an RNA transcript, followed by reverse transcription and integration. They probably originated by reverse transcription from 7SL RNA.

Sequence families similar to *Alu* are characteristic of mammals. They are not known to contribute to the survival of the organism. The most plausible explanation for their presence is that they have evolved a sequence that, making use of the enzymes present in the mammalian cell, is particularly efficient at horizontal multiplication throughout the genome: that is, they are intra-chromosomal parasites. However, their presence does have important effects on mammalian evolution, because interactions between *Alu* sequences at different sites may cause structural rearrangements of the chromosomes. The rapid evolution of chromosome structure in mammals may therefore be caused by the presence of *Alu*-like dispersed sequences. Of course, only a small proportion of the structural mutations that occur, whether or not induced by the presence of *Alu*, will be established in evolution. However, the presence of these elements may generate a larger number of rearrangements as candidates for evolutionary change.

Highly Repetitive DNA

In the grasses, the DNA content of a haploid genome varies between 3.6 and 8.8 pg. Much of it consists of short sequences, with copy numbers that may be greater than a million, occurring in tandemly arranged blocks, distributed over the chromosomes, but concentrated in particular regions. This category of repeated DNA differs from those considered above, not so much in copy number (*Alu* is present in over 10^5 copies), as in its sequence and distribution. The sequences do not seem to show any features, such as terminal repeats and sites for polymerase activity, which function to ensure their transposition. They occur in tandemly arranged blocks, instead of being dispersed singly throughout the genome. Highly repetitive DNA of this kind occurs throughout the animal and plant kingdom, but in very varying amounts.

For any given family of repeated sequences, there is variability within species, but there is variability between species in what families are present. This indicates that new families can arise and spread rapidly through the genome. Unequal crossing over can explain changes in copy number in a given tandemly arranged block, but not the spread of a sequence to different chromosomes. The mechanism of such spread is not known for certain, but is thought to occur by the excision of a short length of DNA from the chromosome, the replication of that DNA to form a tandem array, and its reinsertion at a new chromosomal site.

The Role of Repetitive DNA in Evolution

It is too early to be confident about the evolutionary significance of repeated DNA. Gene clusters have evolved by duplication because of the advantage to the organism of having slightly different versions of a given functional protein. Tandemly repeated sequences with specific functions meet the need for a large amount of gene production (ribosomal genes, histone genes), or for the generation of a high degree of diversity (immunoglobulin genes). In the former case, unequal crossing over can cause concerted evolution, but there is room for doubt whether it operates quickly enough to account for observed sequence homogeneity.

In thinking about middle-repetitive dispersed DNA, and highly repetitive DNA, two general points should be remembered:

1. The interior of the cell, with its battery of enzymes that replicate and recombine nucleic acids, provides the ideal environment for parasitic DNA; we should not, therefore, be surprised if such DNA proves to be widespread.
2. We must distinguish between the selective pressures that have been responsible for the evolution of a molecule, and the effects it may have on evolution: in other words, we must distinguish between the 'function' of a molecule and its 'consequences'.

The case for regarding much middle-repetitive DNA as parasitic is strong. The similarity between copia-like elements and retroviruses is very striking, although

we do not know whether retroviruses have evolved from transposons, or transposons are retroviruses that have abandoned the extra-cellular stage of their life cycle. Either way, copia has structures that adapt it for transposition, as we would expect of a parasite. The fact that it is found at different sites in different flies, but the same site in different tissues of the same fly, is inconsistent with the idea that it plays a role in development. The case for a parasitic role is still stronger for P factors, since we know that flies are apparently quite normal without them. The evidence for self-regulation of copy number in P factors, and the possibility of such regulation for other elements, are also consistent with a parasitic role; an element that transposed too frequently (as P factors do after hybridization to *D* fly) would lower the fitness of the host it infected, and therefore reduce its own chance of leaving descendents. Highly repetitive DNA is harder to interpret, because it is less obvious how its sequence affects its likelihood of spreading through the genome.

Even if repetitive DNA did not evolve because of any advantage it confers on the organism, it will still have consequences for organismic evolution. First, transposition causes both gene mutation and structural changes in chromosomes. Many laboratory mutants of *Drosophila* have been found to have been caused by the presence of a transposable element within the structural gene. Such mutations are unlikely to be important in evolution (except by increasing the genetic load) for two reasons. First, such a mutation would, almost certainly, destroy the function of the gene. Secondly, we know from sequencing studies that structural genes do not differ between species by the presence of larger insertions. If an element is inserted into a structural gene, and later imprecisely excised, this could cause a 'point' mutation that might be established in evolution: if so, transposition is an additional cause of a class of mutation that would happen anyway. More interesting is the fact that the insertion of elements into regions between structural genes can affect the regulation of those genes. Transposition, therefore, is a potential source of regulatory mutations.

Transposable elements may affect the evolution of the number and form of chromosomes ^概i.e. of the **karyotype**. It is interesting to contrast karyotypic evolution in mammals and in plethodontid salamanders. In many mammals, the karyotype evolves rapidly. In contrast, the genus *Plethodon*, although 80 million years old (i.e. as old as the placental mammals), is extremely uniform both in morphology and in karyotype. The DNA content per genome is large and highly variable between species, consisting mainly of dispersed middle-repetitive DNA. It seems that in some lineages there has been a gradual but uniform expansion of all the chromosomes, through the random insertion of additional DNA elements. This has not been accompanied by the evolutionary establishment of inversions or translocations. There are at least three possible explanations for this difference:

1. Mutations causing structural rearrangements are much commoner in mammals, perhaps because of interactions between the *Alu*-like elements that are

characteristic of mammals. Of course, the great majority of such mutations are not established in the population, but the presence of *Alu* elements may generate a much larger number of candidates for evolutionary change. This suggestion is speculative. At present, there is no reason to think that the sequence of *Alu*-like elements makes them particularly likely to generate inversions and translocations, or that the structure of the elements in *Plethodon* favours transposition without causing structural changes. The idea also assumes that the rate of karyotypic evolution is determined by the rate of mutation. This is contrary to what is almost certainly true of morphological evolution, whose rate is determined by the strength of selection. However, this is not unreasonable. Mutations causing specific karyotypic changes are very rare compared to mutations affecting structural genes, or the regulation of those genes. For example, the distribution of gene orders in *Drosophila pseudoobscura* and its relatives is consistent with the view that each inversion arose just once. So the idea that differences in karyotypic evolution between taxa is caused by differences between the nature of the repetitive DNA, although speculative, is not obviously false.

2. Mutants causing karyotypic change occur equally in both groups, but are less likely to cause severe infertility in mammals. A possible reason for this is that a mammal that produces a litter usually implants more embryos than can be brought to term. Hence if some embryos are aneuploid, and die very early, this does not reduce the fertility of the parents.

3. Karyotypic change is necessary for morphological change, and there has been strong selection for morphological evolution in the mammals. The weakness of this view is that there is little evidence of any causal link between morphological and karyotypic change.

It will be apparent from this discussion that we have a very poor understanding of the causes of karyotypic evolution, despite many careful investigations. I find the idea that karyotypic evolution is driven by mutations caused by transposable elements attractive, but it will remain speculative until we know more about the role of transposons in different taxa in generating karyotypic mutations.

It has been suggested that repetitive DNA may be important in speciation, in one of two ways. First, hybrid dysgenesis caused by P factors has been seen as a model for speciation. This does not seem sensible. If hybrids between P^+ and P^- flies were completely inviable or sterile, this would lead to the rapid elimination of the rare form. A newly arisen P factor would therefore have no chance of spreading. As it is, hybrids are not completely sterile, and the result has been the rapid spread of P elements through the species, without causing any dramatic change in the species, and without splitting the species into two. A more plausible suggestion is that repetitive DNA may spread differentially in two geographically isolated populations, causing hybrid sterility when those populations later meet. This may

sometimes happen. However, the DNA difference would have to be large: experiments in both *Drosophila* and grasses have shown that chromosomal pairing and chiasma formation is normal in hybrids between parents differing by up to 50 per cent in DNA content, although still greater differences can disrupt pairing. It is also worth remembering that hybrids between related species, if once formed, are often fertile, and the isolating mechanisms are ecological or behavioural.

A final suggestion is that transposons may transfer genes horizontally between species that are too distantly related to hybridize. This cannot be a common event, even on an evolutionary time-scale, because phylogenies deduced from DNA or amino-acid sequences are similar to those deduced from morphology, and this would not be so if distant gene transfer was common. But it may be occasionally happen. A possible example is the presence of leghaemoglobin in several groups of higher plants (see pp. 206-7).

A final caution is needed. Our knowledge of repeated DNA is recent and incomplete. It is therefore foolish to hold dogmatic views about its role in evolution.

The Evolution of Chromosome Form

A lot is known at the descriptive level about the evolution of the numbers and shapes of chromosomes. That is, of the karyotype, although we know something of the mechanisms of change, and of their consequences on fertility, the general significance of karyotypic evolution is obscure.

The haploid number of chromosomes, n , varies from one in the nematode *Parascaris* to 127 in the hermit crab *Eupagurus*, and over 250 in some ferns. In some taxa, the number is relatively uniform: for example, $n = 13$ in most dragonflies, and $n = 18$ in many snakes. In some groups, **acrocentric** (rod-shaped, with a terminal centromere) and **metacentric** (V-shaped, with a centromere in the middle) chromosomes can be distinguished: the number of chromosome arms may then be more constant than the number of chromosomes. There can, however, be big differences between related species. The most extreme example is the difference between the Assam sub-species of the Indian muntjac *Muntiacus muntjac vaginalis*, with $2n \text{ ♀} = 7$ and $2n \text{ ♂} = 6$, and the Chinese muntjac, *M. reevesi*, with $2n = 46$.

A change in karyotype requires at least two chromosome breaks, followed by a joining of the broken ends: some of the simpler changes are shown in Fig. 11.8. Such changes can be produced by radiation, and by some chemicals. However, it may well be that in nature the commonest cause of breaking and rejoining is recombination between repeated DNA elements: for example, P factors are known to produce chromosome rearrangements in *Drosophila*, and *Alu*-like elements probably do so in mammals. It is thought that the free ends of chromosomes cannot participate in rearrangements: thus inversions involving a single break, with the distal end rejoining in an inverted position, seem not to occur.

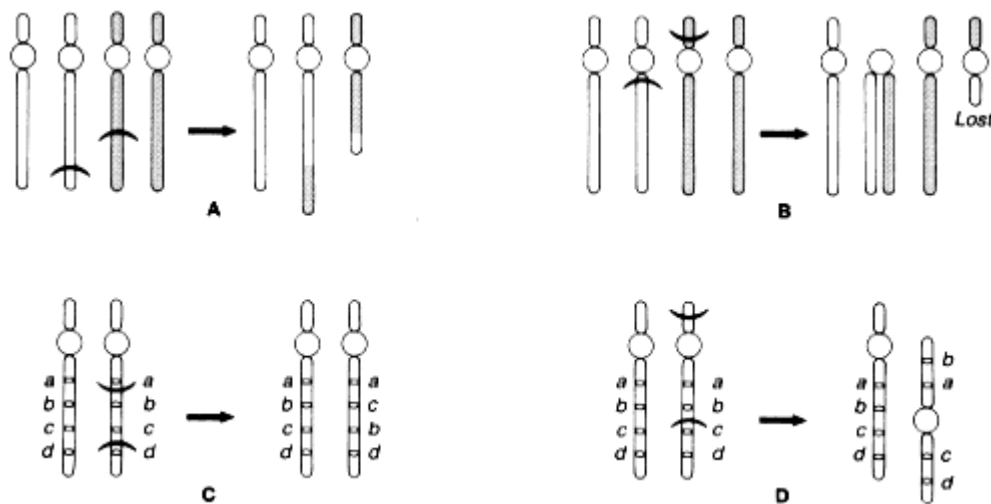


Figure 11.8

Origins of changes in chromosome structure. **A** A translocation; **B** a centric fusion; **C** a paracentric inversion; **D** a pericentric inversion. In **B**, a large metacentric chromosome is formed from two acrocentrics: in **D**, an acrocentric chromosome is converted into a metacentric carrying the same genes. represents non-homologous chromosomes; centromere; points of chromosome breakage or recombination; *a, b, c, d* represent genes.

The most important effect of such changes is a loss of fertility in the heterozygote for the new and old structures, as shown in Figs 11.9 and 11.10. This arises because up to 50 per cent of the gametes produced are aneuploid (i.e. lack some genes altogether, and have other genes present twice). Note that this loss of fertility affects only the heterozygote: homozygotes for the new karyotype are fully fertile.

How, then, can a new karyotype be established in a population? Since the heterozygote is of low fitness, whichever is the rarer form will be eliminated. The likely answer is that the transition from one karyotype to another occurs by chance in a small and partially isolated population. Note that, once a population has acquired a new karyotype, it will not readily lose it again if invaded by immigrants with the old karyotype, which will be locally rare and hence at a disadvantage.

Since structural heterozygotes are usually of low fertility, genetic polymorphism for karyotype is unusual. However, many species of *Drosophila*, and other Diptera, are polymorphic for paracentric inversions. This makes sense, because such inversions do not lower fertility in the Diptera. Thus Fig. 11.10 shows that aneuploid gametes arise from crossing over within the inversion. There is no crossing over in male Diptera. In females, the gametes with two centromeres, or with none, pass to the polar bodies, and the egg pronucleus is euploid.

Another case in which structural heterozygosity leads to no loss of fertility is that of centric fusions in the shrew, *Sorex araneus*. It is clear from Fig. 11.11 that

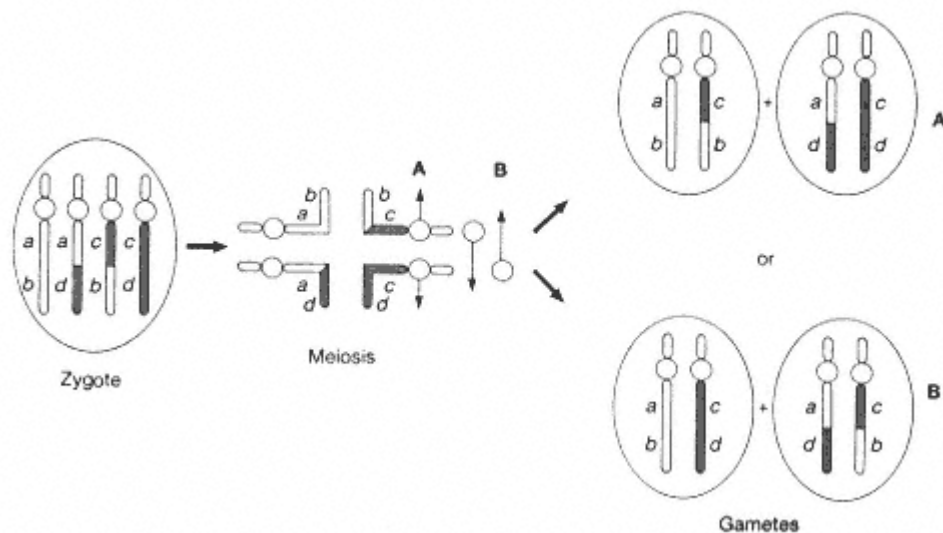


Figure 11.9

Meiosis in a translocation heterozygote. Depending on how the two pairs of centromeres disjoin, meiosis gives rise to: **A** aneuploid gametes, with some genes present twice, and some not at all; or **B** euploid gametes, with all genes represented once only. Notation as in Fig. 11.8.



Figure 11.10

Meiosis in a heterozygote for a paracentric inversion, with a single cross-over within the inversion. The four products are: an inverted and an uninverted chromosome, each with one centromere and a complete set of genes; a chromosome with two centromeres; and a chromosome with no centromere. Note that crossing over involves only two of the four strands.

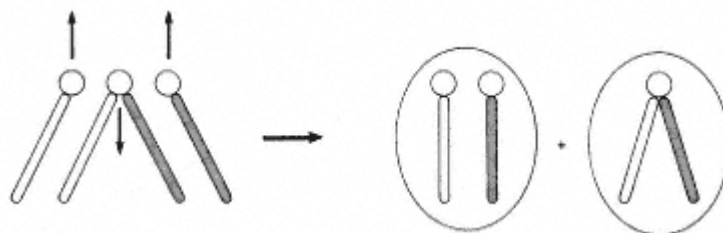


Figure 11.11

Meiosis in a heterozygote for a centric fusion. If, as shown, the centromeres of the two acrocentrics move together to one pole, and the centromere of the metacentric to the other pole, then all the gametes produced are euploid, and no infertility arises.

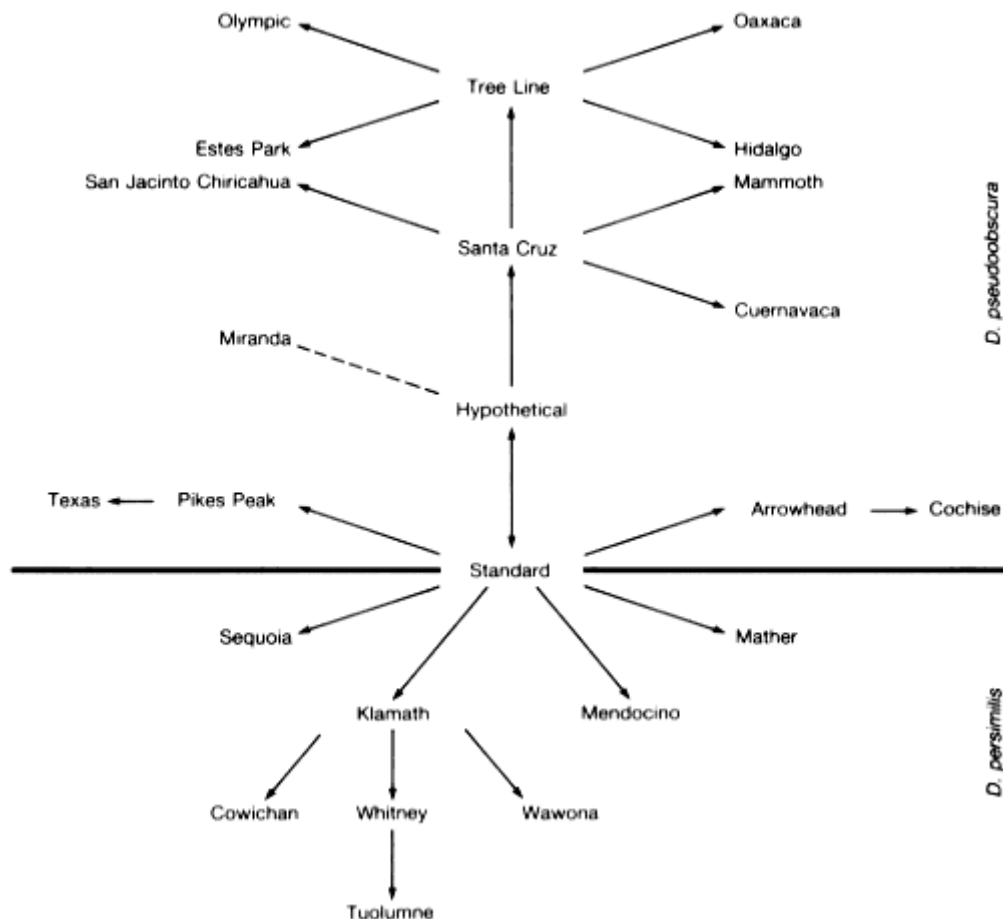


Figure 11.12

The phylogeny of the third chromosome in *Drosophila pseudoobscura* and *D. persimilis*. Arrows connect orders that differ by a single inversion. Standard is the only order present in both species. The Hypothetical order has not been found, but is a necessary link between Standard and Santa Cruz. The chromosome orders in a third species, *D. miranda*, can be derived from Hypothetical. Note that this is an 'unrooted' tree: the direction of evolution is unknown. (Adapted from Dobzhansky 1951.)

there will be no infertility if the centromeres of the two acrocentrics pass to one pole, and the centromere of the metacentric to the other. This is what happens in shrews: not surprisingly, natural populations are polymorphic for centric fusions. In other taxa, centric fusions may cause loss of fertility. For example, the house mouse, *Mus musculus*, has $2n = 40$, and the 'tobacco mouse', *M. poschiavinus*, differs by seven centric fusions, and hence has $2n = 26$. Hybrids between the two are of low fertility.

This mouse example, and the earlier example of the muntjac, illustrate a more general point. It seems that in some lineages, such as that connecting the Indian to the Chinese muntjac, or the house and tobacco mouse, the same kind of structural

change must have occurred repeatedly, affecting many different chromosomes. White (1978) describes a number of such cases, involving different kinds of change. He has called the process **orthoselection**. The term is unfortunate, because we have no idea what selective advantage, if any, is responsible, but the phenomenon is interesting.

In *Drosophila*, it is possible to locate the break points of the inversions with some accuracy, by examining the giant salivary chromosomes. Thus one can construct a phylogenetic tree of the gene orders of chromosome III of *D. pseudoobscura* and *D. persimilis* (Fig. 11.12). Each named order differs from its neighbours in the tree by a single inversion. One sequence, Standard, is common to both species. To complete the tree, one hypothetical order that has not been found must be included: this order is a plausible ancestor for the chromosomes of a third species *D. miranda*. We cannot tell the direction that evolution has taken: in taxonomic jargon, it is an unrooted tree. Two conclusions can be drawn. First, the establishment of each inversion was a unique event: otherwise, the figure would be a network, and not a tree. Secondly, once established, a chromosome order is long lasting: otherwise, there would be more than one gap in the tree.

Further Reading

Bendall, B.S. (ed.) (1983). *Evolution from molecules to men*. Cambridge University Press. (Particularly articles by D.C. Phillips *et al.*, A.J. Jeffreys *et al.*, and W.F. Bodmer.)

Dover, G.A. and Flavell, R.B. (ed.) (1982). *Genome evolution*. Academic Press, London.

White, M.J.D. (1978). *Modes of speciation*. W.H. Freeman, San Francisco.

Computer Projects

1. Unequal crossing over. Write a program to produce data similar to those in Table 11.3, and calculate the genetic loads. (This is not an easy program to write, and probably not for beginners.)

2. Peak shifts in finite populations. In a diploid, the three genotypes at a locus have phenotypes $aa = 1$, $aA = 1 - s$, $AA = 1 + t$; (aA is a chromosomal structural heterozygote of lowered fertility). A population of n males and n females (where n is small; say 5-20) mates randomly. If the initial population consists of one aA (a new mutant) and $2n - 1$ aa individuals, what is the probability that allele A will be fixed? What is the probability that A will be fixed if the initial unique individual is AA (a migrant from an already established AA population)? (You should check that the program is running correctly by answering the question when $s = t = 0$.)

Chapter 12— The Evolution of Genetic Systems: I. Sex and Recombination

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The evolution of genetic systems is perhaps the most difficult and exciting topic in evolutionary genetics. The difficulty arises in part from the complexity of the phenomena that have to be explained, and the bewildering array of facts that have to be borne in mind. More fundamentally, it arises because the genetic system affects not only the fitness of the individual, but also the evolutionary potential of the population: there is therefore a difficulty in deciding at what level, between-individual or between-population, selection has acted.

This chapter discusses why sex and recombination occur at all: the next chapter accepts that sex occurs, and discusses various consequences. Attention is confined to sexual processes as they can be observed today in eukaryotes. The essential feature of the sexual process is that genetic material from different ancestors is brought together in a single descendent. The analogous processes that occur in prokaryotes are so different that they require a separate treatment: they were discussed briefly in Chapter 10. However, in thinking about the origins of eukaryotic sex, it is worth remembering that the enzymes that today mediate genetic recombination during meiosis probably originated because of their role in DNA repair in our prokaryotic ancestors.

The Natural History of Sex

In this section, I describe the main features of the sexual process in eukaryotes, many of which will already be familiar.

Meiosis

In most eukaryotes, there is an alternation between haploid and diploid states (Fig. 12.1). Two haploid cells, or gametes, fuse to form a diploid zygote in the process of **syngamy**: the diploid zygote produces gametes by **meiosis**. The essential characteristics of meiosis are shown in Fig. 12.2. Each chromosome is replicated, to form two **chromatids**. Pairs of homologous chromosomes lie side by side to form

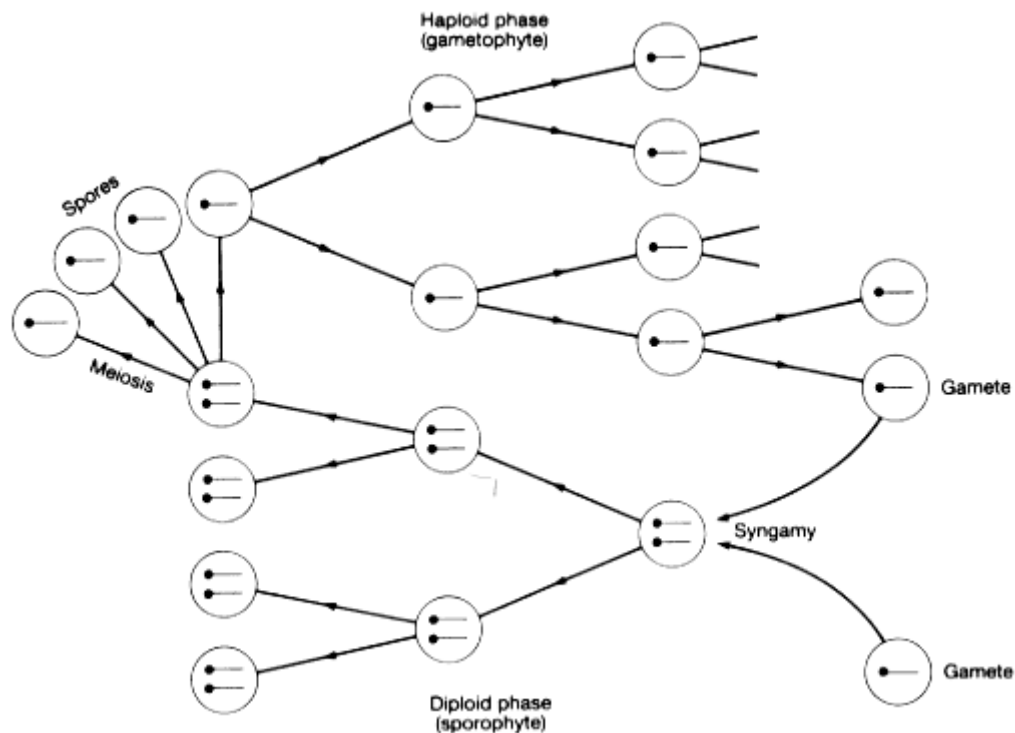


Figure 12.1

Life cycle with alternation of haploid and diploid phases: for simplicity, an organism with a haploid number of one is illustrated.

bivalents consisting of four chromatids, which are then distributed to four haploid nuclei by two meiotic divisions of the nucleus. **Crossing over** takes place at the four-strand stage, by breakage and reunion of two of the four chromatids at exactly homologous points. The result of crossing over is the formation of **chiasmata**, which are essential for proper disjunction: if no chiasmata are formed between two homologous chromosomes, both may pass to the same daughter cell, giving rise to aneuploid gametes. The **centromeres**, by which the chromosomes are attached to the spindle, and by which they are then moved to opposite poles, are not replicated until after the first meiotic division: consequently, the centromeres disjoin at the first meiotic division, as do any genes lying between the centromere and the first chiasma.

Although chiasmata are needed for proper disjunction in most eukaryotes, **achiasmate** meiosis has evolved on several occasions. It is most familiar in male Diptera, including *Drosophila*, but it is also found in female Lepidoptera. The effect is that no recombination occurs between genes on homologous chromosomes. Crossing over is also suppressed if the chiasmata are localized at the ends of chromosomes, as happens in many grasshoppers, and in female *Antechinus*, the

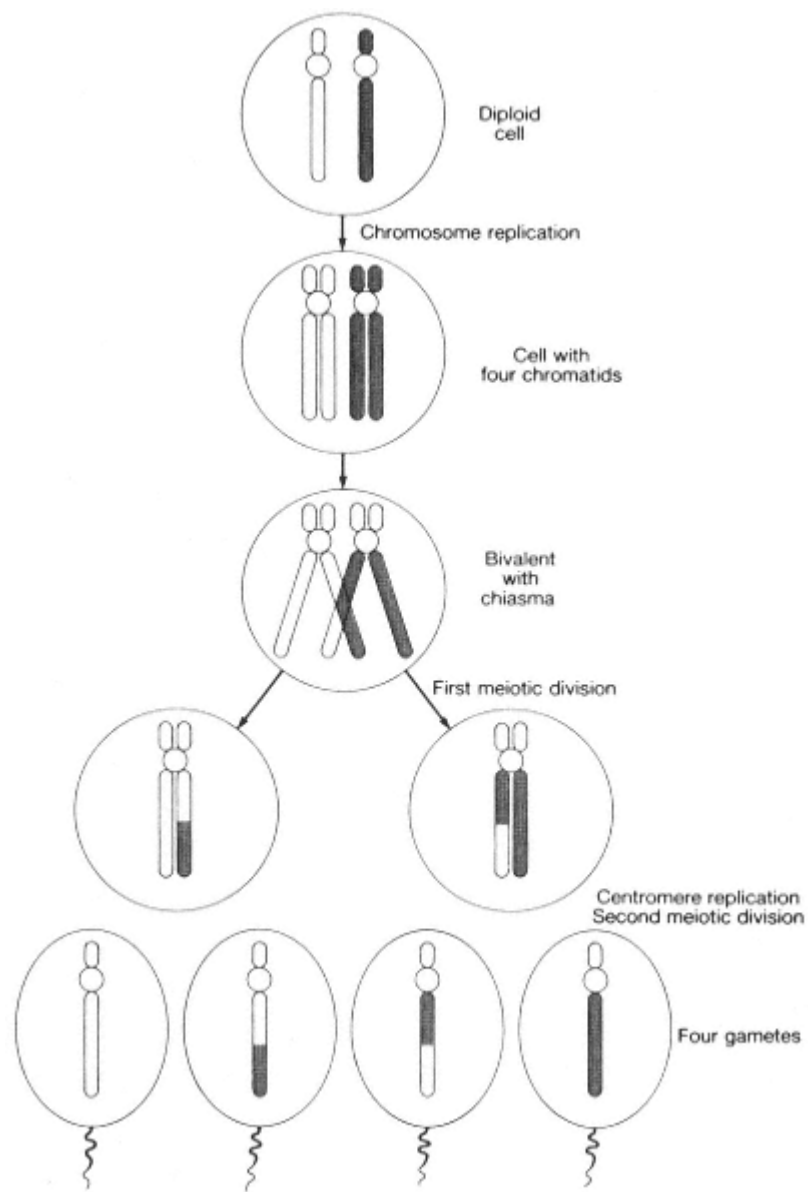


Figure 12.2
The essentials of meiosis.

only marsupial mammal so far investigated. No species is known in which crossing over is suppressed in both sexes.

A haploid-diploid alternation requires a special division in which the chromosome number is halved, but it is not clear why meiosis involves an initial doubling of the chromosome number, followed by two divisions. It is therefore intriguing

that a number of protists (that is, single-celled eukaryotes) have a 'meiosis' which appears to consist of a single cell division, not preceded by chromosome replication. It may be that this represents a primitive state from which typical meiosis has evolved.

Ploidy and Cellular Differentiation

There is great variability in multicellular organisms in the roles played by the haploid and diploid stages of the life history. Figure 12.1 shows diagrammatically the relation between the haploid **gametophyte** (that is, producer of gametes by mitosis) and the diploid **sporophyte**, which produces haploid spores by meiosis. In multicellular green algae (e.g. *Spirogyra*), the typical situation is for the multicellular stage to be haploid, alternating with a single-celled diploid stage. Occasionally, as in *Codium*, this situation is reversed, and the multicellular stage is diploid. On at least three occasions, there has evolved an alternating life history with morphologically identical multicellular haploid and diploid stages: an example is the green seaweed, *Ulva*. A similar alternation of morphologically identical haploid and diploid stages occurs in some brown algae (e.g. *Ectocarpus*), although in the largest species the main vegetative stage is diploid, and the haploid stage is microscopic (e.g. in kelp *Laminaria*) or reduced to a single cell (e.g. in *Fucus*).

In higher plants, there is an association between diploidy and a high degree of cellular differentiation. Thus in the bryophytes (mosses and liverworts) the main vegetative stage is the gametophyte, upon which the sporophyte grows parasitically. In the pteridophytes (ferns), the gametophyte is a flat leaf-like thallus, resembling a liverwort, and the main vascular plant is the sporophyte. Subsequent evolution continued this emphasis on the sporophyte, until in flowering plants the gametophyte was reduced to the pollen tube (male), and the ovule (female). In multicellular animals, the haploid phase has been reduced to the non-dividing gamete (egg or sperm).

These facts suggest that diploidy may be a precondition for the evolution of complex multicellular structures. This is partly confirmed by the situation in the fungi. The thread-like structure of a fungus, the **mycelium**, contains haploid nuclei. Two genetically different mycelia may fuse to form a **heterokaryon**. If so, the two types of nuclei do not fuse to form a diploid nucleus, but both types continue to divide, so that the heterokaryon contains nuclei of two genetically different types. The most complex multicellular structures formed by fungi are the fruiting bodies of basidiomycetes (for example, mushrooms and toadstools). These structures are heterokaryons, and there are special processes that ensure that each cell contains one nucleus of each of the two types (division with so-called clamp connections). Hence the cells of a mushroom, although not containing a diploid nucleus, do contain two sets of genes, one derived from each of the 'parent' mycelia that fused.

Anisogamy.

In many protists and algae, there is no morphological differentiation between male and female gametes. All gametes are small and motile. However, in such **isogamous** ('equal gametes') organisms, the gametes are usually of two types, + and -, and will fuse only with gametes of the opposite type. Higher plants and animals are **anisogamous**: that is, there is differentiation between a small motile gamete, the sperm, and a large non-motile gamete, the egg. In flowering plants, there is no motile sperm, and the 'gamete' is reduced to one nucleus of the gametophyte, or pollen tube.

Intermediates between isogamy and anisogamy exist in some groups of green algae. For example, the Volvocales form spherical multicellular colonies. The smaller species are isogamous, forming motile sperm. There are species of intermediate size, in which some colonies produce small motile gametes, and others larger motile gametes. Finally, in the largest species, some colonies produce sperm, and others large non-motile gametes, or eggs.

Hermaphroditism

In both plants and animals, eggs and sperm may be produced by different individuals, or by the same individual. The terminology is somewhat confusing: it is described in Box 12.1.

Asexual Reproduction

Many plants and animals can reproduce without syngamy. It is useful to distinguish three processes:

1. **Parthenogenesis**: development of a new individual from an unfertilized egg.
2. **Adventitious embryony**: development of a new individual from a single

Box 12.1—

The Terminology of Hermaphroditism

In animals, an individual that reproduces both as a male and a female is a **hermaphrodite**. Hermaphroditism can be simultaneous or sequential: in the latter case, the individual may reproduce first as a male and then as a female, or vice versa. A species that has separate male and female individuals is said to be **gonochoristic**.

In plants, the word hermaphrodite is confined to cases in which male and female functions are present in a single flower. Species in which a single individual has separate male and female flowers are **monoecious** ('one house'); note that to a zoologist a monoecious species would be hermaphrodite. Species with separate male and female individuals (holly, apples, hops) are **dioecious** ('two houses'). In this book, I use the word dioecious for both plants and animals with separate sexes.

somatic cell. Barring mutation, the genotype of the offspring is identical to that of the parent: in this respect it is identical to one form of parthenogenesis (apomixis; see below).

3. **Vegetative reproduction:** development of a new individual from a group of somatic cells, for example plant reproduction by bulbs, corms, rhizomes, or tillers. Again, barring mutation, the offspring are genetically identical to the parent: however, when we allow for mutation, the process differs in its genetic consequences, because a new individual may be a genetic mosaic.

Parthenogenesis is the process that has received most attention from evolutionary biologists. It can take several forms:

1. **Apomixis:** meiosis is suppressed, and the offspring are genetically identical to the parent.

2. **Automixis:** meiosis is retained, and diploidy is restored by fusion of two of the haploid products of meiosis, or of two genetically identical nuclei produced by the mitotic division of the haploid egg nucleus (Fig. 12.3). Typically, the offspring are homozygous at some or all of the loci at which the parent was heterozygous.

3. **Endomitosis:** meiosis is preceded by a round of chromosome replication, giving rise to a tetraploid cell (Fig. 12.4). Pairing then occurs between **sister chromosomes** (that is, chromosomes that have just arisen by the replication of the same parent chromosome), and meiosis then proceeds normally. As in apomixis, the result is to produce offspring genetically identical to the parent. This is the form of parthenogenesis in some lizards of the genera *Cnemidophorus* and *Lacerta*.

This brief list does not exhaust the mechanisms of parthenogenesis. However, the critical distinction is between those processes (apomixis, endomitosis) that produce offspring identical to the parent, and those (most forms of automixis) that produce offspring more homozygous than their parents. It is not hard to see why parthenogenesis of the latter kind is rather rare in nature: it is harder to see why the former kind does not triumph.

Why not be a Parthenogen?

The Cost of Sex

Consider an organism such as a herring, with equal numbers of males and females, and no parental care. In females, a gene *A* suppresses meiosis, and causes the production of diploid eggs that develop without fertilization into females genetically identical to the parent. Figure 12.5 shows that, when rare, such a gene would double in frequency in each generation. This result has been expressed by saying that there is a 'twofold cost of sex', arising from the needless production of males. It is clearer, however, to take a 'gene's eye view': a gene *A* that suppresses meiosis

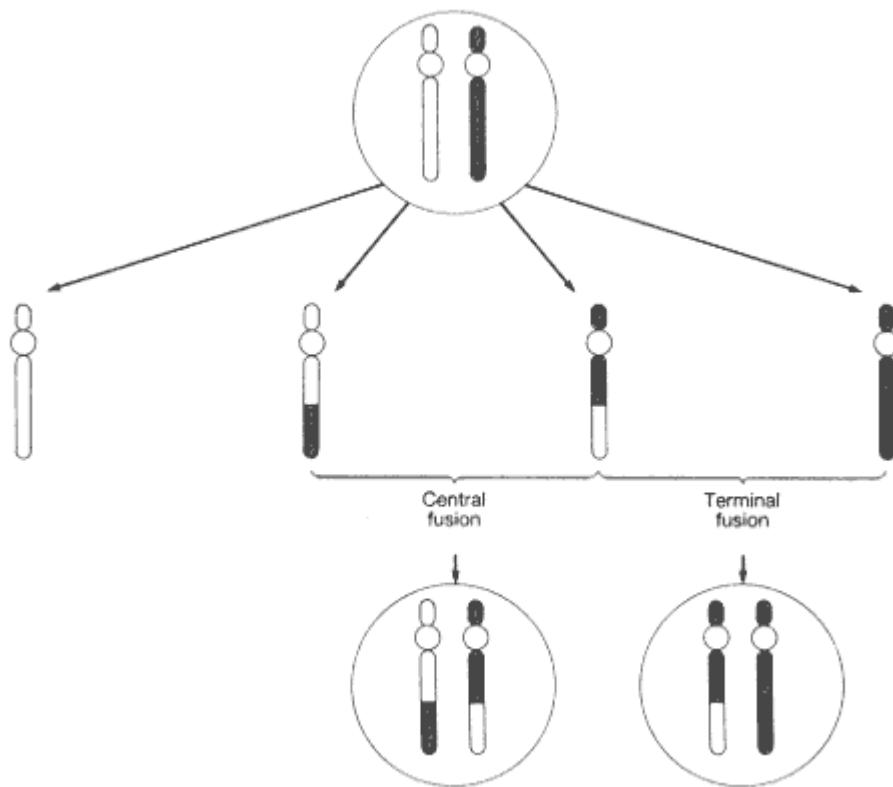


Figure 12.3

Automictic parthenogenesis. If, as in the diagram, diploidy is restored by fusion of two of the four products of meiosis, offspring may be homozygous at loci for which the mother was heterozygous, but, with central fusion, heterozygosity is always maintained for genes between the centromere and the first chiasma. In some automictic parthenogens (for example, *Drosophila mangabieri*), heterozygosity is maintained by combining central fusion with the suppression of crossing over. If diploidy is restored later, by the fusion of two nuclei derived by mitosis of a single product of meiosis, the offspring are homozygous at all loci.

is certain to be transmitted to all the eggs produced by a female, whereas a gene that permits meiosis is transmitted to only half.

The twofold cost of meiosis does not arise in isogamous organisms, so we do not have to allow for it when thinking about the origin of eukaryotic sex. The argument is complicated if there is paternal care, because the number of offspring a female can raise is greater if she has a mate: of course, she could pair with a male but produce eggs parthenogenetically. Figure 12.6 shows the situation in a self-sterile hermaphrodite: there is again a short-term advantage to parthenogenesis.

It is unlikely that a single mutation could be responsible for effective parthenogenesis. The argument in this section shows, however, that a parthenogenetic genotype, perhaps involving many genes, would increase in frequency unless there are counteracting selective forces. The problem is to identify those forces.

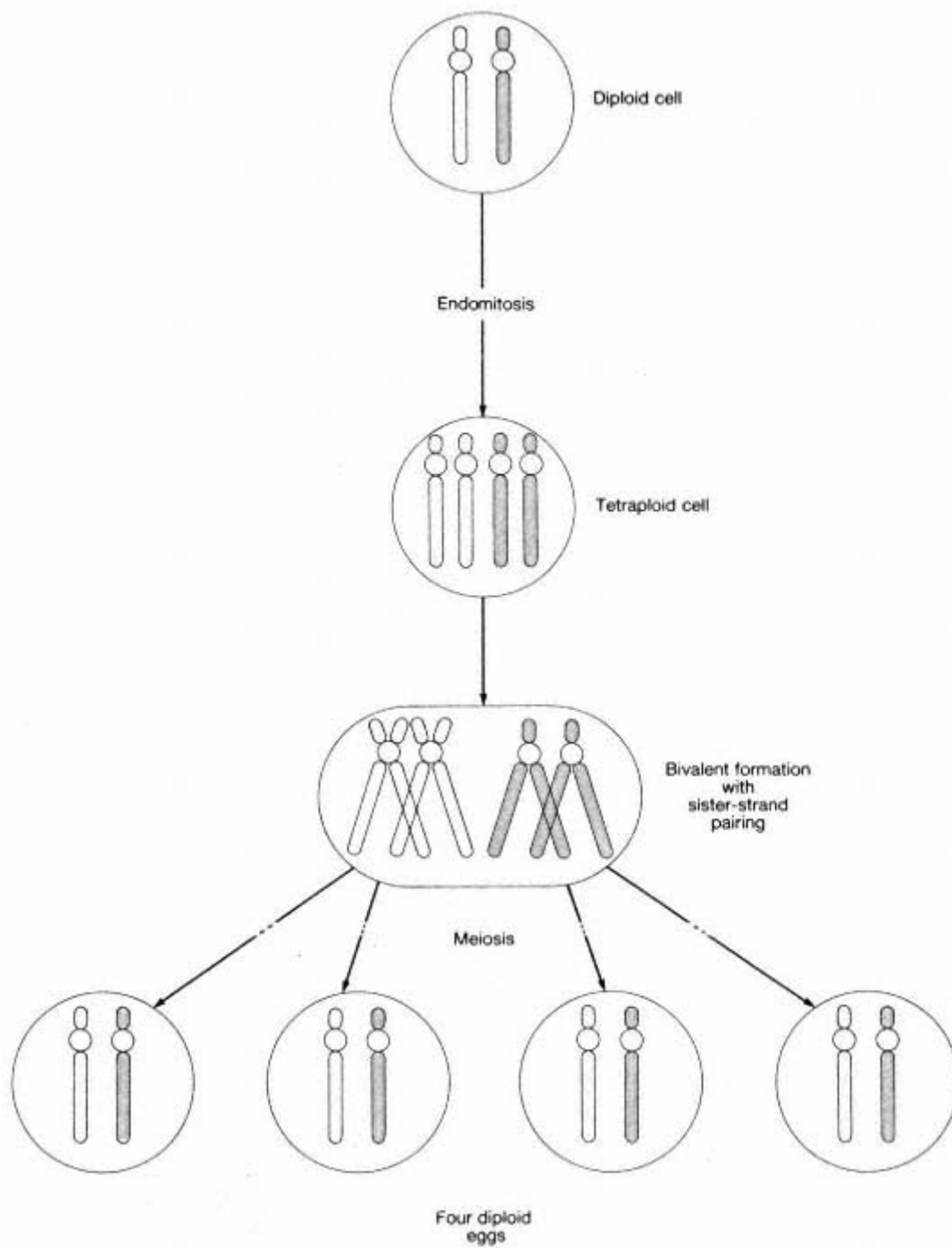


Figure 12.4
Parthenogenesis by endomitosis, followed by sister-strand pairing and meiosis.

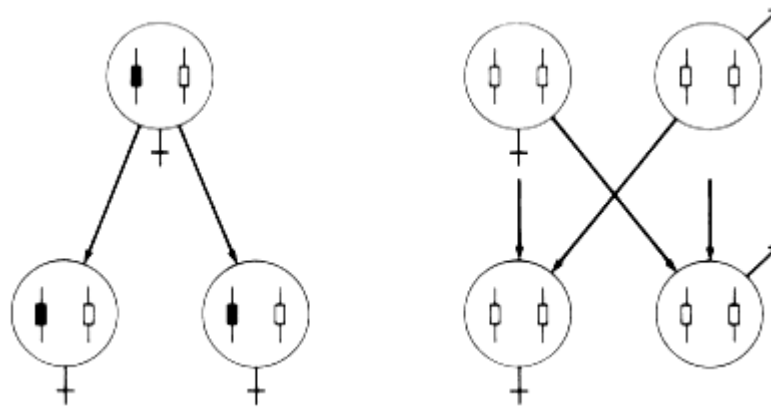


Figure 12.5

The cost of meiosis. ■ represents a gene that suppresses meiosis, causing a female to produce diploid offspring identical to herself. □ represents the sexual allele of ■. If females carrying ■ and □ are equally fecund, allele ■ doubles in frequency in a generation when rare.

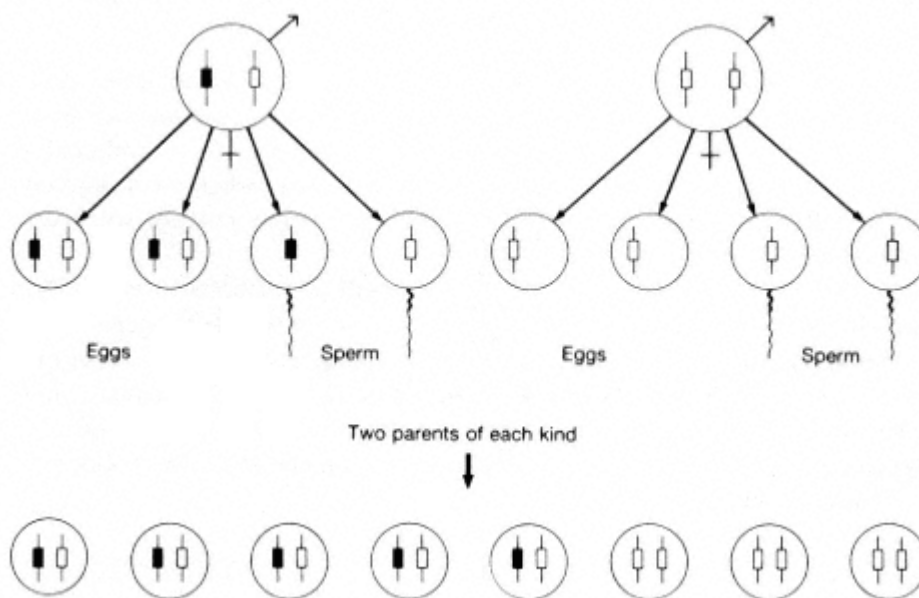


Figure 12.6

Parthenogenesis in a self-sterile hermaphrodite. ■ represents a gene that suppresses meiosis in the eggs, but not in sperm (or pollen). If in one generation half the individuals carried allele ■, in the next generation 5/8 of the individuals would do so.

The Distribution of Parthenogenesis

Parthenogenesis is widely distributed among both plants and animals, although there are some taxa (for example, gymnosperms and mammals) in which it has not been reliably reported. However, most parthenogens have close relatives that are sexual. It is unusual to find a family or higher taxon consisting wholly of partheno-

gens. The bdelloid rotifers, an order in which males have never been reported, constitute an exception. Some large taxa (for example, aphids, and cladocerans) are cyclically parthenogenetic: a number of generations of apomictic parthenogenesis alternate with a sexual generation: it is interesting that, in both the taxa mentioned, there are many varieties that have wholly abandoned sex.

This spotty taxonomic distribution suggests that parthenogenetic varieties may be successful in the short term, but that (with a few exceptions, such as the bdelloids) they have been unable to survive for a long time, or to undergo adaptive radiation. There are two possible explanations for this failure:

1. Parthenogenetic varieties evolve more slowly than sexual populations;
2. Parthenogenetic varieties accumulate harmful mutations.

These possibilities will be discussed in turn.

The Advantages of Sex

Sex Accelerates Evolution

The argument is shown in Fig. 12.7. Suppose that two favourable mutations A and B , occur in different individuals in the same population. Each will increase under selection. In a sexual population AB individuals can then arise by recombination. In an asexual population, an AB individual can arise only when an A mutation occurs in a B individual, or vice versa. Therefore the sexual population will evolve more rapidly.

Note that this argument does not apply to small populations, in which each favourable mutation that occurs will be fixed before the next one happens. Also, it does not apply if fitness interactions are strongly epistatic, so that AB is fitter than ab , but Ab and aB are of low fitness: if this is the case, recombination actually slows down evolution by breaking up favourable gene combinations when they occur.

We can conclude that, although recombination is not necessary for evolution by natural selection, it will often accelerate the process.

Sex and the role of Parasites

The fact that sexual populations can evolve more rapidly can only explain the ubiquity of sex if populations are frequently exposed to directional selection. This has led to the idea that parasites may play a crucial role in the maintenance of sex. An arms race between hosts and their parasites will prevent either population from achieving an optimal genotype. The effect is likely to be particularly strong for the host species, because parasites often have a short generation time, and so are likely to evolve rapidly.

There is empirical support for this idea. Lively *et al.* (1990) compared the intensity of parasite infection in a sexual fish, *Poeciliopsis monacha*, with that in

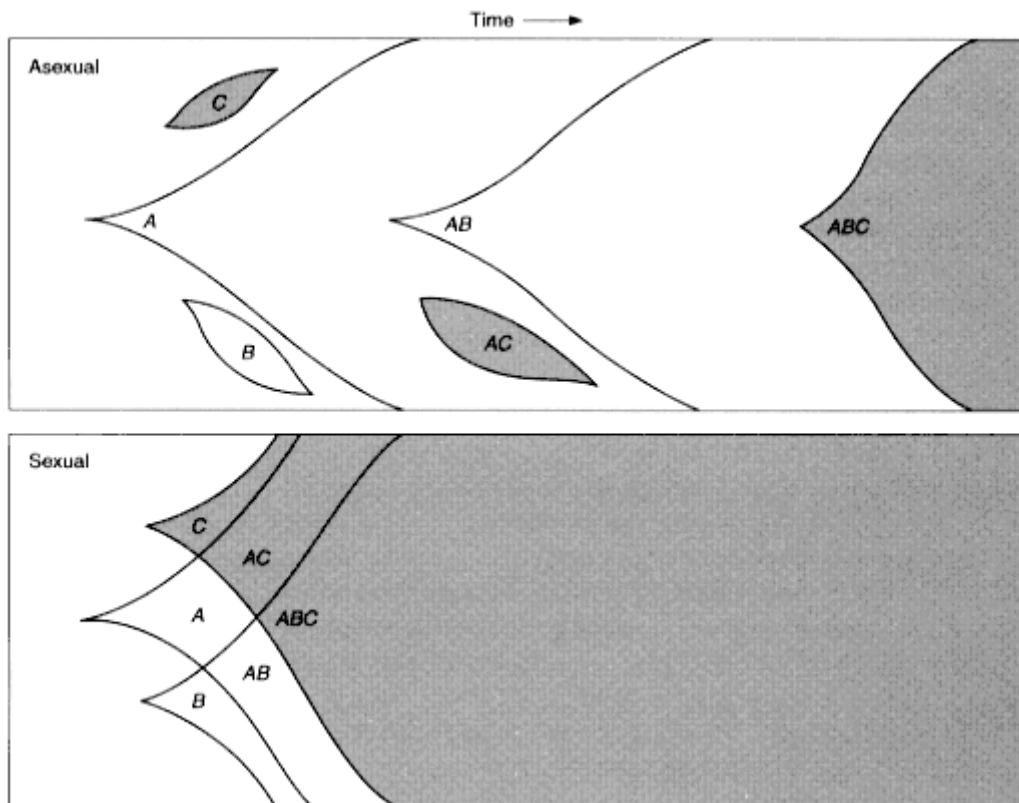


Figure 12.7
Evolution in sexual and asexual populations (after Muller 1932).

two clones of triploid asexual relatives, originating by hybridization between *P. monacha* and the related *P. lucida*. These fishes are infected by trematode larvae, which burrow through the body wall and form externally visible cysts, which can be counted, so that the build-up of infection can easily be measured. The first pool to be studied contained sexual *P. monacha* and members of one asexual clone, MML/I, which were found to be more heavily infected than the sexuals. A second pool was occupied by sexual fish, and by two distinct clones, MML/I and MML/II. The latter clone was more common, and more heavily infected than the sexuals. There was no difference in level of infection between the sexuals and the rarer clone, MML/I, showing that the heavier infection of MML/I in the first pool was due to its high relative abundance there, and not to any intrinsic susceptibility. Finally, in a third pool there had been a local population extinction, followed by reinvasion by a few sexual fish, and also by the clone MML/I. In this pool, the sexual fish were more heavily infected, perhaps because of the deleterious effects of inbreeding. However, after additional sexual fish had been experimentally intro-

duced into the pool, this effect was reversed, and the clonal fish were again more heavily infected.

These results show that (except when the sexual fish were highly inbred) the sexual population was more resistant to parasite infection than the clones. The probable explanation is that the sexual population was genetically more variable. If parasites evolve so as to attack the more common types in the host population, there will be an advantage to rare host genotypes, as suggested by Haldane (1949). This effect may not only favour rare genotypes at the expense of common ones within a sexual population, thus maintaining genetic polymorphism, but, as a number of authors have pointed out, may favour sexual as against asexual populations.

The Accumulation of Deleterious Mutations:

1. 'Muller's Ratchet'.

H.J. Muller pointed out that, in a population without recombination, there is a tendency for slightly deleterious mutations to accumulate. The argument is shown in Fig. 12.8. We classify the population into classes containing 0, 1, 2, . . . k deleterious mutations. Let the number of individuals in the optimal class, with no deleterious mutations, be n_0 . If n_0 is small, then there is in each generation a chance that, despite their high fitness, all n_0 individuals will die without leaving offspring.

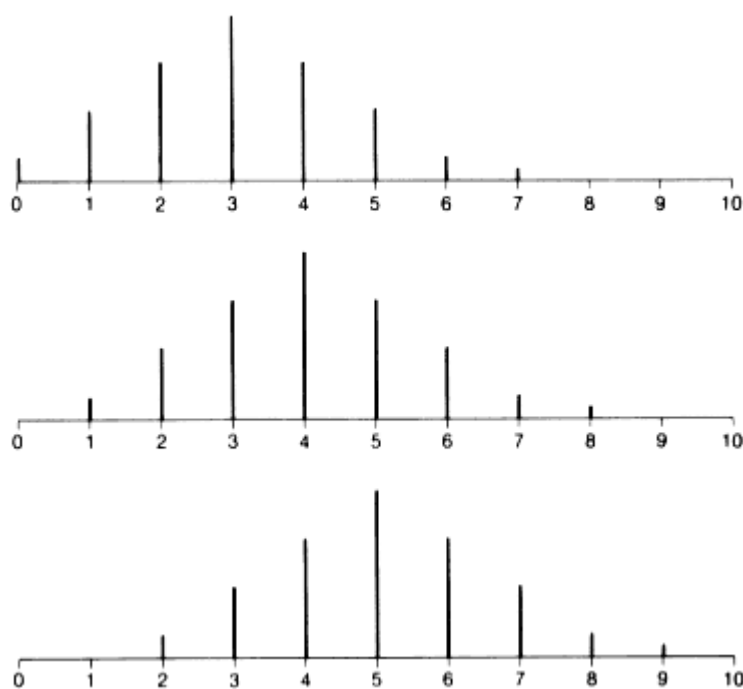


Figure 12.8

Muller's ratchet. The top diagram shows the initial distribution in an asexual population of individuals with 0, 1, 2, 3, . . . slightly deleterious mutations.

The lower two diagrams show the same population after the ratchet has clicked round one, and two, notches.

If so, the optimal class is lost, and can only be reconstituted by back mutation: in a sexual population it can be reconstituted by recombination. In Muller's phrase, the ratchet has clicked round one notch. The new optimal class will carry one deleterious mutation. In time, that class too will be lost, as the ratchet clicks again.

The process is relevant only if n_0 is small. It can be shown (Haigh 1978) that, if fitnesses combine multiplicatively, the equilibrium value of n_0 is

$$n_0 = Ne^{-U/s}, \quad (12.1)$$

where N is the population size, U the expected number of deleterious mutations per genome per generation, and s the selective disadvantage per mutation.

If s is small, then n_0 may be small even if N is large. For example, if $N = 100\,000$, $U = 0.1$, and $s = 0.01$, then $n_0 \cong 4.5$: deleterious mutations would accumulate rapidly. A sexual population with this mutation rate would be in no difficulty: the mean fitness $\bar{W}$, relative to a fitness of 1 for the optimal class, is $\bar{W} = e^{-0.1} = 0.905$.

Muller's argument applies not only to asexual organisms, but to any region of DNA that never recombines: for example, the Y chromosome. It also applies to populations with 100 per cent self-fertilization.

It is hard to decide how important this process has been. One difficulty is that it assumes that many mutations are unconditionally deleterious, and can be corrected only by back mutation. If one supposes that most mutations alter fitness through their effect on quantitative traits that are under normalizing selection, then each mutation can be counterbalanced by mutations in the reverse direction at many other loci. Unfortunately, we have at present no information concerning the accumulation of deleterious mutations in parthenogens.

The Accumulation of Deleterious Mutations:

2. Synergistic Selection

Muller's ratchet operates only in a finite population (although the population need not be small). What if the population is so large that the ratchet does not operate? Will recombination still reduce mutational load? It all depends on how deleterious mutations affect fitness (Kondrashov 1982). Suppose that a single mutation reduces fitness by a factor $1-s$. If n mutations, at separate loci, reduce fitness by a factor $(1-s)^n$, we say that the effects of deleterious mutations are multiplicative. In this case, an asexual and a sexual population would be affected identically, essentially because an infinite asexual population would be in linkage equilibrium.

Suppose, however, that deleterious mutations act **synergistically**; that is, each additional mutation has a greater effect on fitness. Then recombination can reduce the mutational load. The most extreme form of synergistic selection is truncation selection; fitness is unaffected by k mutations, but $k+1$ (or more) mutations are lethal. As shown in Box 12.2, if selection acts in this way, then, for the same rate of deleterious mutation, a sexual population suffers a much lower mutational

**Box 12.2—
Synergistic Selection and Recombination**

Figure 12.9 shows populations with (left) and without (right) recombination. Considering first the population with recombination, the top histogram shows the distribution of individuals with different numbers of mutations, assuming a

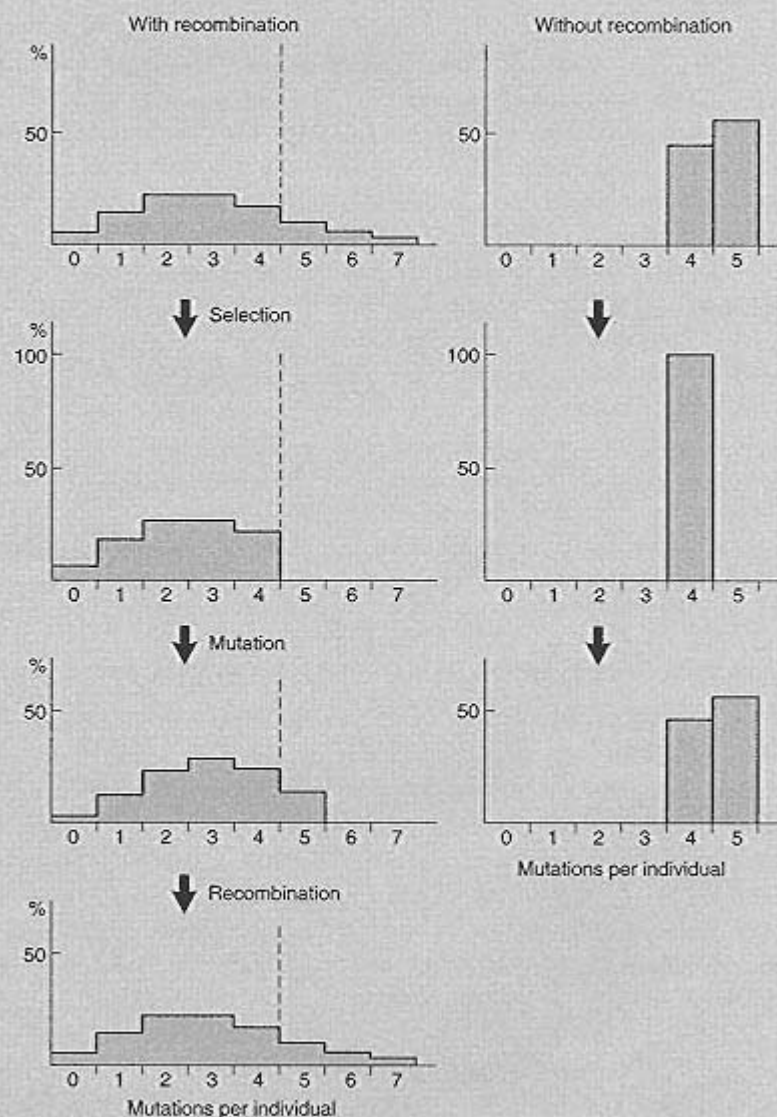


Figure 12.9 Number of mutations per individual in populations with (left) and without (right) recombination.

mean of three mutations per individual, randomly distributed. Truncation selection (the most extreme form of synergistic selection) removes all individuals with five or more mutations, but does not discriminate between individuals with fewer than five mutations. The result is a population with a mean of 2.45 mutations per individual. For each individual, there is then a probability of 0.55 of receiving one additional mutation, generating the population in the third row, again with a mean of three mutations per individual. Finally, recombination redistributes the mutations randomly, without altering the mean, thus restoring the original population distribution.

The population without recombination has the same assumptions about selection (all individuals with five or more mutations die) and mutation (each individual has a probability of 0.55 of receiving an additional mutation).

The crucial point is that 55 per cent of the asexual population dies selectively each generation, but only 15 per cent of the sexual one, even though the mutation rates are identical. The reason is that recombination increases the variance of the number of mutations per individual, and so makes selection more effective in removing them.

load. The reason is that recombination increases the variance in the number of deleterious mutations per individual, and so makes selection more effective.

Interactions between Beneficial and Deleterious Mutations

We have seen that a sexual population is superior to an asexual one both in accumulating beneficial mutations and in eliminating deleterious ones. Peck (1994) pointed out that the advantage of a sexual population is still greater if both types of mutation occur simultaneously. The reason is as follows. In an asexual population, only a small proportion of individuals whose carrying the fewest deleterious mutations have a chance of leaving descendants in the distant future. It follows that only those beneficial mutations that happen to occur in this small proportion of fit individuals have an appreciable chance of establishment. No such restriction applies to a sexual population, which will therefore accumulate favourable mutations more rapidly.

Long-term and Short-term Explanations

A simple explanation of the taxonomic distribution of parthenogens is as follows. Successful parthenogenetic varieties arise rather rarely, because a number of genetic changes are needed (in particular, to suppress meiosis and to initiate development without the stimulus of fertilization). Most such varieties are apomictic (or undergo some process genetically equivalent to apomixis), because automixis produces individuals homozygous for deleterious recessives. Once such a variety

arises, it is likely to be successful in the short run, because it gains the twofold advantage discussed above, and perhaps because apomixis is a means of perpetuating a successful heterozygous genotype. However, in the long run such varieties go extinct, because they cannot evolve so readily to meet changed circumstances, and perhaps because they accumulate slightly deleterious mutations.

This is a 'group selection' explanation: sex is individually disadvantageous, but confers advantages on the population. There are two reasons why such an explanation may be valid in this case:

1. There is reproductive isolation between parthenogenetic varieties and their sexual ancestors.
2. Parthenogenetic varieties arise rarely; therefore the relatively weak force of group selection is sufficient to prevent them from replacing their sexual relatives.

The present distribution of parthenogens makes sense on this hypothesis. For example, there are many asexual 'species' of the lizard *Cnemidophorus* (most of which probably arose by hybridization between sexual species), but they have not wholly replaced the sexual forms. This suggests that there must be some short-term advantage to sex that counterbalances the twofold cost of producing males. The likely explanation is that sex generates a greater number of different genotypes, adapted to a wider range of ecological conditions.

The idea that one or a few clones cannot replace a sexual species because the latter is more widely adapted ecologically is supported by data on the fish, *Poeciliopsis monacha*. Sexual individuals coexist in many Mexican streams with parthenogenetic strains that arise by hybridization with a related species, *P. lucida*. These strains belong to a number of clones (almost certainly representing different origins by hybridization) that differ ecologically and physiologically. Vrijenhoek (1984) found that the proportion of fish in a stream that are sexual decreases as the number of clones increases (Table 12.1). His explanation is that, as the genetic

Table 12.1

The frozen niche hypothesis. The percentage of sexual females at a locality, according to whether one, or more than one, sexual clone is present in the river system (Vrijenhoek 1984).

	Number of river systems	Number of localities	Percentage asexual (mean $\pm$ standard deviation)
Monoclonal rivers	4	33	7.2 .3
Multiclonal rivers	4	15	66.3 1.6

variability of the asexual population increases, the number of habitats within a stream to which sexual genotypes are better adapted decreases. In this particular case, the parthenogens cannot wholly replace the sexuals, because their eggs do not develop unless stimulated by sperm from a sexual male (so-called **pseudogamy**, because the sperm contributes no genes to the new individual). However, the data do show that the ability of parthenogens to replace their sexual relatives will depend on the number of clones present. This number may be small. For example, the asexual 'species' *Cnemidophorus uniparens* has been shown by scale-grafting to consist of a single clone. It is perhaps not surprising that it has not wholly replaced its two parental species

Summary

It is not difficult to think of reasons why sexual populations should have a long-term advantage over asexual ones. They can evolve more rapidly to meet changing circumstances, and they will suffer less from the accumulation of deleterious mutations, either stochastically (Muller's ratchet) or deterministically (if fitness effects are synergistic). This is a group selection explanation, but it is plausible, because there is reproductive isolation between the competing types, and because the origin of new parthenogenetic clones may be a rare event. The taxonomic distribution of parthenogens fits rather well with this hypothesis.

However, there are difficulties with this explanation. In particular, why are facultative parthenogens rare? That is, females that produce some of their offspring sexually and some by parthenogenesis are rare? A gene causing a female to produce some parthenogenetic offspring would have a short-term advantage, similar to the advantage of a gene causing obligate parthenogenesis, yet the population would retain at least some of the advantages of sex. It may be that such a situation would be short lived: once a mechanism for apomictic parthenogenesis had arisen, there would be strong individual selection converting facultative into obligate parthenogenesis. But there is a need for further research, both empirical and theoretical.

Much of the difficulty in understanding the evolution and maintenance of sex arises because it is rare for both sex and parthenogenesis to exist in the same population (although it does occur, particularly in cyclical parthenogens), and we are therefore uncertain whether to seek individual or group advantages. I now turn to a topic where the evolution of recombination rate in sexual populations is which this difficulty does not arise, because there is within-population variation for the trait.

The Evolution of Recombination

Imagine a diploid sexual organism in which the haploid chromosome number is one, and in which no chiasmata are formed in meiosis. There would be no re-

combination: all genes inherited from one parent would be transmitted together. A population of such organisms would not gain the benefits of rapid evolution discussed above, and would accumulate deleterious mutations by Muller's ratchet just as an asexual population. Recombination depends on the independent assortment of genes on different chromosomes, and, for genes on the same chromosome, on the formation of chiasmata.

Changes in chromosome number do occur, as described in Chapter 11, but it is not clear that the changes occur in response to selection for recombination: they may be non-adaptive consequences of the presence of repeated DNA sequences, as discussed on p. 217. The situation is different for changes in the number of chiasmata. The crucial point is that genetic variability has been found within populations for the frequency of recombination between genes on the same chromosome whenever it has been looked for (for example, in *Drosophila*, *Schistocerca*, *Tribolium*, and *Neurospora*). A trait for which there is genetic variability within populations will respond to selection. There are reasons, both theoretical and experimental, to think that there is selection tending to reduce recombination. Theoretically, if there are epistatic fitness interactions, the gene combinations present in adult organisms, which have survived selection, will be superior to those generated by recombination. There is empirical evidence for this prediction (Charlesworth and Charlesworth 1975). If, as therefore seems to be the case, there is sometimes selection tending to reduce recombination, there must be counterbalancing selection tending to increase it. Otherwise we would find localized chiasmata, or achiasmate meiosis, to be the rule rather than the exception. What is the nature of selection favouring increased recombination?

Suppose that there are two alleles at a locus R , favouring a high rate of recombination, and r , favouring a low rate. Why should selection sometimes favour allele R ? There are two possible opinions. The first is that R alleles are selected because they repair damaged DNA, or play some other essential role in meiosis. For example, in resetting the developmental programme. A number of such ideas are discussed in Michod and Levin (1988). None seem to me convincing, essentially because they cannot explain the occurrence of syngamy and meiosis, but only of diploidy. The second opinion is that R alleles are selected because they give rise to new combinations of genes in future generations. If so, selection is indirect. The alleles have no effect on the fitness of individuals in which they find themselves. Instead R alleles, by causing recombination, tend to occur in association with genes at other loci that are of high fitness. This is a hitch-hiking explanation: R alleles are given a lift in frequency because selection is favouring alleles at other loci with which they are associated. I think that the existence of crossing over in eukaryotes requires a hitch-hiking explanation.

Consider first the explanation shown in Fig. 12.10. It suggests that R alleles spread because they produce one good chromosome out of two bad ones. It can be called the 'engine and gearbox' model, because one good motorcar can be made

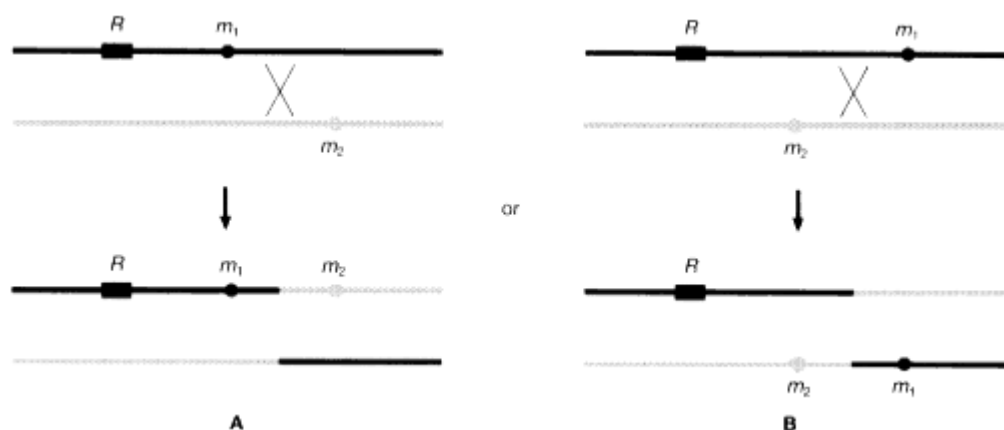


Figure 12.10

The 'engine and gearbox' model of recombination. A gene R mediates recombination between two lethal mutations, m_1 and m_2 . In case **A**, gene R is no worse off, because it was already linked to a lethal; in case **B**, gene R is better off. Selection is supposed to act on the haploid phase.

from two corks by taking the engine from one and the gearbox from the other. In this simple form, I think the argument is fallacious. The figure suggests that the R allele always starts out linked to a deleterious gene. But, as shown in Fig. 12.11, if genes are in linkage equilibrium, an R allele is just as likely to start out linked to no deleterious genes and to finish up linked to such a gene, as it is to start out (as in Fig. 12.10) linked to a deleterious gene and to finish up linked to no such gene. In short, it is as likely to lose as to gain by causing recombination.

The moral is that selection will alter the frequency of R alleles if, and only if, there is linkage disequilibrium. In seeking a hitch-hiking explanation, therefore, the first question is: what is the cause of the linkage disequilibrium? There are two possible answers:

1. chance: that is, the effects of finite population size;
2. selection: that is, epistatic fitness effects.

Consider first the effects of chance. In fact, we have already discussed two examples. The argument illustrated in Fig. 12.7 does assume a finite population. Thus, let the initial frequencies of the favourable alleles A and B be P_A and P_B , respectively. If the population were infinite, and fitnesses multiplicative, the initial frequency of the optimal genotype AB , would be $P_A P_B$. The frequency of this genotype would increase at a rate unaffected by recombination. But suppose the population were finite. The expected frequency of AB would be $P_A P_B$, as before, but, if P_A and P_B are small, the actual number of AB individuals would probably be zero. Therefore evolution in the absence of recombination would have to wait for a B mutation in an A individual, or vice versa. Similarly, the Muller's ratchet mechanism also depends on finite population size.

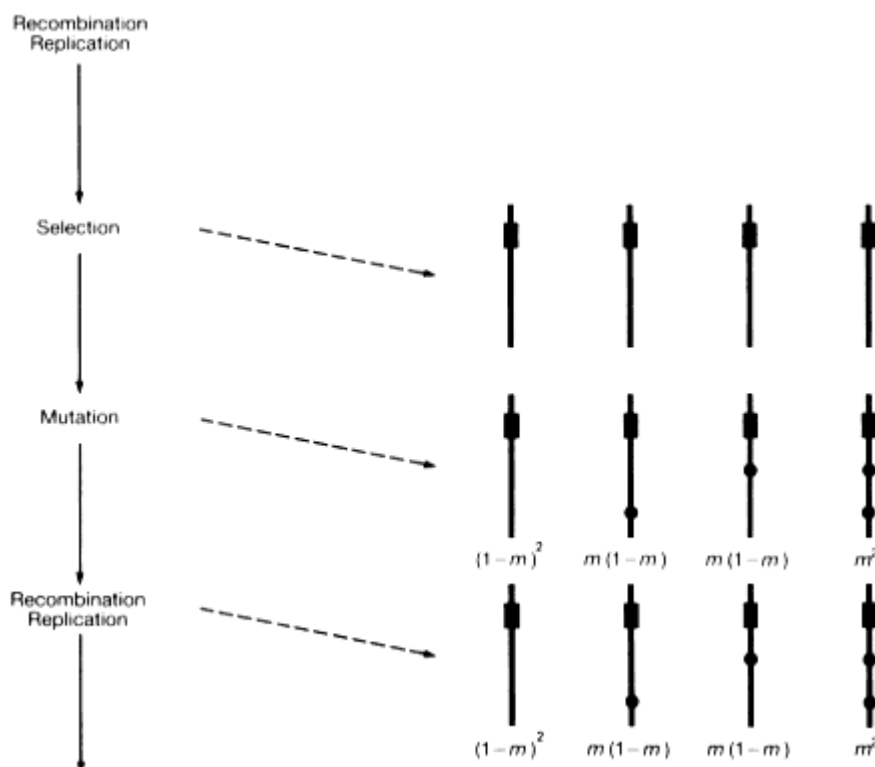


Figure 12.11

Recombination and linkage equilibrium. ■ represents a gene R mediating recombination. ● represents lethal mutations. After selection, there are no lethals. Mutation then produces lethals in linkage equilibrium. Therefore recombination does not alter the probability that R will be linked to a lethal, because all it does is to bring the frequencies closer to linkage equilibrium.

Thus one possible answer to our question is that linkage disequilibrium arises because of chance events in finite populations, and that R alleles are favoured in consequence. One possible objection to this argument is that the advantages of high evolution rate, or of avoiding the accumulation of deleterious mutations, were couched in terms of group advantage, and not of selection favouring one allele rather than another within a population. This objection need not be fatal. Thus, suppose that the high recombination allele, R , is recessive, and that no recombination occurs in Rr or rr genotypes. There would then be effective genetic isolation between chromosomes carrying the R and r alleles, and the 'group' advantage of that part of the population carrying R would result in an increase in frequency of R relative to r . However, we cannot get over the difficulty so easily if some recombination occurs in Rr genotypes.

I turn now to the possibility that the source of the relevant linkage disequilibrium is selection on genes with epistatic effects on fitness. We have seen

already (p. 89) that normalizing selection generates linkage disequilibrium between polygenes. However, normalizing selection would also favour alleles that suppress recombination. In contrast, if a population is under directional selection for a polygenic trait, then genes for high recombination will increase in frequency, provided that they are linked to the genes determining the selected trait. In brief, the mechanism is as follows:

1. Directional selection generates negative (that is, $+ - + -$) linkage disequilibrium, which is less rec^+ chromosomes (that is, chromosomes carrying the high recombination allele CH) than in rec^- chromosomes. This is not intuitively obvious, because $++++$ chromosomes are the most favoured, so one might expect positive linkage disequilibrium: the reason for negative disequilibrium is given in Box 12.3.

2. In consequence, the rec^+ chromosomes contribute more to the genetic variance of the selected trait. Therefore they respond more to selection: that is, they accumulate selected alleles more rapidly.

Box 12.3—

Directional Selection Generates Linkage Disequilibrium

For simplicity, consider a haploid with two alleles at each of two loci, with additive effects on the selected phenotype, so that $ab = 0$; $aB, Ab = 1$; $AB = 2$. Let the initial gene frequencies be 0.5, and the linkage disequilibrium $D = 0$. Suppose that fitness increases linearly with phenotype, so that $ab = 1 - s$; $aB, Ab = 1$; $AB = 1 + s$. Table 12.2 shows that, after one generation of selection, $D = -s^2/16$, which is negative.

If fitness increases multiplicatively with phenotype $ab = 1$; $aB, Ab = 1 + s$; $AB = (1 + s)^2$ it is easy to see that D would remain at zero. Although this has been shown only for a numerical case, it is in general true that if fitnesses combine multiplicatively, then, if D is initially zero, it will remain at zero. But if, as will usually be the case, the increase in fitness is less than multiplicative, D will become negative.

Table 12.2

Directional selection generates linkage disequilibrium

Genotype	ab	aB	Ab	AB
Frequency	0.25	0.25	0.25	0.25
Fitness	$1 - s$	1	1	$1 + s$
Frequency after selection	$0.25 (1 - s)$	0.25	0.25	$0.25 (1 + s)$

Before selection, $D = 0$

After selection, $D = (1-s)(1+s)/16 - 1/16 = -s^2/16$.

3. Therefore, the rec^+ chromosomes increase in frequency: that is, the allele for high recombination CH , increases relative to the allele for low recombination CL .

Figure 12.12 shows a computer simulation illustrating this process.

A final possibility is that, even in a constant environment, deleterious mutations

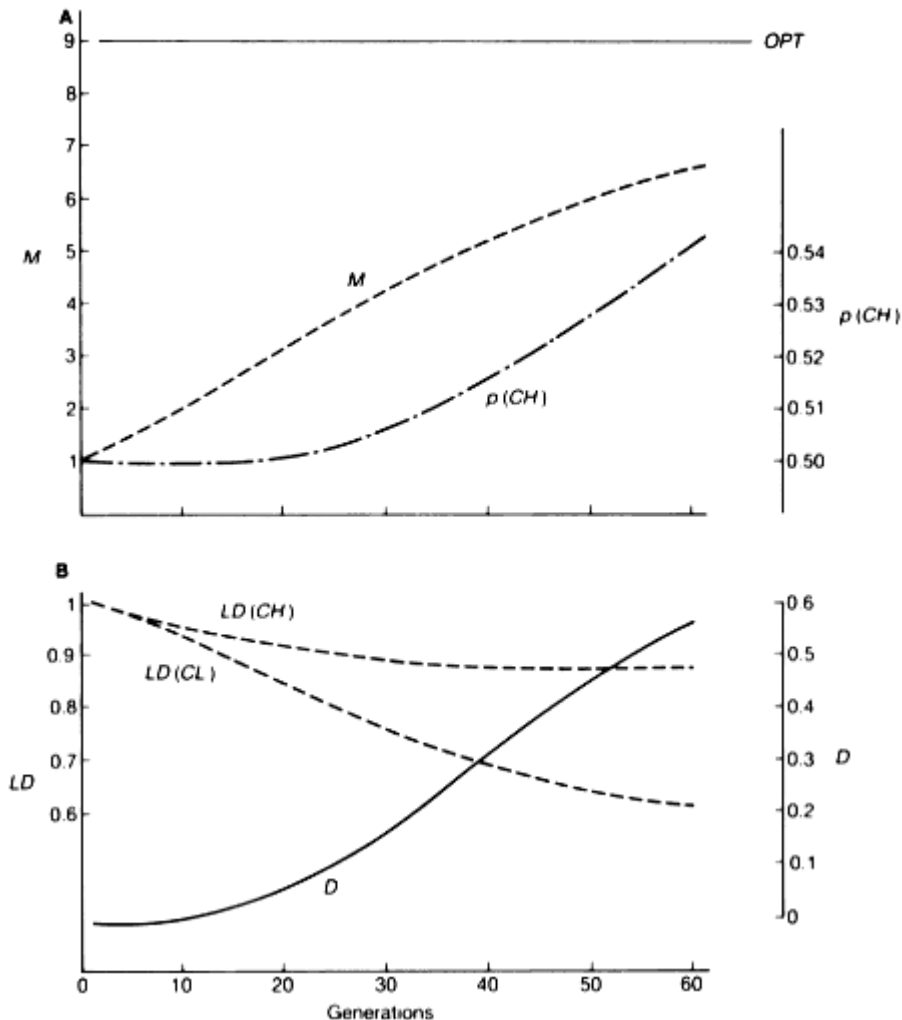


Figure 12.12

Simulation of the effects of directional selection on recombination (Maynard Smith 1988), in a six-locus polygenic model. **A:** OPT , optimal phenotype; M , population mean; $p(CH)$, frequency of allele for high recombination. **B:** D , difference in phenotype between individuals carrying only the high recombination (CH) or the low recombination (CL) chromosomes. $LD(CH)$ and $LD(CL)$ are measures of the linkage disequilibrium in the CH and CL chromosomes. $LD(CH) = V/V_{LE}$, where V is the actual variance of a population carrying CH chromosomes, and V_{LE} is the variance of a population carrying chromosomes with the same allele frequencies, but in linkage equilibrium. Thus $LD(CH) = 1$ if there is linkage equilibrium, and $LD(CH) < 1$ if there is negative (+ - + -) disequilibrium. $LD(CL)$ has a similar meaning for the CL chromosomes.

		Eliminating bad genes	Combining good genes
Origin of linkage disequilibrium	Chance	Muller's ratchet	Fisher-Muller
	Selection	Kondrashov, Feldman <i>et al.</i>	Parasites, shifting optimum

Figure 12.13

A classification of models in which selection favours alleles for high recombination.

will occur. If the effects of these mutations on fitness are multiplicative, then recombination is irrelevant (except in a finite population, because of the ratchet). But if the effects of deleterious mutations are synergistic, so that to have several such mutations has a greater deleterious effect than would be expected from the action of each by itself, then it can be shown (Feldman *et al.* 1980; Kondrashov 1982) that selection again favours alleles for recombination.

The argument of this section is summarized in Fig. 12.13. Models in which selection favours alleles for higher recombination rates can be classified by two criteria:

1. Is the source of the relevant linkage disequilibrium chance in a finite population, or selection on genes with epistatic fitness effects?
2. Is selection favouring the spread of new allele combinations in a changing environment, or eliminating unfavourable alleles in a constant one?

This two-way classification gives four types of model. Examples of each have been discussed, and are summarized in the figure.

The relative importance of these different processes is hard to evaluate. One possible approach is illustrated by the data in Fig. 12.14. The 'excess chiasma number' N , is the average number of chiasmata per meiosis, minus the haploid chromosome number. The logic of this is that one chiasma is needed per bivalent to ensure proper disjunction. It is clear that N increases with age at maturity. This suggests that models involving evolution in a changing environment may be the relevant ones, because selection may be more intense if generation time is long. The suggestion is confirmed by the fact that N is unusually high in domesticated animals, which have been exposed to intense artificial selection. No correlation was found between N and fecundity, although such a correlation would be

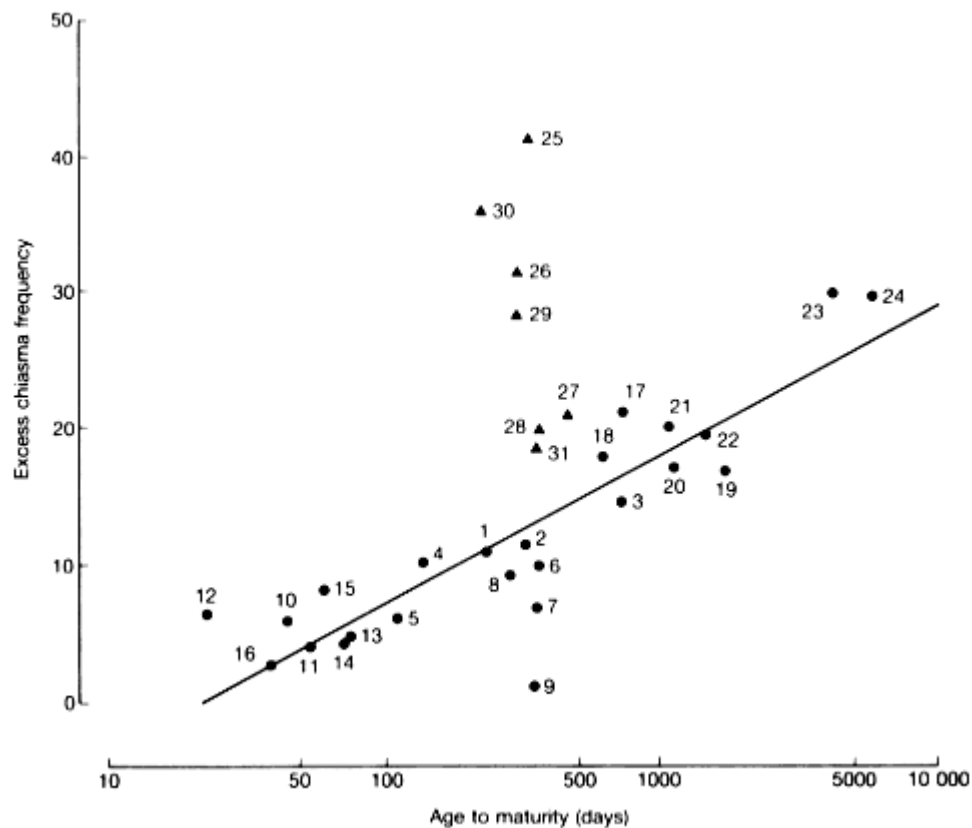


Figure 12.14

Chiasma number in male mammals, plotted against age to maturity. Excess chiasma number is the number of chiasmata, minus the haploid chromosome under. ■, Seven species of domestic mammal (not used in calculating the regression line). (After Burt and Bell 1987.)

expected if sib competition is an important process. No correlation was found between haploid chromosome number and either age at maturity or fecundity. This suggests that haploid chromosome number, unlike N , has not been adjusted to match life history variables.

Further Reading

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Chapter 13—
The Evolution of Genetic Systems:
II. Some Consequences of Sex

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In this chapter, I assume sexual reproduction, and discuss some consequences.

The Sex Ratio.

It is a familiar fact that, in most dioecious species, there are approximately equal numbers of males and females. Why should this be so? One answer would be that, in most species, sex is determined by the segregation of X and Y chromosomes in the meiosis of the heterogametic sex (usually the male), and that meiosis generates a 1:1 ratio. This is true enough as far as it goes. If it were selectively advantageous to do so, however, organisms would surely have evolved some mechanism that produced a sex ratio different from 1:1. We therefore want a selective explanation for this ratio.

For the present, assume that the sex ratio is determined by genes acting in the parent: for example, genes could act in the heterogametic sex by altering the ratio of male- and female-determining gametes produced, or, in the homogametic sex, by altering the success of the two types of gamete in fertilization. However, I assume that a gene in a parent cannot alter its own likelihood of being transmitted to a gamete.

First, consider a verbal explanation of the 1:1 ratio. Suppose that there were more females in the population than males. Then males will have, on average, more children than females. Therefore, a gene acting in a parent would be transmitted to more descendants if it could cause that parent to produce the rarer sex—that is, sons. Similarly, if there were more males in the population than females, a gene causing parents to produce females would spread. Hence the only evolutionarily stable sex ratio would be 1:1, because only then is the reproductive value of a son equal to that of a daughter. This argument, originally due to R.A. Fisher, is made more precise in Box 13.1. In fact, it is shown that it is not the numbers of males and females that are equal at equilibrium, but the parental expenditure on males and females: if the costs of a son and a daughter are equal, equal expenditure implies a 1:1 ratio.

Box 13.1—
The Evolution of Stable Sex Ratios

Suppose that a pair can produce m sons and f daughters, where

$$m + kf = C. \quad (13.1)$$

In this equation, C represents the total possible 'expenditure' on offspring, and a daughter costs k times as much as a son.

Consider a random-mating population in which typical pairs produce m^* sons and f^* daughters. A rare dominant gene M causes females to produce m sons and f daughters, and is not expressed in males (an exactly similar conclusion follows for a gene expressed only in males). The frequency of $M/+$ females is P , and of $M/+$ males is p . P and p are small, so we can ignore M/M genotypes, and matings of $M/+ \times M/+$. If we ignore terms in P^2 , Pp , and p^2 , we have the values in Table 13.1.

Table 13.1

Evolution of the sex ratio in diploids

Mating		Frequency	Offspring per pair			
♀	♂		male $M/+$	$+/+$	female $M/+$	$+/+$
$M/+$	$+/+$	$P(1-p) \approx P$	$m/2$	$m/2$	$f/2$	$f/2$
$+/+$	$M/+$	$p(1-P) \approx p$	$m^*/2$	$m^*/2$	$f^*/2$	$f^*/2$
$+/+$	$+/+$	$(1-P)(1-p) \approx 1 - P - p$	—	m^*	—	f^*
Hence, among the offspring:						
$M/+$ males		$Pm/2 + pm^*/2$				
$M/+$ females		$Pf/2 + pf^*/2$				
total males		$m^*(1-P) + mP \approx m^*$				
total females		$f^*(1-P) + fP \approx f^*$				

Hence, if P' and p' are the frequencies of $M/+$ females and males, respectively, in the next generation, we have

$$P' = \frac{1}{2}P \frac{f}{f^*} + \frac{1}{2}p,$$

$$p' = \frac{1}{2}P \frac{m}{m^*} + \frac{1}{2}p, \quad (13.2)$$

Adding these equations gives

$$(P' + p') = \frac{1}{2}P \left(\frac{f}{f^*} + \frac{m}{m^*} \right) + p, \quad (13.3)$$

$$= (P + p) + RP,$$

where

$$R = \frac{1}{2} \left(\frac{f}{f^*} + \frac{m}{m^*} \right) - 1. \quad (13.4)$$

Note that when $f = f^*$ and $m = m^*$, $R = 0$; that is, $(P + p)$ neither increases nor decreases. This is as expected, since gene M has no effect.

From Equation 13.1, $f = (C - m)/k$, and $f^* = (C - m^*)/k$. Substituting in Equation 13.4 and rearranging gives

$$R = \frac{(C - 2m^*)(m - m^*)}{2m^*(C - m^*)}. \quad (13.5)$$

Now if $R > 0$, the mutant M increases in frequency. Hence, if $m^* < C/2$, the mutant M increases in frequency if $m > m^*$, and if $m^* > C/2$, M increases in frequency if $m < m^*$. That is to say, if $m^* < C/2$, mutants increasing m^* will invade, and if $m^* > C/2$, mutants reducing m^* will invade. Hence the evolutionarily stable sex ratio is $m^* = C/2$. This implies that $m^* = kf^*$, and hence that expenditure on sons and daughters is equal. The best way of seeing what this means in practice is to work through the examples at the end of this chapter.

The stable sex ratio for a haplo-diploid species can be found in the same way. Table 13.2 lists the possible matings. Hence, in the next generation,

$$P' = \frac{P}{2} \frac{f}{f^*} + p,$$

$$p' = \frac{P}{2} \frac{m}{m^*}.$$

Table 13.2

Evolution of the sex ratio in haplo-diploids

Mating		Frequency	Offspring per pair			
♀	♂		male		female	
			$M/+$	+	$M/+$	$+/+$
$M/+$	+	$P(1-p) = P$	$m/2$	$m/2$	$f/2$	$f/2$
$+/+$	M	$p(1-P) = p$	—	m^*	f^*	—
$+/+$	+	$(1-P)(1-p) = 1 - P - p$	—	m^*	—	f^*

Hence, among the offspring:

M males	$Pm/2$,
$M/+$ females	$Pf/2 + pf^*$,
total males	m^* ,
total females	f^* .

Adding these equations gives

$$(P' + p') = \frac{1}{2}P\left(\frac{f}{f^*} + \frac{m}{m^*}\right) + p.$$

This is identical to Equation 13.3, so we again conclude that, at equilibrium, expenditure on sons and daughters is equal.

Exceptions to the rule are of particular interest. In some populations of the mosquito *Aedes aegypti*, there is a great excess of males, caused by a Y-linked gene, *M*. Males carrying this gene produce an excess of sons, because X chromosomes are broken during meiosis, and sperm carrying deficient X chromosomes degenerate. If this were the whole story, natural populations of *Aedes aegypti* would consist mainly of males, and would be close to extinction. However, there are resistant X chromosomes, able to suppress the action of *M*. As one would expect, X chromosomes from Africa and central America, where *M* is commonly found, tend to be resistant.

In this example, Fisher's prediction of a 1:1 ratio fails, because one of its basic assumptions does not hold: the gene *M* does affect its own likelihood of being transmitted. This is an example of **meiotic drive**. Another assumption of Fisher's argument is that all males have an equal chance of mating, as do all females. If this assumption is false, this too can lead to a distorted sex ratio. The phenomenon, first described by Hamilton (1967), is the case of **local mate competition**. Consider a parasitic insect that lays several eggs in a host caterpillar. These eggs hatch and develop within the host, and pupate there. They then emerge and mate with one another before dispersing. If only one female lays eggs in each host, then females are mated by their brothers. What sex ratio should a female produce if she is to maximize the number of genes she transmits to future generations? Clearly, she should produce one son, and the rest of her offspring should be female.

Do parasitic animals with this life history produce female-biased sex ratios? The question has been studied mainly in haplo-diploid organisms, for two reasons. First, many haplo-diploids (particularly hymenoptera, and some mites) have the appropriate life history. Secondly, females of haplo-diploid species can choose the sex of each offspring, by fertilizing, or not fertilizing, each egg. Hence haplo-diploids are ideal for testing ideas about sex ratio evolution. In fact, parasitic haplo-diploids often produce highly female-biased sex ratios, as the theory predicts. The extreme example is a mite, *Acarophenax*, in which the female produces live young: the single male in the litter hatches, mates with his 15 or so sisters, and dies before he is born.

Before we accept these data as confirming theories about local mate competition, we must be satisfied that the biased sex ratios are not a direct adaptation

to haplo-diploidy. Box 13.1 shows that Fisher's prediction of a 1:1 ratio does apply to random-mating haplo-diploids. In fact, haplo-diploids that are neither parasitic nor social do have approximately equal sex ratios.

Selfing and Outcrossing

This section shows that there is strong selection favouring selfing in hermaphrodites. Since, in fact, selfing is the exception rather than the rule, there must also be strong selection against it: the obvious disadvantage is the low fitness of inbred offspring.

Consider first a hermaphroditic plant species, containing selfing and outcrossing individuals. Suppose that an outcrossing individual has, on average, n surviving offspring as the seed parent, and N offspring as the pollen parent. A selfing individual will transmit genes to the next generation in three ways:

- (1) via its own seed;
- (2) via its own pollen that fertilizes its own seed; and
- (3) via its pollen that fertilizes seed from other plants.

If, as seems reasonable, the quantity of pollen needed to fertilize its own seed is negligible compared to the total pollen produced, the selfing plant will have, by route (3), the same number of offspring N , as an outcrossing plant. Hence, in comparing the fitnesses of the two types of plant, we need only take into account routes (1) and (2). If selfing and outcrossing plants produce the same number of seeds, and if those seeds are equally likely to survive to become adults, the selfer will transmit $2n$ genes for every n genes transmitted by the outcrosser. Clearly, selfing would increase in frequency very rapidly.

If the survival chances of offspring produced by selfing is $1 - s$ times that of offspring produced by outcrossing, then selfing will increase in frequency if $2(1 - s) > n$, or $1 - s > 1/2$. This implies that the deleterious effects of inbreeding must be very severe if selfing is not to spread. The real situation is somewhat more complex, because the fitness of selfed progeny will depend on the number of generations of selfing (see Fig. 6.6, p. 104). It seems that, in most hermaphrodite species, the effects of inbreeding are severe enough to prevent the spread of selfing, but that, occasionally, the crisis of inbreeding illustrated in Fig. 6.6 is passed, and a reasonably viable selfing species evolves. Once it has arisen, it is unlikely to revert to outcrossing. Instead, investment in pollen, and in attractive flowers, is reduced.

Hermaphroditism

Why are some species hermaphroditic and others dioecious? To answer this question, we will ask whether a gene for hermaphroditism could invade a dioecious population. This is equivalent to asking whether the reproductive success of a

hermaphrodite, when rare, is greater than that of males and females. In what follows, I assume that hermaphrodites are effectively self-sterile.

Let R_m and R_f be the reproductive success of males and females, respectively. At equilibrium, if there is a 1:1 sex ratio, males and females have equal numbers of offspring: that is $R_m = R_f$. Let the reproductive success of a hermaphrodite be αR_m as a male, and βR_f as a female: α and β represent the potential reproductive success of a hermaphrodite. Clearly, since $R_m = R_f$, hermaphroditism can invade if $\alpha + \beta > 1$. Different hermaphrodites could allocate their resources between male and female functions differently, and so have different values of α and β . The set of all possible hermaphrodites can be represented by the set of all possible pairs of values of α and β . If this set is convex (Fig. 13.A), hermaphroditism will replace dioecy: if the set is concave (Fig. 13.B), dioecy will be evolutionarily stable.

What features will tend to make the set either convex or concave? Probably the main factor making for a convex set is a 'law of diminishing returns' on investment in gametes of a given sex. This supposes that, in a hermaphroditic plant, the production of pollen and of seeds is limited, at least in part, by the same resources. If so, a hermaphrodite might produce exactly half as much pollen as a male, and half as many seeds as a female. However, it is likely that a male that produced twice as much pollen as a hermaphrodite would not pollinate twice as many seeds, because of competition between its own pollen: if a plant produced enough pollen to fertilize all the seeds of a neighbour, it would not gain by producing twice as

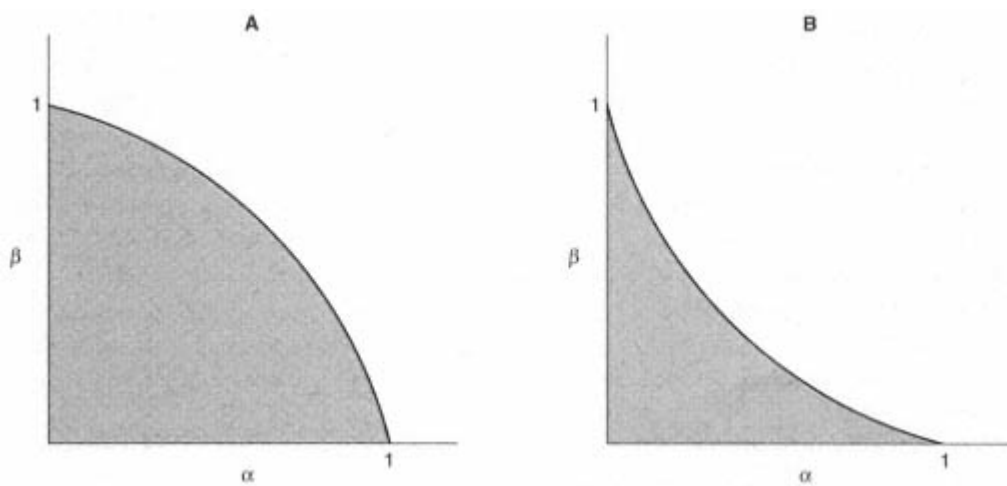


Figure 13.1

The evolution of hermaphroditism. α is a measure of the success of a hermaphrodite as a male, relative to a pure male; β is the breeding success of a hermaphrodite as a female, relative to a pure female.

The shaded areas represent the set of possible phenotypes. For a convex set, **A**, hermaphroditism is the evolutionarily stable state; for a concave set, **B**, dioecy is stable.

much pollen. Because of pollen competition, $\alpha > 1/2$. There may be a similar law of diminishing returns affecting seed production, if the seeds from a single plant compete with one another. If $\alpha\beta > 1/2$. Now if either $\alpha > 1/2$, $\beta = 1/2$; or $\alpha = 1/2$, $\beta > 1/2$; the set is convex, and hermaphroditism is favoured.

A second factor favours hermaphroditism in animal-pollinated plants: the same expenditure on petals and nectar serves both male and female functions. The effect of this is to produce a convex set (see Box 13.2). This may explain why dioecy is rarer in animal-pollinated than in wind-pollinated plants.

Box 13.2—

Resource Allocation in Hermaphrodites

Suppose that a single flower can produce n_1 seeds, each at a cost s , and N_1 pollen grains, each at a cost t , and that the cost of petals, nectar, etc. is u . Then the total cost of a hermaphrodite flower is $u + n_1s + N_1t$.

If the total expenditure by the plant is C , then a female plant produces $Cn_1/(u + n_1s)$ seeds, a male plant produces $CN_1/(u + N_1t)$ pollen grains, and a hermaphrodite produces $Cn_1/(u + n_1s + N_1t)$ seeds, and $CN_1/(u + n_1s + N_1t)$ pollen grains. Hence

$$\alpha = \frac{u + n_1s}{u + n_1s + N_1t} \quad \text{and} \quad \beta = \frac{u + N_1t}{u + n_1s + N_1t}.$$

Hence

$$\alpha + \beta = \frac{2u + n_1s + N_1t}{u + n_1s + N_1t}.$$

Thus $\alpha + \beta > 1$, and the set is convex. Further, the degree of convexity increases with the expenditure on attracting pollinators.

The main factor tending to make the set concave is the existence of organs which serve only one of the two sexual functions. For example, a male red deer which invested half as much in male weapons and excess size would probably obtain fewer than half as many matings: if $\alpha < 1/2$.

It is not surprising that most higher plants are hermaphrodites (strictly, the word hermaphrodite is used only for a plant whose flowers contain both male and female organs; however, monoecious plants, with separate male and female flowers on the same individual, obtain some of the resource-allocation advantages of hermaphrodites, and have the additional advantage of being less likely to self-fertilize). Indeed, it is surprising at first sight that any plants should be dioecious. One reason may be that dioecy prevents self-fertilization. The 'law of diminishing returns' argument is supported by the fact that, in both gymnosperms and angiosperms (Table 13.3), dioecy is commoner in plants with animal-dispersed than wind-dispersed seeds: with wind dispersal (or often, non-dispersal), competition between

Table 13.3

The relation between breeding system and method of seed dispersal in angiosperms; numbers and percentages of angiosperm families (after Givnish 1982)

	Dioecy	Monoecy	Hermaphroditism
Seeds swallowed by animals	17 (27%)	3 (5%)	43 (68%)
Other modes of dispersal	10 (7%)	15 (11%)	117 (82%)

offspring of the same parent is likely to be more intense. Givnish (who collected the data in Table 13.3) has also argued that, in a species that attracts frugivores, a plant that produces twice as many fruits will attract more than twice as many animals, an increasing return on investment that would favour dioecy.

Sexual Selection

Darwin suggested the process of sexual selection to meet a difficulty that confronted his concept of natural selection. This difficulty is that some characteristics of animals do not seem to contribute either to their own survival, or that of their offspring. Such traits are the antlers of deer, and the plumage of male pheasants, peacocks, and birds of paradise. One might argue that antlers can be used in defence against predators: but if so one would expect them to be present in females also. Darwin suggested that these characteristics evolved because of selection operating during the acquisition of mates. He saw this as happening in one of two ways:

1. Competition between members of the same sex for access to mates. Usually, but not always, there is competition between males for access to females. This is because, in many species, the fecundity of a female is limited by her ability to lay eggs (or, in mammals, to bear and nurse her young), so that her fecundity would not be increased by access to more males, although, as Darwin pointed out, the fitness of her offspring could be affected by which particular male she mated with. In contrast, in a species with no paternal care, a male's fitness is increased by access to more females. Hence male competition for access to females is common. This analysis is borne out by the fact that in some species, such as sea horses and jacanas (wading birds), in which parental care is by the males only, females compete for access to males.
2. Choice by members of one sex of particular members of the opposite sex. In polygamous species, the choice is by females of males, but in monogamous species it may involve both sexes.

The first of these two processes does not present any special difficulty. The 'size game' described in Chapter 7 was originally developed as a model of male-male competition. The process can lead to an increase in size or weapons in one sex well

beyond the level that would evolve in the absence of such intra-sex competition. It is also likely to lead to sexual dimorphism in size. Figure 13.2 shows that, in primates, sexual dimorphism in size is greater in species in which a male can have access to more than one female than it is in monogamous species. The figure also shows that, in polygynous species, the degree of sexual dimorphism increases with size. It is not obvious why this should be so. One possible explanation is as follows. In polygynous species, male-male competition for mates leads to an increase in male size. Since most genes affecting size will affect both males and females, there will be a size increase in both sexes. But, because natural selection favours smaller size, and some genes have sex-limited effects, there will also be an increase in dimorphism. Hence both size and dimorphism will increase in evolutionary time: the species lying on the sloping regression line represent different stages in an evolutionary progression.

There are greater difficulties with Darwin's idea of choice. By female choice we mean any structure or behaviour in a female that makes her more likely to mate

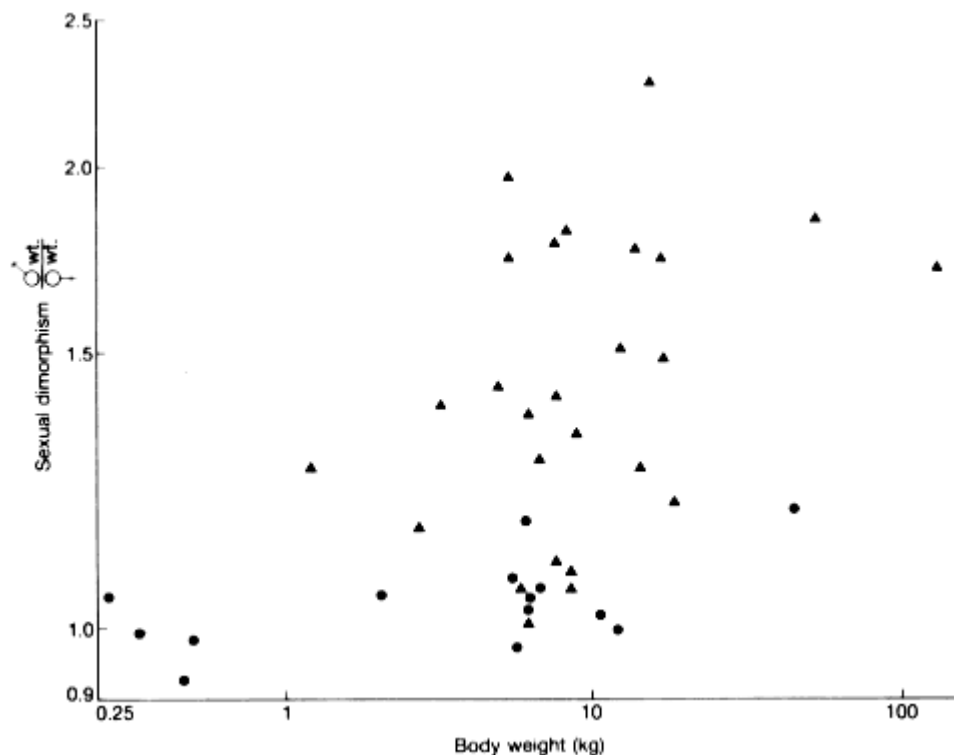


Figure 13.2

Sexual dimorphism and body weight in primates. The 'socionomic sex ratio' is the mean number of adult females per adult male in breeding groups. Species with a ratio of one (usually, the breeding group is a monogamous pair); ▲, species with a ratio greater than one. (Adapted from Clutton-Brock *et al.* 1977.)

with one male than another. I first give some evidence that choice in this sense exists, and then discuss why it has evolved. In *Drosophila subobscura*, when an unmated outbred female is placed with a young outbred male, mating almost always takes place within an hour, and usually within 15 minutes. If the male was inbred, however, mating took place in only about 50 per cent of cases within one hour (Maynard Smith 1956). This was not because the inbred males did not court or attempt to mount the females: they made repeated attempts to do so. The explanation lies in the courtship dance of the female. When approached by a male, she moves rapidly from side to side. The male attempts to move so as to keep facing her. If he succeeds, she stands still and allows the male to mate. If he lags too far behind, she moves away, and kicks off the male if he attempts to mount (as inbred males often did). Inbred males failed to mate because they failed to keep facing the female. This conclusion is confirmed by the fact that old males, and males carrying mutations affecting their locomotion or vision, also fail to mate.

The courtship dance of *D. subobscura* meets the definition of choice given above: it is a behaviour that makes a female more likely to mate with some males than others. However, the experiment did not demonstrate any genetic variance in choice: this is important, because a trait will only evolve if it is heritable. The existence of heritable variability in choice has been demonstrated in the ladybird *Adalia bipunctata*, by Majerus *et al.* (1983). In natural populations, melanic males have a higher mating success than non-melanics. Selection experiments, in which females that mated with melanics were chosen as parents of the next generation, demonstrated that there are genetic differences between females in the degree of preference they show for melanic males.

There is no reason to doubt, then, that female choice exists, and that it is variable. But why should it evolve? Darwin himself did not answer this question. There are a number of possible answers:

1. The choice is an unselected consequence of characteristics that evolved for other reasons. For example, female natterjack toads move towards calling males. By using loudspeakers, Arak (1983) showed that females move up a sound gradient. This would lead them to mate preferentially with the loudest males, but the behaviour could have evolved merely to ensure that the female mates with some male.
2. Females have been selected to mate with members of their own rather than other species. This is discussed further on p. 270. There can be little doubt that selection for species recognition occurs, and may have unselected consequences on choice between conspecifics. But it is hard to believe that such selection could, by itself, lead to the extreme elaboration of secondary sexual characters that worried Darwin.
3. Females have been selected to mate with males of high viability. If, for example, males differ in their load of deleterious genes, then a female that was able to

detect these differences, and mate with a male with few deleterious genes, would have offspring of higher-than-average fitness. Hence a gene causing her to choose would be transmitted to more descendants. It is tempting to interpret the behaviour of female *D. subobscura* in this way, although there is no evidence that, in nature, males with the athletic ability required for the courtship dance have, on average, fitter offspring. However, many animal courtships have the effect of testing the sensory and locomotor skills of the participants, and female choice of males of high viability is a plausible explanation.

4. If males with some trait, T , are, for whatever reason, preferred by females, then they get more matings, and are therefore fitter than males without the trait. Therefore it pays females to mate with T males. Hence both T , and the preference for T , are selected. This idea is due to R.A. Fisher, who thought that it could lead to a runaway process, in which both trait and preference were further elaborated. Fisher recognized that his process had to get started in some way: that is, there had to be some female preference initially present to prime the process. This initial preference could arise for any one of the three reasons listed above, but once it was present, Fisher argued, it could lead to extreme exaggeration of both the selected trait and the preference.

Fisher's argument is hard to follow. A number of attempts have been made to model it (e.g. O'Donald 1967; Lande 1981a; Kirkpatrick 1982). Box 13.3 describes Kirkpatrick's model. The consensus of a number of recent papers seems to be as follows. Given some initial preference for a trait T , sexual selection can cause it to evolve to a degree different from that favoured by viability selection alone. There is still debate about the significance of the line of equilibria, which is a feature of the models of Lande and Kirkpatrick. There is also debate about the extent to which an association will evolve between sexually selected traits, and traits that confer high viability in other contexts.

There is one feature common to these models of sexual selection that it is important to grasp. The gene or genes responsible for female choice do not increase the fitness of females that carry them. Either the choice genes have no effect on female fitness, as in the early models referred to above, or, if there is a cost to choice, in time expended or in other ways, the choice genes actually lower female fitness. They are not expressed in males at all. How, then, can selection increase their frequency? The answer is that, because of assortative mating, the choice genes are in linkage disequilibrium with genes that do affect fitness (remember that linkage disequilibrium does not require that the genes be on the same chromosome). For example, in Kirkpatrick's model, the choice gene P , and the gene determining the selected trait, T , will be positively associated, because P females tend to mate with T males. If T males are fitter than non- T (because of their high mating success), then as T increases in frequency, P will increase also because of linkage disequilibrium.

Box 13.3— A Model of Sexual Selection

Kirkpatrick (1982) considers a sexual haploid population with two loci, and two alleles at each locus. Allele T determines the presence of some trait—say a long tail—and t its absence. The locus is not expressed in females. T males have a viability $1 - s$ times that of t males.

Females with allele P mate preferentially with T males. p females mate randomly. The locus is not expressed in males. Preference works as follows: if p_T is the frequency of T , and $1 - p_T$ of t , then a P female mates with a T male with probability $\alpha p_T / (\alpha p_T + 1 - p_T)$. Then $\alpha > 1$ implies a preference for T .

This model can be treated analytically: it is also easy to simulate. The results are shown in Fig. 13.3.

Lande (1981a) analysed a polygenic model. He also concluded that there would be a line of equilibria. However, in his model, the line could be unstable.

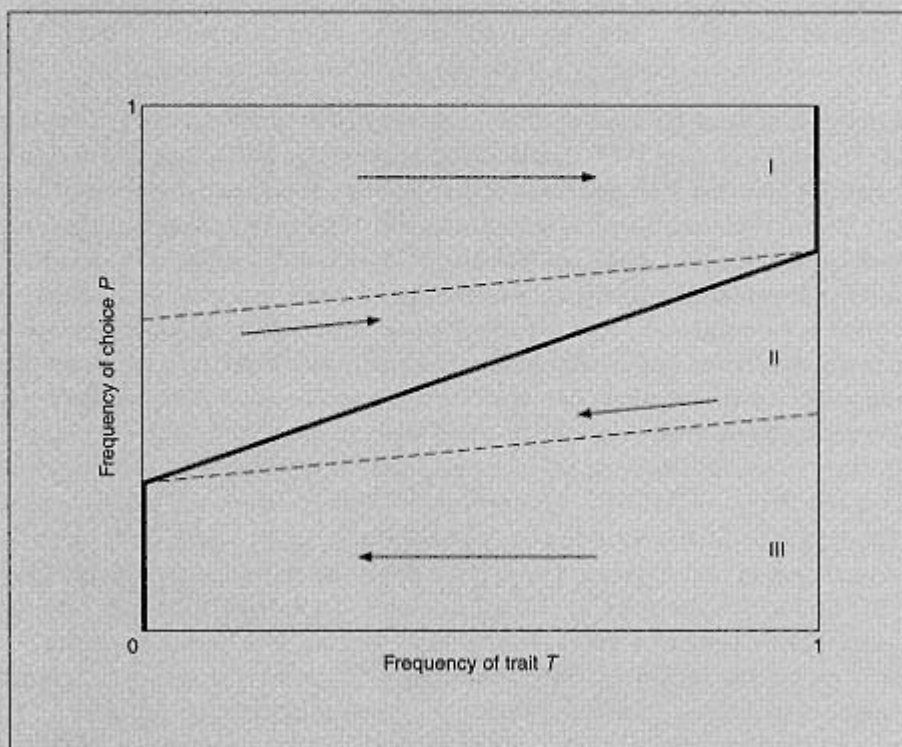


Figure 13.3 The dynamic behaviour of Kirkpatrick's model. The bold line represents the possible equilibrium states. If initial conditions lie in region I (choice common), then T is fixed; if they lie in region III (choice rare), then the more viable genotype, t , is fixed; if initial conditions lie in the intermediate region II, the population evolves to lie on the line of equilibria: if it is then disturbed,